

Electronic Companion - Technical Proofs

(Saving Kermit: Dynamic Assortment Planning in a Boiling Market)

Guide to the proofs

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E.C.1 Proofs for Section 4

E.C.1.1 Proof of Lemma 1

Proof. Fix an assortment strategy $\pi \in \mathcal{P}$ and two paths of preferences $F^{(\mathbb{N})}, G^{(\mathbb{N})} \in \mathcal{F}$. Let $i^{\pi,t} \in S^{\pi,t} \cup \{0\}$ be the purchase outcome in period t , where $i^{\pi,t} = 0$ represents no purchase. Let $\mathcal{H}^{\pi,t} := ((S^{\pi,s}, i^{\pi,s}) : s \in [t-1])$ be the sales history up to period $t-1$. Under a fixed strategy π , the assortment $S^{\pi,t}$ is selected from the past history $\mathcal{H}^{\pi,t}$. Thus, when comparing $F^{(\mathbb{N})}$ and $G^{(\mathbb{N})}$, the same assortment-selection rule is used under both paths. Conditional on $\mathcal{H}^{\pi,t}$, both paths induce the same assortment $S^{\pi,t}$ up to some randomization mechanism. The only difference in the period t sales-history distribution therefore comes from the conditional distribution of the purchase outcome $i^{\pi,t}$ given $S^{\pi,t}$. We formalize this observation below.

Fix a realization $h^{T+1} = ((S^t, i^t) : t \in [T])$ of the sales history up to period T . Under $F^{(\mathbb{N})}$, and by using an argument similar to that in Lemma 14, the probability of observing that exact realization of the sales history is given by the following identity:

$$\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}[\mathcal{H}^{\pi,T+1} = h^{T+1}] = \prod_{t=1}^T \pi_t(S^t \mid h^t) p_{i^t}(S^t, F^t),$$

where $\pi_t(S^t \mid h^t)$ corresponds to the probability of offering assortment $S^t \in \mathcal{S}$ to customer t conditional on history h^t up to customer $t-1$. An analogous expression holds under $G^{(\mathbb{N})}$.

Since the assortment-selection term is the same under the two paths, the strategy terms cancel in the likelihood ratio. For every sales history with positive probability under $\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}$, we obtain:

$$\frac{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}[\mathcal{H}^{\pi,T+1} = h^{T+1}]}{\mathbb{P}_{G^{(\mathbb{N})}}^{\pi,T}[\mathcal{H}^{\pi,T+1} = h^{T+1}]} = \prod_{t=1}^T \frac{p_{i^t}(S^t, F^t)}{p_{i^t}(S^t, G^t)}.$$

Taking logarithms on both sides gives the following identity:

$$\log \left(\frac{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}[\mathcal{H}^{\pi,T+1} = h^{T+1}]}{\mathbb{P}_{G^{(\mathbb{N})}}^{\pi,T}[\mathcal{H}^{\pi,T+1} = h^{T+1}]} \right) = \sum_{t=1}^T \log \left(\frac{p_{i^t}(S^t, F^t)}{p_{i^t}(S^t, G^t)} \right).$$

By the definition of KL divergence, we obtain:

$$D_{\text{KL}}(\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T} \parallel \mathbb{P}_{G^{(\mathbb{N})}}^{\pi,T}) = \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} \left[\log \left(\frac{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}[\mathcal{H}^{\pi,T+1}]}{\mathbb{P}_{G^{(\mathbb{N})}}^{\pi,T}[\mathcal{H}^{\pi,T+1}]} \right) \right].$$

Substituting the likelihood-ratio decomposition and using linearity of expectation gives:

$$D_{\text{KL}}(\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T} \parallel \mathbb{P}_{G^{(\mathbb{N})}}^{\pi,T}) = \sum_{t=1}^T \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} \left[\log \left(\frac{p_{i^{\pi,t}}(S^{\pi,t}, F^t)}{p_{i^{\pi,t}}(S^{\pi,t}, G^t)} \right) \right].$$

Fix $t \in [T]$ and $Z_t \equiv \log(p_{i^{\pi,t}}(S^{\pi,t}, F^t)) - \log(p_{i^{\pi,t}}(S^{\pi,t}, G^t))$. We condition on the assortment $S^{\pi,t}$. For $S \in \mathcal{S}$ with $\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}[S^{\pi,t} = S] > 0$, the conditional distribution of $i^{\pi,t}$ under $\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}$ given $S^{\pi,t} = S$ is the categorical distribution on $S \cup \{0\}$ with probabilities $(p_i(S, F^t) : i \in S \cup \{0\})$. Hence:

$$\mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} [Z_t \mid S^{\pi,t} = S] = \sum_{i \in S \cup \{0\}} p_i(S, F^t) \log \left(\frac{p_i(S, F^t)}{p_i(S, G^t)} \right) =: \mathcal{K}_{F^{(\mathbb{N})}, G^{(\mathbb{N})}}^t(S).$$

Therefore, one has $\mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} [Z_t \mid S^{\pi,t}] = \mathcal{K}_{F^{(\mathbb{N})}, G^{(\mathbb{N})}}^t(S^{\pi,t})$. Using the law of total expectations (Jacod

and Protter 2004), we obtain the following identities:

$$\mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} [Z_t] = \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} \left[\mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} [Z_t \mid S^{\pi,t}] \right] = \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} [\mathcal{K}_{F^{(\mathbb{N})},G^{(\mathbb{N})}}^t(S^{\pi,t})].$$

Substituting this identity for each $t \in [T]$ and then summing gives the following identity:

$$D_{\text{KL}}(\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T} \parallel \mathbb{P}_{G^{(\mathbb{N})}}^{\pi,T}) = \sum_{t=1}^T \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} [\mathcal{K}_{F^{(\mathbb{N})},G^{(\mathbb{N})}}^t(S^{\pi,t})] = \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi,T}} \left[\sum_{t=1}^T \mathcal{K}_{F^{(\mathbb{N})},G^{(\mathbb{N})}}^t(S^{\pi,t}) \right],$$

which proves the result. ■

E.C.1.2 Proofs for Remark 1

We clarify what the magnitude $\mathcal{M}(F^{(\mathbb{N})}, T)$ captures by comparing it with the variation measure developed by Foussoul et al. (2023) specifically for evolving MNL preferences. Their measure tracks changes in the vector of attraction parameters, or equivalently in the utility parameters, whereas ours records only the resulting changes in purchasing distributions.

Let $\mathcal{F}_{\text{MNL}}$ be the set of paths in which customers' preferences follow an MNL choice model in every period. For a path $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$, let $\nu^t \equiv \nu(F^t) := (\nu_i^t : i \in \mathcal{N})$ be the attraction vector at time t , so that $p_i(S, F^t) = \nu_i^t (1 + \sum_{j \in S} \nu_j^t)^{-1}$. The variation measure of Foussoul et al. (2023) is defined by $\Delta_{\infty}^K(F^{(\mathbb{N})}, T) := \sum_{t=2}^T \max \{ \sum_{i \in S} |\nu_i^t - \nu_i^{t-1}| : |S| \leq 2K \}$.

Proposition 4. *Fix a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$ and suppose that $\nu_i^t \geq \underline{\nu} > 0$ for every product $i \in \mathcal{N}$ and every customer $t \in [T]$. Then, $\mathcal{M}(F^{(\mathbb{N})}, T) \leq (\Delta_{\infty}^K(F^{(\mathbb{N})}, T))^2 / 2\underline{\nu}^2$.*

Proof. Fix a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$, characterized by attraction parameters $(\nu_i^t : t \geq 1, i \in \mathcal{N})$ satisfying the condition in the statement of the proposition, i.e., $\nu_i^t \geq \underline{\nu} > 0$ for all $t \in [T]$ and $i \in \mathcal{N}$. For each time period $t \in [T]$ and product $i \in \mathcal{N}$, define $\mu_i^t = \log \nu_i^t$. For a feasible assortment $S \in \mathcal{S}$, define $A(S, \mu) = \log(1 + \sum_{i \in S} \exp(\mu_i))$. Under the MNL choice model, the purchasing probability of a product $i \in S$ satisfies:

$$p_i(S, F^t) := p_i(S, \mu^t) = \frac{\exp(\mu_i^t)}{1 + \sum_{j \in S} \exp(\mu_j^t)} = \exp(\mu_i^t - A(S, \mu^t)),$$

and the no-purchase probability is:

$$p_0(S, F^t) := p_0(S, \mu^t) = \frac{1}{1 + \sum_{j \in S} \exp(\mu_j^t)} = \exp(-A(S, \mu^t)).$$

For $t \geq 2$, we introduce $h^t = \mu^t - \mu^{t-1}$. We first expand $\mathcal{K}^t(S)$. By definition of the KL divergence, we obtain the following identity:

$$\mathcal{K}^t(S) = \sum_{i \in S} p_i(S, F^t) \log \left(\frac{p_i(S, F^t)}{p_i(S, F^{t-1})} \right) + p_0(S, F^t) \log \left(\frac{p_0(S, F^t)}{p_0(S, F^{t-1})} \right).$$

For every product $i \in S$, the following identity is satisfied:

$$\log \left(\frac{p_i(S, F^t)}{p_i(S, F^{t-1})} \right) = (\mu_i^t - A(S, \mu^t)) - (\mu_i^{t-1} - A(S, \mu^{t-1})) = h_i^t + A(S, \mu^{t-1}) - A(S, \mu^t).$$

Similarly, the following identity is satisfied:

$$\log \left(\frac{p_0(S, F^t)}{p_0(S, F^{t-1})} \right) = -A(S, \mu^t) + A(S, \mu^{t-1}).$$

Substituting these two identities into the definition of $\mathcal{K}^t(S)$ gives us:

$$\begin{aligned}\mathcal{K}^t(S) &= \sum_{i \in S} p_i(S, F^t) (h_i^t + A(S, \mu^{t-1}) - A(S, \mu^t)) + p_0(S, F^t) (A(S, \mu^{t-1}) - A(S, \mu^t)) \\ &= \sum_{i \in S} p_i(S, F^t) h_i^t + \left(\sum_{i \in S} p_i(S, F^t) + p_0(S, F^t) \right) (A(S, \mu^{t-1}) - A(S, \mu^t)) \\ &= \sum_{i \in S} p_i(S, F^t) h_i^t + A(S, \mu^{t-1}) - A(S, \mu^t),\end{aligned}$$

where the last equality follows from the fact that $\sum_{i \in S \cup \{0\}} p_i(S, \mu^t) = 1$. Since $\nabla_i A(S, \mu^t) = p_i(S, \mu^t)$, for all $i \in S$, we obtain the following identity:

$$\mathcal{K}^t(S) = A(S, \mu^{t-1}) - A(S, \mu^t) + \nabla A(S, \mu^t)^\top h^t(S),$$

where $h^t(S) \in \mathbb{R}^S$ corresponds to the vector induced by h^t such that $h_i^t(S) = h_i^t$ for all $i \in S$.

Next, given products $i, j \in S$, the purchasing probabilities satisfy:

$$\frac{\partial p_i(S, \mu)}{\partial \mu_j} = p_i(S, \mu) (\mathbf{1}\{i = j\} - p_j(S, \mu)).$$

Therefore, $\nabla^2 A(S, \mu) = \Sigma(S, \mu) = \text{diag}(p(S, \mu)) - p(S, \mu)p(S, \mu)^\top$. Next, we introduce the function $g : [0, 1] \rightarrow \mathbb{R}$ by $g(\theta) = A(S, \mu^{t-1} + \theta h^t)$, for all $\theta \in [0, 1]$. By the chain rule, the first and second derivatives of g are given by the following two identities:

$$g'(\theta) = \nabla A(S, \mu^{t-1} + \theta h^t)^\top h^t(S) \quad \text{and} \quad g''(\theta) = h^t(S)^\top \Sigma(S, \mu^{t-1} + \theta h^t) h^t(S),$$

respectively. Using Taylor's theorem with integral remainder, we obtain the following identity:

$$g(0) - g(1) + g'(1) = \int_0^1 \theta g''(\theta) d\theta,$$

and recalling that $\mathcal{K}^t(S) = A(S, \mu^{t-1}) - A(S, \mu^t) + \nabla A(S, \mu^t)^\top h^t(S)$, we obtain the following identity:

$$\mathcal{K}^t(S) = \int_0^1 \theta h^t(S)^\top \Sigma(S, \mu^{t-1} + \theta h^t) h^t(S) d\theta.$$

By definition of the matrix $\Sigma(S, \mu)$, one has $(h^t(S))^\top \Sigma(S, \mu) h^t(S) \leq \|h^t(S)\|_2^2$. Therefore, $\mathcal{K}^t(S) \leq \|h^t(S)\|_2^2/2$. Moreover, since $x \mapsto \log x$ is $1/\underline{\nu}$ -Lipschitz on $[\underline{\nu}, \infty)$, for every $i \in \mathcal{N}$,

$$|h_i^t| = \left| \log \nu_i^t - \log \nu_i^{t-1} \right| \leq \frac{|\nu_i^t - \nu_i^{t-1}|}{\underline{\nu}}.$$

It follows that the following inequality is satisfied:

$$\mathcal{K}^t(S) \leq \frac{1}{2} \sum_{i \in S} |h_i^t|^2 \leq \frac{1}{2\underline{\nu}^2} \sum_{i \in S} |\nu_i^t - \nu_i^{t-1}|^2 \leq \frac{1}{2\underline{\nu}^2} \left(\sum_{i \in S} |\nu_i^t - \nu_i^{t-1}| \right)^2 \leq \frac{1}{2\underline{\nu}^2} (d^t)^2,$$

where we denote by $d^t := \max \{ \sum_{i \in S} |\nu_i^t - \nu_i^{t-1}| : S \in \mathcal{S} \}$.

Taking the maximum over $S \in \mathcal{S}$ and summing over $t \in \{2, \dots, T\}$ gives us:

$$\mathcal{M}(F^{(\mathbb{N})}, T) = \sum_{t=2}^T \max \{ \mathcal{K}^t(S) : S \in \mathcal{S} \} \leq \frac{1}{2\underline{\nu}^2} \sum_{t=2}^T (d^t)^2.$$

Finally, applying the inequality $\sum_t a_t^2 \leq (\sum_t a_t)^2$ to the non-negative sequence $(d^t : t \geq 1)$, we obtain $\sum_{t=2}^T (d^t)^2 \leq (\sum_{t=2}^T d^t)^2 \leq (\Delta_\infty^K(F^{(\mathbb{N})}, T))^2$, where we use that $\max \{ \sum_{i \in S} |\nu_i^t - \nu_i^{t-1}| : S \in \mathcal{S} \} \leq \max \{ \sum_{i \in S} |\nu_i^t - \nu_i^{t-1}| : |S| \leq 2K \}$. Combining the last two inequalities proves the claim. ■

Proposition 4 relates our magnitude measure to the variation measure of Foussoul et al. (2023).

For MNL preferences, variation in the utility parameters, as measured by $\Delta_\infty^K(F^{(\mathbb{N})}, T)$, controls the induced variation in purchasing distributions and therefore upper-bounds $\mathcal{M}(F^{(\mathbb{N})}, T)$. The converse does not generally hold: $\Delta_\infty^K(F^{(\mathbb{N})}, T)$ can assign different values to paths of customers' preferences with the same magnitude $\mathcal{M}(F^{(\mathbb{N})}, T)$.

Proposition 5. *Assume that $K = 1$ and $T \geq 3$. There exist two paths $F_1^{(\mathbb{N})}, F_2^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$ with the same number of changes such that $\Delta_\infty^1(F_1^{(\mathbb{N})}, T) = \Delta_\infty^1(F_2^{(\mathbb{N})}, T)$, but $\mathcal{M}(F_1^{(\mathbb{N})}, T) < \mathcal{M}(F_2^{(\mathbb{N})}, T)$.*

Proof. Fix $a > 0$, $\varepsilon > 0$, and an integer $2 \leq m \leq T - 1$. We consider the following subclass of MNL preferences. Fix $K = 1$ so that $\mathcal{S} = \{\{i\} : i \in \mathcal{N}\}$. Then, let $\tilde{\mathcal{F}}_{\text{MNL}}(a, \varepsilon, m) \subseteq \mathcal{F}_{\text{MNL}}$ be the set of paths $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$ such that only product 1's attraction changes, and $\nu_1(F^1) = a$, $\nu_1(F^{m+1}) = a + \varepsilon$ with $\nu_1(F^{t+1}) = \nu_1(F^t)$, for all $t \geq m + 1$. All other product attractions are constant over time, i.e., $\nu_i(F^t) = \nu_j(F^t) = \nu_j(F^{t+1}) < a$ for all $t \in \mathbb{N}$ and for all $i, j \in \mathcal{N} \setminus \{1\}$. Next, we introduce the function $\ell : \mathbb{R}_{>0} \times \mathbb{R}_{>0} \rightarrow \mathbb{R}$ as follows. For $x, y > 0$, we define $\ell(x, y)$ by:

$$\begin{aligned} \ell(x, y) &= \frac{x}{1+x} \log\left(\frac{x/(1+x)}{y/(1+y)}\right) + \frac{1}{1+x} \log\left(\frac{1/(1+x)}{1/(1+y)}\right) \\ &= \frac{x}{1+x} \log\left(\frac{x}{y}\right) + \log\left(\frac{1+y}{1+x}\right). \end{aligned}$$

Thus, $\ell(x, y)$ essentially represents the KL divergence between the purchasing distribution induced by offering a single product with attraction x and with attraction y .

We first prove an exact inequality related to ℓ . For every $0 < x < y < z$, one has:

$$\ell(z, x) - \ell(z, y) - \ell(y, x) = \left(\frac{z}{1+z} - \frac{y}{1+y}\right) \log\left(\frac{y}{x}\right) > 0.$$

Indeed, the equality follows by substituting the expression for $\ell(\cdot, \cdot)$. The inequality is strict because $z > y > x > 0$, the map $u \mapsto u/(1+u)$ is strictly increasing on $(0, \infty)$, and $\log(y/x) > 0$.

Define the path $F_1^{(\mathbb{N})} \in \tilde{\mathcal{F}}_{\text{MNL}}(a, \varepsilon, m)$ by $\nu_1(F_1^t) = a + (t-1)\varepsilon/m$ for $t \in [m+1]$, $\nu_1(F_1^{t+1}) = \nu_1(F_1^t)$ for all $t \geq m+1$, and keep all other product attractions constant over time. Next, fix $\delta > 0$ such that $0 < \delta < \varepsilon/(m-1)$. Define the path $F_2^{(\mathbb{N})}(\delta)$ by $\nu_1(F_2^1(\delta)) = a$, $\nu_1(F_2^2(\delta)) = a + \varepsilon - (m-1)\delta$ and, for every $t \in \{3, \dots, m+1\}$, $\nu_1(F_2^t(\delta)) = a + \varepsilon - (m+1-t)\delta$. Again, $\nu_1(F_2^{t+1}(\delta)) = \nu_1(F_2^t(\delta))$ for all $t \geq m+1$ and all other product attractions are constant over time. Both paths belong to $\tilde{\mathcal{F}}_{\text{MNL}}(a, \varepsilon, m)$, both paths start with the same attraction parameters, both paths end with the same attraction parameters, and both paths have exactly m non-zero changes in their attraction parameters.

Because only product 1's attraction changes, the definition of Δ_∞^1 gives:

$$\Delta_\infty^1(F_1^{(\mathbb{N})}, T) = \sum_{t=2}^T |\nu_1(F_1^t) - \nu_1(F_1^{t-1})| = \sum_{t=2}^{m+1} \frac{\varepsilon}{m} = \varepsilon.$$

Similarly, the following equality is satisfied:

$$\Delta_\infty^1(F_2^{(\mathbb{N})}(\delta), T) = (\varepsilon - (m-1)\delta) + \sum_{t=3}^{m+1} \delta = (\varepsilon - (m-1)\delta) + (m-1)\delta = \varepsilon.$$

We now compare the magnitude. Since all singleton assortments except $\{1\}$ induce time-invariant

purchasing distributions, every non-zero term in the maximum defining the magnitude $\mathcal{M}(\cdot, T)$ is attained at the assortment $\{1\}$. Therefore, we obtain the following identity:

$$\mathcal{M}(F_1^{(\mathbb{N})}, T) = \sum_{r=1}^m \ell(a + \frac{r}{m}\varepsilon, a + \frac{r-1}{m}\varepsilon).$$

Applying the strict inequality with ℓ recursively to the chain $a < a + \frac{1}{m}\varepsilon < \dots < a + \varepsilon$ gives:

$$\ell(a + \varepsilon, a) > \sum_{r=1}^m \ell(a + \frac{r}{m}\varepsilon, a + \frac{r-1}{m}\varepsilon) = \mathcal{M}(F_1^{(\mathbb{N})}, T).$$

For the second path of customers' preferences, one has:

$$\mathcal{M}(F_2^{(\mathbb{N})}(\delta), T) = \ell(a + \varepsilon - (m-1)\delta, a) + \sum_{r=1}^{m-1} \ell(a + \varepsilon - (m-1-r)\delta, a + \varepsilon - (m-r)\delta).$$

The function ℓ is continuous on $(0, \infty)^2$. Hence,

$$\lim_{\delta \downarrow 0} \mathcal{M}(F_2^{(\mathbb{N})}(\delta), T) = \ell(a + \varepsilon, a) + \sum_{r=1}^{m-1} \ell(a + \varepsilon, a + \varepsilon) = \ell(a + \varepsilon, a).$$

Combining the preceding two inequalities, we obtain $\lim_{\delta \downarrow 0} \mathcal{M}(F_2^{(\mathbb{N})}(\delta), T) > \mathcal{M}(F_1^{(\mathbb{N})}, T)$. Therefore, there exists $\bar{\delta} > 0$ such that, for every $0 < \delta < \min\{\bar{\delta}, \varepsilon/(m-1)\}$, one has $\mathcal{M}(F_2^{(\mathbb{N})}(\delta), T) > \mathcal{M}(F_1^{(\mathbb{N})}, T)$. Choosing such a δ and setting $F_2^{(\mathbb{N})} = F_2^{(\mathbb{N})}(\delta)$ completes the proof. $\blacksquare$

Proposition 5 shows that variation in utility parameters can assign the same value to paths of customers' preferences that impose different learning burdens on the retailer. Dynamic regret guarantees indexed by $\Delta_{\infty}^K(F^{(\mathbb{N})}, T)$, of the form $\mathcal{O}(T^{\alpha}(\Delta_{\infty}^K(F^{(\mathbb{N})}, T))^{\beta})$, would therefore classify these paths as equally difficult. By contrast, the magnitude $\mathcal{M}(F^{(\mathbb{N})}, T)$ distinguishes them, leading to different regret orders in bounds of the form $\mathcal{O}(T^{\alpha}(\mathcal{M}(F^{(\mathbb{N})}, T))^{\beta})$.

E.C.1.3 Proof of Remark 2

Lemma 2. $\min\{\mathcal{K}^{\tau}(S) \mid S \in \mathcal{S}\} \leq D_{\text{KL}}(\mathbb{P}_{F^{(\mathbb{N})}}^{\pi, T} \parallel \mathbb{P}_{F^{(\mathbb{N}-1)}}^{\pi, T}) \leq \max\{\mathcal{K}^{\tau}(S) \mid S \in \mathcal{S}\}.$

Proof. Fix an admissible strategy $\pi \in \mathcal{P}$ and a path of abruptly changing preferences $F^{(\mathbb{N})} \in \mathcal{F}_A(\gamma_T)$ with change time τ . We restrict our attention to the case $\tau \in \{2, \dots, T\}$. Given $F^{(\mathbb{N})}$, define the one-period shifted preferences path $F^{(\mathbb{N}-1)} \equiv G^{(\mathbb{N})} := (G^t : t \in \mathbb{N})$ by $G^1 = F^1$ and $G^t = F^{t-1}$ for $t \geq 2$. By applying Lemma 1, one has $D_{\text{KL}}(\mathbb{P}_{F^{(\mathbb{N})}}^{\pi, T} \parallel \mathbb{P}_{F^{(\mathbb{N}-1)}}^{\pi, T}) \equiv \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi, T}} \left[\sum_{t=2}^T \mathcal{K}^t(S^{\pi, t}) \right].$

Since $F^{(\mathbb{N})} \in \mathcal{F}_A(\gamma_T)$ has a single abrupt change at time τ , preferences are constant before the change and constant after the change. Hence, for every $t \neq \tau$ and every $S \in \mathcal{S}$, one has $F^t = F^{t-1}$ so that $\mathcal{K}^t(S) = 0$. Therefore, we obtain $D_{\text{KL}}(\mathbb{P}_{F^{(\mathbb{N})}}^{\pi, T} \parallel \mathbb{P}_{F^{(\mathbb{N}-1)}}^{\pi, T}) = \mathbb{E}_{\mathbb{P}_{F^{(\mathbb{N})}}^{\pi, T}} [\mathcal{K}^{\tau}(S^{\pi, \tau})]$. By definition, for every feasible assortment $\tilde{S} \in \mathcal{S}$, the following two inequalities are satisfied:

$$\min\{\mathcal{K}^{\tau}(S) : S \in \mathcal{S}\} \leq \mathcal{K}^{\tau}(\tilde{S}) \leq \max\{\mathcal{K}^{\tau}(S) : S \in \mathcal{S}\}.$$

Applying this inequality to $S = S^{\pi, \tau}$ and taking expectations under $\mathbb{P}_{F^{(\mathbb{N})}}^{\pi, T}$ gives:

$$\min\{\mathcal{K}^{\tau}(S) : S \in \mathcal{S}\} \leq D_{\text{KL}}(\mathbb{P}_{F^{(\mathbb{N})}}^{\pi, T} \parallel \mathbb{P}_{F^{(\mathbb{N}-1)}}^{\pi, T}) \leq \max\{\mathcal{K}^{\tau}(S) : S \in \mathcal{S}\}.$$

These inequalities prove in turn the result. $\blacksquare$

E.C.2 Proofs for Section 5

E.C.2.1 Proof of Proposition 1

The proof follows the logic of Besbes and Zeevi (2011) for dynamic pricing with abruptly changing demand. The assortment setting, however, requires a different construction because the retailer chooses from a discrete set of feasible assortments rather than from a continuous price range.

Proof. Let $\pi \in \mathcal{P}$ be an admissible strategy characterized by the (possibly random) assortment mapping $\psi_t(\mathcal{H}^t) \in \mathcal{S}$ for each $t \in [T]$. Fix a path of preferences $F^{(\mathbb{N})} \in \mathcal{F}_D(\gamma_T, \varepsilon)$ in which the change time $\tau \geq 2$ is to be specified later. We denote by F^1 the preferences before the change occurs and F^τ those after the change occurs. Specifically, for all $t \in \mathbb{N}$, one has $F^t \equiv F^1$ if $t < \tau$ and $F^t \equiv F^\tau$ otherwise.

Let $\mathcal{J}^\pi(F^{(\mathbb{N})}, T)$ be the random variable defined over the probability space $(\Omega, \mathcal{B}, \mathbb{P})$ that captures the strategy's realized revenue (see the discussion in Lemma 13). For $\eta > 0$, we introduce the event:

$$B_\eta := \left\{ \omega \in \Omega : J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T) < \eta \right\}.$$

We define $j_0 := \lceil \eta / \gamma_T \rceil$. Moreover, we introduce $\hat{k}_1, \dots, \hat{k}_T$, which are defined as follows:

$$\hat{k}_1 := \min \left\{ 1 \leq t \leq T : \left\{ \psi_t(\mathcal{H}^t) = S^*(F^\tau) \right\} \text{ or } \{t = T\} \right\},$$

and, for $i \geq 1$, we construct $\hat{k}_{i+1}$ as:

$$\hat{k}_{i+1} := \begin{cases} \min \left\{ \hat{k}_i < t \leq T : \left\{ \psi_t(\mathcal{H}^t) = S^*(F^\tau) \right\} \cup \{t = T\} \right\}, & \text{if } \hat{k}_i < T, \\ T, & \text{if } \hat{k}_i \geq T. \end{cases}$$

Next, we construct a stopping rule $\hat{k}^* := \hat{k}_{j_0}$ to estimate the change time τ from strategy π .

Lemma 3. *For any $\omega \in B_\eta$, we have: $0 \leq \hat{k}^* - \tau \leq 2j_0$.*

By leveraging Lemma 3, we establish the following chain of set inclusions:

$$B_\eta \subseteq \left\{ \omega \in \Omega : 0 \leq \hat{k}^* - \tau \leq 2j_0 \right\} \subseteq \left\{ \omega \in \Omega : |\hat{k}^* - \tau| \leq 2j_0 \right\}.$$

Consequently, $\mathbb{P}_\tau^\pi[B_\eta] \leq \mathbb{P}_\tau^\pi[|\hat{k}^* - \tau| \leq 2j_0]$ and, by considering the complement of B_η , denoted by B_η^c , we obtain $\mathbb{P}_\tau^\pi[B_\eta^c] \geq \mathbb{P}_\tau^\pi[|\hat{k}^* - \tau| > 2j_0]$ for all $2 \leq \tau \leq T$.

Hence, since the former inequality holds for any $2 \leq \tau \leq T$, the following inequality is satisfied:

$$\max \left\{ \mathbb{P}_\tau^\pi[B_\eta^c] : 2 \leq \tau \leq T \right\} \geq \max \left\{ \mathbb{P}_\tau^\pi[|\hat{k}^* - \tau| > 2j_0] : 2 \leq \tau \leq T \right\}.$$

Lemma 4. *There exists $\tilde{C} \equiv \tilde{C}(\varsigma) > 0$ and $\alpha(\varsigma) > 0$ for which any admissible stopping rule $\hat{k}$ with respect to the history $(\mathcal{H}^t)_{t=1}^T$ satisfies $\max \left\{ \mathbb{P}_\tau^\pi[|\hat{k} - \tau| > \tilde{C} \log T] : 2 \leq \tau \leq T \right\} \geq \alpha$.*

We fix $\eta := (C_1 \log T - \gamma_T)^+$, where $C_1 = \tilde{C} \gamma_T / 2$, and $\tilde{C} \equiv \tilde{C}(\varsigma)$ is the constant from Lemma 4. Then, we derive the following sequence of inequalities:

$$2j_0 = 2\lceil \eta / \gamma_T \rceil = 2\lceil (C_1 \log T - \gamma_T)^+ / \gamma_T \rceil = 2\lceil (\frac{\tilde{C}}{2} \log T - 1)^+ \rceil \leq \tilde{C} \log T.$$

By using Lemma 4 with $\hat{k}^*$, we obtain:

$$\begin{aligned} \max \left\{ \mathbb{P}_\tau^\pi[B_\eta^c] : 2 \leq \tau \leq T \right\} &\geq \max \left\{ \mathbb{P}_\tau^\pi[|\hat{k}^* - \tau| > 2j_0] : 2 \leq \tau \leq T \right\} \\ &\geq \max \left\{ \mathbb{P}_\tau^\pi[|\hat{k}^* - \tau| > \tilde{C} \log T] : 2 \leq \tau \leq T \right\} \geq \alpha. \end{aligned}$$

Finally, by using that $\mathbb{P}[B_\eta] + \mathbb{P}[B_\eta^c] = 1$, and by definition of the event B_η , we obtain:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\geq \max \left\{ \eta \mathbb{P}_\tau^\pi[B_\eta^c] : 2 \leq \tau \leq T \right\} \\ &\geq \max \left\{ \alpha(C_1 \log T - \gamma_T)^+ : 2 \leq \tau \leq T \right\} \stackrel{(a)}{\geq} c \log T, \end{aligned}$$

where inequality (a) holds for all $T \geq T_0(\varsigma) \equiv \exp(\max\{4/\tilde{C}(\varsigma), \log(3)\})$ for $c \equiv c(\gamma_T, \varsigma) := \alpha \gamma_T \tilde{C}(\varsigma)/4$. Moreover, c depends only on γ_T and ς , while α depends only on ς . Hence, both constants are independent of $F^{(\mathbb{N})}$. Taking the supremum over $F^{(\mathbb{N})} \in \mathcal{F}_D(\gamma_T, \varepsilon)$ completes the proof. $\blacksquare$

Proof of Lemma 3. We establish the statement using a proof by contradiction. Assume, for the sake of contradiction, that either $\hat{k}^* - \tau < 0$ or $\hat{k}^* - \tau > 2j_0$. We analyze these two cases separately.

(Case 1) Suppose that $\hat{k}^* - \tau > 2j_0$. Then, the following inequality holds:

$$\sum_{t=\tau}^{\hat{k}^*} \mathbf{1}\{\psi_t(\mathcal{H}^t) \neq S^*(F^\tau)\} = \sum_{t=\tau}^{\hat{k}_{j_0}} \mathbf{1}\{\psi_t(\mathcal{H}^t) \neq S^*(F^\tau)\} \stackrel{(a)}{\geq} \hat{k}_{j_0} - \tau + 1 - j_0 \stackrel{(b)}{>} 1 + j_0 \stackrel{(c)}{\geq} \frac{\eta}{\gamma_T},$$

where (a) follows from the fact that the strategy π offers at most j_0 times assortment $S^*(F^\tau)$ in the time horizon $[\hat{k}_{j_0}]$ and (b) from the fact that $\hat{k}_{j_0} - \tau > 2j_0$. Moreover, (c) follows from $j_0 = \lceil \eta/\gamma_T \rceil$.

Next, recall that $\omega \in B_\eta$. Consequently, the following sequence of inequalities hold:

$$\begin{aligned} \eta > J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\geq \gamma_T \sum_{t=\tau}^T \mathbf{1}\{\psi_t(\mathcal{H}^t) \neq S^*(F^\tau)\} \\ &\geq \gamma_T \sum_{t=\tau}^{\hat{k}^*} \mathbf{1}\{\psi_t(\mathcal{H}^t) \neq S^*(F^\tau)\} > \gamma_T \frac{\eta}{\gamma_T} = \eta, \end{aligned}$$

which is clearly a contradiction. Therefore, we must have $\hat{k}^* - \tau \leq 2j_0$.

(Case 2) Assume that $\hat{k}^* - \tau < 0$. Then, the following sequence of inequalities holds:

$$\sum_{t=1}^{\hat{k}^*} \mathbf{1}\{\psi_t(\mathcal{H}^t) \neq S^*(F^1)\} \geq \sum_{t=1}^{\hat{k}^*} \mathbf{1}\{\psi_t(\mathcal{H}^t) = S^*(F^\tau)\} = \sum_{t=1}^{\hat{k}_{j_0}} \mathbf{1}\{\psi_t(\mathcal{H}^t) = S^*(F^\tau)\} \stackrel{(a)}{\geq} j_0 \geq \frac{\eta}{\gamma_T},$$

where (a) follows from the construction of the indices $\hat{k}_i$ for all $i \in [T]$ and that $\hat{k}^* < \tau \leq T$.

Next, since $\omega \in B_\eta$, the following sequence of inequalities holds:

$$\begin{aligned} \eta > J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\geq \gamma_T \sum_{t=1}^{\tau-1} \mathbf{1}\{\psi_t(\mathcal{H}^t) \neq S^*(F^1)\} \\ &\geq \gamma_T \sum_{t=1}^{\hat{k}^*} \mathbf{1}\{\psi_t(\mathcal{H}^t) \neq S^*(F^1)\} \geq \gamma_T \frac{\eta}{\gamma_T} = \eta, \end{aligned}$$

which is clearly a contradiction. Therefore, we have that $0 \leq \hat{k}^* - \tau \leq 2j_0$. $\blacksquare$

Proof of Lemma 4. Our proof closely follows the approach by Besbes and Zeevi (2011), which itself draws inspiration from Korostelev (1988). Let $\hat{k}$ denote any admissible stopping rule based on the filtration $(\mathcal{H}^t)_{t=1}^T$. We first introduce the largest one-period log-likelihood change between the preferences before and after the change (F^1 and F^τ , respectively). Specifically, we define:

$$\phi(F^1, F^\tau) := \max\{|\log F^1(z | S) - \log F^\tau(z | S)| : z \in \{0, 1\}^N, \|z\|_1 \leq 1, z_i = 0 \text{ for all } i \notin S, S \in \mathcal{S}\}.$$

We slightly abuse notation in the definition above. The distributions F^1 and F^τ are originally defined over customers' utilities, as in Section 3. Here, $F^1(z | S)$ and $F^\tau(z | S)$ denote the induced distributions over purchase outcomes when assortment S is offered, where z is the multivariate binary encoding of either the purchase of a product in S or no purchase.

We verify that this quantity is strictly positive and uniformly bounded. By the non-degeneracy condition on $\mathcal{F}$ from Section 3, every purchasing distribution belongs to $[\varsigma, 1 - \varsigma]$. Hence, for every feasible assortment $S \in \mathcal{S}$ and every feasible purchase realization $z \in \{0, 1\}^N$, one has $\frac{\varsigma}{1-\varsigma} \leq F^1(z | S)/F^\tau(z | S) \leq \frac{1-\varsigma}{\varsigma}$. By setting $\vartheta := \max\{-\log \frac{\varsigma}{1-\varsigma}, \log \frac{1-\varsigma}{\varsigma}\}$, we then obtain $\phi(F^1, F^\tau) \leq \vartheta < +\infty$.

It remains to prove strict positivity. Since the path $F^{(\mathbb{N})}$ belongs to $\mathcal{F}_D(\gamma_T, \varepsilon)$, the preferences before and after the change are passively detectable. Specifically, for every feasible assortment S , and hence for at least one feasible assortment, one has $\mathcal{K}(F^\tau, F^1; S) \geq \varepsilon$. On the other hand, for any feasible assortment $S \in \mathcal{S}$, the following inequalities are satisfied:

$$\mathcal{K}(F^\tau, F^1; S) = \sum_z F^\tau(z | S) \log \frac{F^\tau(z | S)}{F^1(z | S)} \leq \sum_z F^\tau(z | S) \left| \log \frac{F^1(z | S)}{F^\tau(z | S)} \right| \leq \phi(F^1, F^\tau).$$

Therefore, $\phi(F^1, F^\tau) \geq \varepsilon > 0$. Combining the two bounds gives $0 < \varepsilon \leq \phi(F^1, F^\tau) \leq \vartheta < +\infty$. Thus, $\phi(F^1, F^\tau)$ is well defined and uniformly bounded above by a constant depending only on ς .

Moreover, the definition of $\phi(F^1, F^\tau)$ implies that, for every feasible assortment S , every feasible purchase realization z , and every period t , the following inequality is satisfied:

$$-\vartheta \leq \log \frac{F^1(z | S)}{F^\tau(z | S)} \leq \vartheta.$$

For an interval $I \subseteq [T]$ of size Δ and $(Z^t : t \in I)$ with $Z^t \in \{0, 1\}^N$, the log-likelihood ratio satisfies:

$$-\Delta\vartheta \leq \sum_{t \in I} \log \frac{F^1(Z^t | \psi_t(\mathcal{H}^t))}{F^\tau(Z^t | \psi_t(\mathcal{H}^t))} \leq \Delta\vartheta.$$

We use this bound in the change-of-measure argument below.

Next, we fix $\beta \in (0, 1)$ arbitrarily, and define the constant $C = (2\vartheta)^{-1}$. Also, we introduce:

$$B(\vartheta, x) := e^{-\vartheta} \frac{x^{1-C\vartheta}}{2C \log(x) + 2} = e^{-\vartheta} \frac{x^{\frac{1}{2}}}{2C \log(x) + 2}.$$

which is increasing for x (after decreasing ς , if necessary, assume $\varsigma < (e + 1)^{-1}$), and such that

$\lim_{x \rightarrow +\infty} B(\vartheta, x) = +\infty$. Given $\beta \in (0, 1)$, we construct $n_0 > 0$ such that $n_0 = \min\{j \in \mathbb{N}_{\geq 1} : B(\vartheta, j) \geq \beta^{-2}\}$. Then, we define $g(x) = x(C \log(x) + 1)^{-1}$. Observe that $g(\cdot)$ is an increasing function (after decreasing ς , if necessary, assume $\varsigma < (e + 1)^{-1}$) and that $\lim_{x \rightarrow +\infty} g(x) = +\infty$.

Moreover, we construct $n_1 \geq n_0$ such that $n_1 = \min\{j \in \mathbb{N}_{\geq 1} : j \geq n_0, g(j) \geq 6\}$.

(**Case 1**) Assume that $T \geq n_1$. Define $\Delta = \lceil C \log T \rceil$ and $\tilde{T} = \lfloor (T - 2)/\Delta \rfloor + 1$. Then, one has:

$$T/\Delta = T(\lceil C \log T \rceil)^{-1} \geq T(C \log T + 1)^{-1} = g(T) \geq g(n_1) \geq 6,$$

which then implies that, $\tilde{T} \geq 2$.

Next, define $\ell_j = 2 + (j - 1)\Delta$ for $j \in [\tilde{T}]$, so that $\ell_{\tilde{T}} \leq T$. Moreover, we denote by $Z := (Z^1, \dots, Z^T) \in \{0, 1\}^{\mathcal{N} \times T}$ the random vector corresponding to the customer's purchasing decisions over the T time period. Next, we introduce a new random variable $\tilde{Z}_j$, for each $j \in [\tilde{T} - 1]$, defined by:

$$\tilde{Z}_j = \sum_{t=\ell_j}^{\ell_{j+1}-1} (\log F^1(Z^t \mid \psi_t(\mathcal{H}^t)) - \log F^\tau(Z^t \mid \psi_t(\mathcal{H}^t))).$$

The next step consists of showing that the following inequality holds:

$$\max \left\{ \mathbb{P}_{\ell_j}^\pi [|\hat{k} - \tau| > \Delta/3] : 1 \leq j \leq \tilde{T} - 1 \right\} \geq 1 - \beta.$$

We assume for the sake of contradiction that this inequality does not hold. That is:

$$\max \left\{ \mathbb{P}_{\ell_j}^\pi [|\hat{k} - \tau| > \Delta/3] : 1 \leq j \leq \tilde{T} - 1 \right\} < 1 - \beta.$$

Then, we define the event $A_j := \{\omega \in \Omega : |\hat{k} - \ell_j| \leq \Delta/3\}$ for $j \in [\tilde{T} - 1]$. Note that the events A_j are disjoint as each segment $\{\ell_j, \dots, \ell_{j+1} - 1\}$ is of size Δ , and $\{\omega \in \Omega : |\hat{k} - \ell_{\tilde{T}}| > \Delta/3\} \supset \bigcup_{j=1}^{\tilde{T}-1} A_j$. Thus, the following chain of inequalities hold:

$$\mathbb{P}_{\ell_{\tilde{T}}}^\pi [|\hat{k} - \tau| > \Delta/3] \geq \mathbb{P}_{\ell_{\tilde{T}}}^\pi \left[\bigcup_{j=1}^{\tilde{T}-1} A_j \right] = \sum_{j=1}^{\tilde{T}-1} \mathbb{P}_{\ell_{\tilde{T}}}^\pi [|\hat{k} - \ell_j| \leq \Delta/3] = \sum_{j=1}^{\tilde{T}-1} \mathbb{E}_{\mathbb{P}_{\ell_{\tilde{T}}}^\pi} [\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}].$$

Since $\hat{k}$ is a stopping rule adapted to $(\mathcal{H}^t)_{t=1}^T$, the event $\{|\hat{k} - \ell_j| \leq \Delta/3\}$ is $\mathcal{H}^{\ell_{j+1}}$ -measurable. Its probability is therefore determined by the sales history observed before period ℓ_{j+1} . Changes occurring at or after ℓ_{j+1} do not affect the law of this history, so the following equality holds:

$$\mathbb{E}_{\mathbb{P}_{\ell_{\tilde{T}}}^\pi} [\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}] = \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} [\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}].$$

Next, let V be an integrable $\mathcal{H}^{\ell_{j+1}}$ -measurable random variable. Since V is $\mathcal{H}^{\ell_{j+1}}$ -measurable, there exists a measurable map v such that $V = v(\mathcal{H}^{\ell_{j+1}})$. For a realization of the history $h^{\ell_{j+1}} = \{(S^{\pi,t}, z^t) : t \in [\ell_{j+1} - 1]\}$ up to time ℓ_{j+1} , define $L_j(h^{\ell_{j+1}}) = \prod_{t=\ell_j}^{\ell_{j+1}-1} \frac{F^1(z^t \mid S^{\pi,t})}{F^\tau(z^t \mid S^{\pi,t})}$. Equivalently, $L_j(\mathcal{H}^{\ell_{j+1}}) = \exp(\tilde{Z}_j)$. The ratio is well defined on the support of $\mathbb{P}_{\ell_j}^\pi$, because the non-degeneracy condition gives strictly positive purchasing probabilities for every purchase or no-purchase outcome.

By Lemma 14, the law of the partial history $\mathcal{H}^{\ell_{j+1}}$ under the candidate change time ℓ_j satisfies:

$$\mathbb{P}_{\ell_j}^\pi [\mathcal{H}^{\ell_{j+1}} = h^{\ell_{j+1}}] = \prod_{t=1}^{\ell_j-1} \pi_t(S^{\pi,t} \mid h^t) F^1(z^t \mid S^{\pi,t}) \prod_{t=\ell_j}^{\ell_{j+1}-1} \pi_t(S^{\pi,t} \mid h^t) F^\tau(z^t \mid S^{\pi,t}).$$

Similarly, under the candidate change time ℓ_{j+1} , all periods $t \in \{1, \dots, \ell_{j+1} - 1\}$ are generated under F^1 , and therefore the following equality holds:

$$\mathbb{P}_{\ell_{j+1}}^\pi [\mathcal{H}^{\ell_{j+1}} = h^{\ell_{j+1}}] = \prod_{t=1}^{\ell_{j+1}-1} \pi_t(S^{\pi,t} \mid h^t) F^1(z^t \mid S^{\pi,t}).$$

Multiplying the first expression by $L_j(h^{\ell_{j+1}})$, all factors involving F^τ on the interval $\{\ell_j, \dots, \ell_{j+1} - 1\}$ are replaced by the corresponding factors involving F^1 . Hence, for every realization of the history

$h^{\ell_{j+1}}$, one has $L_j(h^{\ell_{j+1}})\mathbb{P}_{\ell_j}^\pi[\mathcal{H}^{\ell_{j+1}} = h^{\ell_{j+1}}] = \mathbb{P}_{\ell_{j+1}}^\pi[\mathcal{H}^{\ell_{j+1}} = h^{\ell_{j+1}}]$. It follows that:

$$\mathbb{E}_{\mathbb{P}_{\ell_j}^\pi}[e^{\tilde{Z}_j}V] = \sum_{h^{\ell_{j+1}}} L_j(h^{\ell_{j+1}})v(h^{\ell_{j+1}})\mathbb{P}_{\ell_j}^\pi[\mathcal{H}^{\ell_{j+1}} = h^{\ell_{j+1}}] = \sum_{h^{\ell_{j+1}}} v(h^{\ell_{j+1}})\mathbb{P}_{\ell_{j+1}}^\pi[\mathcal{H}^{\ell_{j+1}} = h^{\ell_{j+1}}] = \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi}[V].$$

Moreover, since $\tilde{Z}_j \geq -\Delta\vartheta$, the following sequence of inequalities holds:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{\ell_j}^\pi}[e^{\tilde{Z}_j}\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}] &\geq \mathbb{E}_{\mathbb{P}_{\ell_j}^\pi}[e^{-\Delta\vartheta}\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}] \\ &\geq \mathbb{E}_{\mathbb{P}_{\ell_j}^\pi}[e^{-(1+C\log T)\vartheta}\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}] = \frac{e^{-\vartheta}}{T^{C\vartheta}}\mathbb{E}_{\mathbb{P}_{\ell_j}^\pi}[\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}]. \end{aligned}$$

Consequently, we derive the following sequence of inequalities:

$$\begin{aligned} \mathbb{P}_{\ell_{\tilde{T}}}^\pi[|\hat{k} - \tau| > \Delta/3] &\geq \sum_{j=1}^{\tilde{T}-1} \mathbb{E}_{\mathbb{P}_{\ell_{\tilde{T}}}^\pi}[\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}] \\ &\geq \sum_{j=1}^{\tilde{T}-1} \frac{e^{-\vartheta}}{T^{C\vartheta}} \mathbb{E}_{\mathbb{P}_{\ell_j}^\pi}[\mathbf{1}\{|\hat{k} - \ell_j| \leq \Delta/3\}] \\ &\geq \frac{(\tilde{T}-1)e^{-\vartheta}}{T^{C\vartheta}} \min\left\{\mathbb{P}_{\ell_j}^\pi[|\hat{k} - \ell_j| \leq \Delta/3] : 1 \leq j \leq \tilde{T}-1\right\}. \end{aligned}$$

Since $\tilde{T}-1 \geq T/(2\Delta)$, we can derive the following sequence of inequalities:

$$e^{-\vartheta}T^{-C\vartheta}(\tilde{T}-1) \geq e^{-\vartheta}T^{-C\vartheta}\frac{T}{2\Delta} \geq e^{-\vartheta}T^{1-C\vartheta}(2\lceil C\log T \rceil)^{-1} \geq B(\vartheta, T) \geq \frac{1}{\beta^2},$$

which holds as $T \geq n_1 \geq n_0$.

Consequently, we obtain the following inequalities:

$$\mathbb{P}_{\ell_{\tilde{T}}}^\pi[|\hat{k} - \tau| > \Delta/3] \geq \frac{1}{\beta^2} \min\left\{\mathbb{P}_{\ell_j}^\pi[|\hat{k} - \ell_j| \leq \Delta/3] : 1 \leq j \leq \tilde{T}-1\right\} \geq \frac{\beta}{\beta^2} = \frac{1}{\beta} > 1,$$

which is clearly a contradiction. Therefore, we conclude:

$$\max\left\{\mathbb{P}_{\ell_j}^\pi[|\hat{k} - \tau| > \Delta/3] : 1 \leq j \leq \tilde{T}-1\right\} \geq 1 - \beta.$$

That is, for all $T \geq n_1$, we obtain the following inequality:

$$\max\left\{\mathbb{P}_\tau^\pi[|\hat{k} - \tau| > \lceil C\log T \rceil/3] : 2 \leq \tau \leq T\right\} \geq 1 - \beta,$$

where $C = (2\vartheta)^{-1}$. That is, both n_1 , as well as C only depends on the parameter ϑ .

(Case 2) In the second case, we assume that $T < n_1$ (but $T \geq 3$ by assumption in Proposition 1).

Then, we define $\tau_1 = T-1$ and $\tau_2 = T$. Suppose, first, that $\mathbb{P}_{\tau_1}^\pi[|\hat{k} - \tau_1| = 0] \geq \frac{1}{2}$. Next, observe that:

$$\begin{aligned} \mathbb{P}_{\tau_2}^\pi[|\hat{k} - \tau_1| = 0] &= \mathbb{E}_{\mathbb{P}_{\tau_2}^\pi}[\mathbf{1}\{|\hat{k} - \tau_1| = 0\}] \\ &\stackrel{(a)}{=} \mathbb{E}_{\mathbb{P}_{\tau_1}^\pi}\left[\exp\left(\log\left(\frac{F^1(Z^{T-1} \mid \psi_{T-1}(\mathcal{H}^{T-1}))}{F^\tau(Z^{T-1} \mid \psi_{T-1}(\mathcal{H}^{T-1}))}\right)\right)\mathbf{1}\{|\hat{k} - \tau_1| = 0\}\right], \end{aligned}$$

where (a) is obtained through a change-of-measure argument, similar to the one employed in the Case 1. Hence, we have that:

$$\mathbb{P}_{\tau_2}^\pi[|\hat{k} - \tau_1| = 0] \geq e^{-\vartheta}\mathbb{E}_{\mathbb{P}_{\tau_1}^\pi}[\mathbf{1}\{|\hat{k} - \tau_1| = 0\}] \geq e^{-\vartheta}\frac{1}{2}.$$

Therefore, we obtain the following inequalities:

$$\mathbb{P}_{\tau_2}^\pi[|\hat{k} - \tau_2| \geq 1] = \mathbb{P}_{\tau_2}^\pi[|\hat{k} - T| \geq 1] \geq \mathbb{P}_{\tau_2}^\pi[|\hat{k} - (T-1)| = 0] \geq \frac{1}{2}e^{-\vartheta}.$$

On the other hand, if $\mathbb{P}_{\tau_1}^\pi[|\hat{k} - \tau_1| = 0] < \frac{1}{2}$, then we have:

$$\mathbb{P}_{\tau_1}^\pi[|\hat{k} - \tau_1| \geq 1] > \frac{1}{2} \geq \frac{1}{2}e^{-\vartheta}.$$

Therefore, we obtain the following inequality:

$$\max \left\{ \mathbb{P}_\tau^\pi[|\hat{k} - \tau| \geq 1] : \tau \in \{\tau_1, \tau_2\} \right\} \geq \frac{1}{2}e^{-\vartheta}.$$

Since $1 \geq T/n_1 \geq \log(T)/n_1$, we have that:

$$\max \left\{ \mathbb{P}_\tau^\pi[|\hat{k} - \tau| \geq \log(T)/n_1] : 2 \leq \tau \leq T \right\} \geq \frac{1}{2}e^{-\vartheta}.$$

Finally, by combining both Case 1 and Case 2, we obtain the following result:

$$\max \left\{ \mathbb{P}_\tau^\pi[|\hat{k} - \tau| \geq \tilde{C} \log T] : 2 \leq \tau \leq T \right\} \geq \alpha,$$

where $\tilde{C} := \min\{C/6, 1/(2n_1)\}$ and $\alpha := \min\{1 - \beta, e^{-\vartheta}/2\}$. Moreover, both $\tilde{C} \equiv \tilde{C}(\varsigma)$ and $\alpha \equiv \alpha(\varsigma)$ only depend on ς through ϑ , which concludes the proof. $\blacksquare$

E.C.2.2 Proof of Theorem 1

We first formally introduce the two-sample test used in Algorithm 1. Recall that $\mathcal{H}^t = \{(S^u, i^u) : 1 \leq u < t\}$ denotes the sales history observed prior to period t . Fix $S \in \mathcal{S}$, and let $\mathcal{T}_I, \mathcal{T}_J \subseteq [T]$ be two disjoint sets of periods, each indexed in increasing order, such that $S^u = S$ for all $u \in \mathcal{T}_I \cup \mathcal{T}_J$. We denote by $\mathcal{I} = (i^u : u \in \mathcal{T}_I)$ and $\mathcal{J} = (i^u : u \in \mathcal{T}_J)$ the corresponding sales histories. The sequence $\mathcal{J}$ is referred to as the *reference history*, while $\mathcal{I}$ is the *test history*. For $i \in S \cup \{0\}$, let $\hat{p}_i^J(S, \mathcal{J})$ and $\hat{p}_i^I(S, \mathcal{I})$ be the empirical purchasing distribution over the two sales histories. The pooled empirical distribution is given by $\hat{p}_i^0(S, \mathcal{J}, \mathcal{I}) = (|\mathcal{J}|\hat{p}_i^J(S, \mathcal{J}) + |\mathcal{I}|\hat{p}_i^I(S, \mathcal{I})) / (|\mathcal{J}| + |\mathcal{I}|)$.

We use the two-sample statistic to test equality of the two purchasing distributions:

$$G(S, \mathcal{J}, \mathcal{I}) = |\mathcal{J}|\mathcal{K}(\hat{p}^J(S, \mathcal{J}), \hat{p}^0(S, \mathcal{J}, \mathcal{I})) + |\mathcal{I}|\mathcal{K}(\hat{p}^I(S, \mathcal{I}), \hat{p}^0(S, \mathcal{J}, \mathcal{I})).$$

The statistic is small when the two sales histories are consistent with the same purchasing distribution, and large when sales under S provide evidence of a change. We declare a change in preferences whenever $G(S, \mathcal{J}, \mathcal{I}) > \Gamma(|\mathcal{J}|, |\mathcal{I}|, T)$, with $\Gamma(|\mathcal{J}|, |\mathcal{I}|, T) = 8(K+1) \log(|\mathcal{J}| + |\mathcal{I}|) + 8 \log(T+1)$. The threshold is not arbitrary, and we refer to Lemma 10 for the concentration inequality used below.

Proof. Let $\pi \equiv \pi(\mathcal{A}, T)$ be the passive monitoring strategy from Algorithm 1. Fix an arbitrary path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_D(\gamma_T, \varepsilon)$. Denote by τ its change time, so that $F^1 = \dots = F^{\tau-1}$ and $F^\tau = \dots = F^T$ when $2 \leq \tau \leq T$, with $\tau = T+1$ to indicate $F^t \equiv F^1$ for all $t \in [T]$. In particular, the definition of the risk profile $\mathcal{F}_D(\gamma_T, \varepsilon)$ gives $\min\{\mathcal{K}^\tau(S) : S \in \mathcal{S}\} \geq \varepsilon$.

Step 1 (Partition of the horizon). Algorithm 1 partitions the horizon into intervals during which either $\mathcal{A}$ is run or monitoring is performed. We introduce the notation needed to describe these intervals. To begin, let $h_T = \lceil \log \log(T) \rceil$, $q_T = \lceil \log(T) \rceil h_T$, $m_T = 8q_T$, and $H_T = m_T + q_T = 9q_T$. The strategy first partitions the horizon into $\mathcal{T}_0^B \subseteq [T]$ and $\mathcal{T}_0^J \subseteq [T]$, with $|\mathcal{T}_0^B| \leq m_T$ and $|\mathcal{T}_0^J| \leq q_T$. It then refreshes $\mathcal{A}$. For every subsequent cycle $k \geq 1$, provided no change in preferences

has been detected, the strategy partitions the horizon into intervals $\mathcal{T}_k^B \subseteq [T]$, $\mathcal{T}_k^J \subseteq [T]$, and $\mathcal{T}_k^I \subseteq [T]$, in that order, with $|\mathcal{T}_k^B| \leq m_T$, $|\mathcal{T}_k^J| \leq q_T$, and $|\mathcal{T}_k^I| \leq q_T$.

On $\mathcal{T}_k^B$, the strategy runs $\mathcal{A}$. If $\mathcal{T}_k^B \neq \emptyset$, then the strategy draws uniformly $U_k \sim \text{Unif}(\mathcal{T}_k^B)$ and defines $\tilde{S}_k = S^{U_k}$, where S^{U_k} is the assortment offered by the strategy π at period U_k . The set $\mathcal{T}_k^J$ is generated by offering $\tilde{S}_k$, whereas $\mathcal{T}_k^I$ is generated by offering $\tilde{S}_{k-1}$. After $\mathcal{T}_k^I$ has been collected, the strategy compares the histories $\mathcal{T}_{k-1}^J$ and $\mathcal{T}_k^I$ under the common assortment $\tilde{S}_{k-1}$. A change is detected at the end of cycle k whenever $G(\tilde{S}_{k-1}, \mathcal{T}_{k-1}^J, \mathcal{T}_k^I) > \Gamma(|\mathcal{T}_{k-1}^J|, |\mathcal{T}_k^I|, T)$. We let $\hat{\tau}$ denote the first period at which such a detection occurs, with $\hat{\tau} = T+1$ if no change is detected by the strategy π .

Step 2 (Clean comparisons). Assume that $\tau \leq T - q_T$. We say that a *comparison* (of sales histories) is clean when the two sales histories compared by the statistical test are generated under the same assortment, with one sales history collected before the change and the other collected after the change. Therefore, the purchasing distributions being compared are those induced by $F^1 \equiv F^{\tau-1}$ and F^τ . By Lemma 9, there exists a constant $c \equiv c(\varsigma) > 0$, such that, for every $S \in \mathcal{S}$:

$$\mathcal{K}^\tau(S) \leq c \left(\sum_{i \in S \cup \{0\}} |p_i(S, F^\tau) - p_i(S, F^{\tau-1})| \right)^2.$$

Since $F^{(\mathbb{N})} \in \mathcal{F}_D(\gamma_T, \varepsilon)$, one has $\min \{ \mathcal{K}^\tau(S) : S \in \mathcal{S} \} \geq \varepsilon$. Therefore, for every $S \in \mathcal{S}$,

$$\sum_{i \in S \cup \{0\}} |p_i(S, F^\tau) - p_i(S, F^{\tau-1})| \geq \sqrt{\varepsilon/c}.$$

Next, define $C \equiv C(\varepsilon, \varsigma, K) := \max \{ 40c(K+1)^2/\varepsilon, 512c(K+2)/\varepsilon, 1 \}$. Choose T_0 large enough so that, for all $T \geq T_0$, we have $q_T \geq C \log(T)$, $q_T \leq T$, and $T \geq \max\{e^e, 4(K+1)\}$. Fix $S \in \mathcal{S}$. Let $\mathcal{J}$ and $\mathcal{I}$ be two sales histories, each of length q_T , generated under the categorical laws induced by $F^{\tau-1}$ and F^τ , respectively, when assortment S is offered. Define $d_S = |S \cup \{0\}| \leq K+1$ and $a = (1/4)\sqrt{\varepsilon/c}$. For every $i \in S \cup \{0\}$, Hoeffding's inequality (Boucheron et al. 2013) gives:

$$\mathbb{P} \left[|\hat{p}_i^J(S, \mathcal{J}) - p_i(S, F^{\tau-1})| \geq a/d_S \right] \leq 2 \exp(-2q_T a^2/d_S^2),$$

and the following inequality:

$$\mathbb{P} \left[|\hat{p}_i^I(S, \mathcal{I}) - p_i(S, F^\tau)| \geq a/d_S \right] \leq 2 \exp(-2q_T a^2/d_S^2).$$

Thus, a union bound gives the following inequality:

$$\begin{aligned} \mathbb{P} \left[\sum_{i \in S \cup \{0\}} |\hat{p}_i^J(S, \mathcal{J}) - p_i(S, F^{\tau-1})| \geq a \text{ or } \sum_{i \in S \cup \{0\}} |\hat{p}_i^I(S, \mathcal{I}) - p_i(S, F^\tau)| \geq a \right] &\leq 4d_S \exp(-2q_T a^2/d_S^2) \\ &\leq 4(K+1) \exp(-q_T \varepsilon / (8c(K+1)^2)) \leq T^{-4}. \end{aligned}$$

On the complementary event, the triangle inequality gives:

$$\begin{aligned} \sum_{i \in S \cup \{0\}} |\hat{p}_i^I(S, \mathcal{I}) - \hat{p}_i^J(S, \mathcal{J})| &\geq \sum_{i \in S \cup \{0\}} |p_i(S, F^\tau) - p_i(S, F^{\tau-1})| \\ &\quad - \sum_{i \in S \cup \{0\}} |\hat{p}_i^J(S, \mathcal{J}) - p_i(S, F^{\tau-1})| \\ &\quad - \sum_{i \in S \cup \{0\}} |\hat{p}_i^I(S, \mathcal{I}) - p_i(S, F^\tau)| \geq \frac{1}{2} \sqrt{\varepsilon/c}. \end{aligned}$$

For samples of equal length q_T , the pooled empirical distribution satisfies $\hat{p}^0 = (\hat{p}^J + \hat{p}^I)/2$. Pinsker's inequality (Tsybakov 2003) then gives:

$$\begin{aligned} G(S, \mathcal{J}, \mathcal{I}) &\geq \frac{q_T}{2} \left(\sum_{i \in S \cup \{0\}} |\hat{p}_i^J(S, \mathcal{J}) - \hat{p}_i^0(S, \mathcal{J}, \mathcal{I})| \right)^2 + \frac{q_T}{2} \left(\sum_{i \in S \cup \{0\}} |\hat{p}_i^I(S, \mathcal{I}) - \hat{p}_i^0(S, \mathcal{J}, \mathcal{I})| \right)^2 \\ &= \frac{q_T}{4} \left(\sum_{i \in S \cup \{0\}} |\hat{p}_i^J(S, \mathcal{I}) - \hat{p}_i^I(S, \mathcal{J})| \right)^2 \geq \frac{q_T \varepsilon}{16c}. \end{aligned}$$

Moreover, by definition and Step 1, one has:

$$\Gamma(q_T, q_T, T) = 8(K+1) \log(2q_T) + 8 \log(T+1) \leq 8(K+1) \log(2T) + 8 \log(T+1) \leq 16(K+2) \log(T).$$

Since $q_T \geq C \log(T)$ and $C\varepsilon/(16c) \geq 32(K+2)$, we obtain the following inequality:

$$G(S, \mathcal{J}, \mathcal{I}) \geq \frac{q_T \varepsilon}{16c} \geq 32(K+2) \log(T) > \Gamma(q_T, q_T, T).$$

Thus, whenever a reference history is generated under $F^{\tau-1}$ and a test history is generated under F^τ , the test rejects the null hypothesis that the two purchasing distributions induced by S are equal with probability at least $1 - T^{-4}$, uniformly over $S \in \mathcal{S}$.

Step 3 (False alarms). For $k \geq 1$, let $\mathcal{J}_{k-1}$ denote the reference history tested against $\mathcal{I}_k$. For $1 \leq \ell \leq q_T$, let $\mathcal{I}_k^\ell$ denote the first ℓ observations of $\mathcal{I}_k$; the case $\ell < q_T$ can occur only when the horizon ends before the strategy π performs the statistical test. Define the (random) event:

$$\mathcal{N}_{k,\ell} = \{|\mathcal{T}_k^I| = \ell\} \cap \left[\left\{ \max\{u : u \in \mathcal{T}_{k-1}^J \cup \mathcal{T}_k^I\} < \tau \right\} \cup \left\{ \min\{u : u \in \mathcal{T}_{k-1}^J \cup \mathcal{T}_k^I\} \geq \tau \right\} \right].$$

On $\mathcal{N}_{k,\ell}$, all observations entering $(\mathcal{J}_{k-1}, \mathcal{I}_k^\ell)$ are generated under the same preferences. Define:

$$\mathcal{R}_{k,\ell} = \{|\mathcal{T}_k^I| = \ell\} \cap \{G(\tilde{\mathcal{S}}_{k-1}, \mathcal{J}_{k-1}, \mathcal{I}_k) > \Gamma(|\mathcal{J}_{k-1}|, \ell, T)\},$$

as the event on which the statistical test used in π rejects the hypothesis of no change in preferences between cycles $k-1$ and k when the test history is restricted to the first ℓ observations. The event that the strategy falsely declares a change is contained in:

$$\mathcal{E}_{\text{false}} = \bigcup_{k \geq 1} \bigcup_{\ell=1}^{q_T} (\mathcal{N}_{k,\ell} \cap \mathcal{R}_{k,\ell}).$$

Fix $k \geq 1$ and $1 \leq \ell \leq q_T$. Let $\mathcal{G}_{k,\ell} = \sigma(\tilde{\mathcal{S}}_{k-1}, \mathcal{T}_{k-1}^J, \mathcal{T}_k^I)$ be the σ -field generated by the assortment $\tilde{\mathcal{S}}_{k-1}$ and the strategy randomization. On $\mathcal{N}_{k,\ell}$, conditional on $\mathcal{G}_{k,\ell}$, the samples $\mathcal{J}_{k-1}$ and $\mathcal{I}_k^\ell$ are independent draws from a common categorical distribution on $\tilde{\mathcal{S}}_{k-1} \cup \{0\}$. Using the law of total expectation (Jacod and Protter 2004) and then applying Lemma 10 give:

$$\mathbb{P}_\tau^\pi[\mathcal{N}_{k,\ell} \cap \mathcal{R}_{k,\ell}] = \mathbb{E}_{\mathbb{P}_\tau^\pi}[\mathbb{P}_\tau^\pi[\mathcal{N}_{k,\ell} \cap \mathcal{R}_{k,\ell} | \mathcal{G}_{k,\ell}]] \leq \frac{1}{(|\mathcal{J}_{k-1}| + \ell)^{2(K+1)}(T+1)^4} \leq \frac{1}{(T+1)^4}.$$

There can be at most T cycles k that satisfy $\mathcal{T}_k^I \neq \emptyset$ before the horizon. Hence, one has:

$$\mathbb{P}_\tau^\pi[\mathcal{E}_{\text{false}}] \leq \sum_{k \geq 1} \sum_{\ell=1}^{q_T} \mathbb{P}_\tau^\pi[\mathcal{N}_{k,\ell} \cap \mathcal{R}_{k,\ell}] \leq \frac{Tq_T}{(T+1)^4} \leq T^{-2},$$

where the last inequality holds for all $T \geq T_0$, since T_0 is chosen so that $q_T \leq T$; recall Step 1.

Step 4 (Detection delay). Assume $H_T < \tau \leq T$. Since all observations before τ are generated under $F^{\tau-1}$, any rejection before τ is based on a test that compares two sales histories generated

under the same static preferences. Hence, on $\mathcal{E}_{\text{false}}^c$, one has $\hat{\tau} \geq \tau$. On this event, the strategy does not declare a change before τ , so there exists a unique index $k_\tau \geq 1$ such that $\tau \in \mathcal{T}_{k_\tau}^B \cup \mathcal{T}_{k_\tau}^J \cup \mathcal{T}_{k_\tau}^I$. We define the following three events $\mathcal{E}_B = \{\tau \in \mathcal{T}_{k_\tau}^B\}$, $\mathcal{E}_J = \{\tau \in \mathcal{T}_{k_\tau}^J\}$, and $\mathcal{E}_I = \{\tau \in \mathcal{T}_{k_\tau}^I\}$.

On $\mathcal{E}_B$, the history $\mathcal{J}_{k_\tau-1}$ is generated under $F^{\tau-1}$, while $\mathcal{I}_{k_\tau}$ is generated under F^τ , provided the horizon does not end first. Moreover, the end of $\mathcal{I}_{k_\tau}$ occurs at most $m_T + 2q_T \leq 10q_T$ periods after τ . On $\mathcal{E}_J$, the pair $(\mathcal{J}_{k_\tau-1}, \mathcal{I}_{k_\tau})$ has a reference history generated under $F^{\tau-1}$ and a test history generated under F^τ , provided the horizon does not end first, and the end of $\mathcal{I}_{k_\tau}$ occurs at most $2q_T$ periods after τ . On $\mathcal{E}_I$, $\mathcal{I}_{k_\tau}$ may contain observations from both preferences. However, $\mathcal{J}_{k_\tau}$ is generated before τ , while $\mathcal{I}_{k_\tau+1}$ is generated after τ , provided the horizon does not end first. The end of $\mathcal{I}_{k_\tau+1}$ occurs at most $q_T + m_T + 2q_T \leq 11q_T$ periods after τ .

We now define the set of “clean” detection cycles, namely cycles in which all sales in the reference history are generated under the preferences before the change, while all sales in the test history are generated under the preferences after the change. Formally, let:

$$\mathcal{E}_{\text{clean}} = \{k \geq 1 : |\mathcal{J}_{k-1}| = q_T, |\mathcal{I}_k| = q_T, \max\{u : u \in \mathcal{T}_{k-1}^J\} < \tau \leq \min\{u : u \in \mathcal{T}_k^I\}\}.$$

For every $k \in \mathcal{E}_{\text{clean}}$, the pair $(\mathcal{J}_{k-1}, \mathcal{I}_k)$ consists of a reference history collected before the change and a test history collected after the change. Define $\mathcal{G}_k = \sigma(\tilde{S}_{k-1}, \mathcal{T}_{k-1}^J, \mathcal{T}_k^I, \mathbf{1}\{k \in \mathcal{E}_{\text{clean}}\})$, where the indicator makes membership in $\mathcal{E}_{\text{clean}}$ measurable. Conditional on $\mathcal{G}_k$ and on $\{k \in \mathcal{E}_{\text{clean}}\}$, the assortment $\tilde{S}_{k-1}$ and the tested time indices are fixed. The observations in $\mathcal{J}_{k-1}$ are generated under $F^{\tau-1}$, whereas the observations in $\mathcal{I}_k$ are generated under F^τ . Hence, by Step 2,

$$\mathbb{P}_\tau^\pi \left[G(\tilde{S}_{k-1}, \mathcal{J}_{k-1}, \mathcal{I}_k) \leq \Gamma(|\mathcal{J}_{k-1}|, |\mathcal{I}_k|, T) \mid \mathcal{G}_k \right] \mathbf{1}\{k \in \mathcal{E}_{\text{clean}}\} \leq T^{-4} \mathbf{1}\{k \in \mathcal{E}_{\text{clean}}\}.$$

Therefore, by using the law of total expectations (Jacod and Protter 2004), we obtain:

$$\begin{aligned} & \mathbb{P}_\tau^\pi \left[\{k \in \mathcal{E}_{\text{clean}}\} \cap \{G(\tilde{S}_{k-1}, \mathcal{J}_{k-1}, \mathcal{I}_k) \leq \Gamma(|\mathcal{J}_{k-1}|, |\mathcal{I}_k|, T)\} \right] \\ &= \mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\mathbf{1}\{k \in \mathcal{E}_{\text{clean}}\} \mathbb{P}_\tau^\pi \left[G(\tilde{S}_{k-1}, \mathcal{J}_{k-1}, \mathcal{I}_k) \leq \Gamma(|\mathcal{J}_{k-1}|, |\mathcal{I}_k|, T) \mid \mathcal{G}_k \right] \right] \leq T^{-4} \mathbb{P}_\tau^\pi[k \in \mathcal{E}_{\text{clean}}] \leq T^{-4}. \end{aligned}$$

Then, define the event:

$$\mathcal{E}_{\text{det}} = \bigcap_{k \geq 1} \left(\{k \notin \mathcal{E}_{\text{clean}}\} \cup \left\{ G(\tilde{S}_{k-1}, \mathcal{J}_{k-1}, \mathcal{I}_k) > \Gamma(|\mathcal{J}_{k-1}|, |\mathcal{I}_k|, T) \right\} \right).$$

Thus, $\mathcal{E}_{\text{det}}^c$ is the event that the strategy may not detect a change during a clean detection cycle. Since at most T cycles can be generated before the horizon, a union bound gives:

$$\mathbb{P}_\tau^\pi[\mathcal{E}_{\text{det}}^c] \leq \sum_{k \geq 1} \mathbb{P}_\tau^\pi \left[\{k \in \mathcal{E}_{\text{clean}}\} \cap \{G(\tilde{S}_{k-1}, \mathcal{J}_{k-1}, \mathcal{I}_k) \leq \Gamma(|\mathcal{J}_{k-1}|, |\mathcal{I}_k|, T)\} \right] \leq T^{-3}.$$

Define $\mathcal{E}_{\text{good}} = \mathcal{E}_{\text{false}}^c \cap \mathcal{E}_{\text{det}}$. By Step 3 and the preceding inequality, for all $T \geq T_0$, $\mathbb{P}_\tau^\pi[\mathcal{E}_{\text{good}}^c] \leq 2T^{-2}$. Moreover, on $\mathcal{E}_{\text{good}} \cap \{H_T < \tau \leq T\}$, the beginning of Step 4 gives $\hat{\tau} - \tau + 1 \leq 11q_T$. If $\tau = T + 1$, then no change occurs before the horizon, so every rejection before T compares histories generated under the same preferences. In that case, on $\mathcal{E}_{\text{good}}$, one necessarily has $\hat{\tau} = T + 1$.

Step 5 (Discarding initialization and detection periods). Define the initialization periods

$\mathcal{Q}_T = \mathcal{T}_0^B \cup \mathcal{T}_0^J \cup \mathcal{T}_1^I$. Then, $|\mathcal{Q}_T| \leq m_T + 2q_T = 10q_T$. Next, we define:

$$\mathcal{W}_T = \begin{cases} \emptyset, & \tau \leq H_T \text{ or } \tau = T + 1, \\ \{\tau, \tau + 1, \dots, \min\{\hat{\tau}, T\}\}, & H_T < \tau \leq T. \end{cases}$$

By Step 4, $|\mathcal{W}_T| \leq 11q_T$ on $\mathcal{E}_{\text{good}}$. For every t , one-period regret is bounded by $\|w\|_\infty$. Hence, on $\mathcal{E}_{\text{good}}$, the following inequality is satisfied:

$$\sum_{t \in \mathcal{Q}_T \cup \mathcal{W}_T} \left(r(S^*(F^t), F^t) - r(S^{\pi, t}, F^t) \right) \leq 21q_T \|w\|_\infty.$$

Step 6 (Regret before detection). Define $\mathcal{T}_T = \{t \in [T] : t \notin \mathcal{Q}_T \cup \mathcal{W}_T, t < \hat{\tau}\}$. On $\mathcal{E}_{\text{good}}$, all purchases during periods in $\mathcal{T}_T$ are generated under the same static preferences, which we denote by F^1 by abuse of notation; the case in which they are generated under F^τ is identical. For $S \in \mathcal{S}$, define $\Delta(S, F^1) = r(S^*(F^1), F^1) - r(S, F^1)$. We split $\mathcal{T}_T$ into periods during which $\mathcal{A}$ is run $\mathcal{T}_T^B = \mathcal{T}_T \cap \bigcup_{k \geq 1} \mathcal{T}_k^B$ and monitoring periods $\mathcal{T}_T \cap \bigcup_{k \geq 0} (\mathcal{T}_k^J \cup \mathcal{T}_{k+1}^I)$; recall Step 1. For $k \geq 0$, define $\mathcal{R}_k^R = \mathcal{T}_T \cap (\mathcal{T}_k^J \cup \mathcal{T}_{k+1}^I)$. Then, the sets $(\mathcal{R}_k^R : k \geq 0)$ are disjoint and satisfy $\mathcal{T}_T^R = \bigcup_{k \geq 0} \mathcal{R}_k^R$.

We first show that, on $\mathcal{E}_{\text{good}}$, if $\mathcal{R}_k^R \neq \emptyset$, then $\mathcal{T}_k^B \subseteq \mathcal{T}_T$ and $|\mathcal{T}_k^B| = m_T$. If $\mathcal{R}_k^R \neq \emptyset$, then at least one period in $\mathcal{T}_k^J \cup \mathcal{T}_{k+1}^I$ belongs to $\mathcal{T}_T$, and therefore occurs before $\hat{\tau}$. Hence, all m_T periods in the preceding interval $\mathcal{T}_k^B$ occur before that period, so $|\mathcal{T}_k^B| = m_T$. Moreover, $\mathcal{T}_k^B$ cannot intersect $\mathcal{Q}_T$. If $k = 0$, then $\mathcal{T}_k^J \cup \mathcal{T}_{k+1}^I = \mathcal{T}_0^J \cup \mathcal{T}_1^I \subseteq \mathcal{Q}_T$, which contradicts $\mathcal{R}_k^R \subseteq \mathcal{T}_T$. Thus, $k \geq 1$, and all periods in $\mathcal{T}_k^B$ occur after the periods in $\mathcal{Q}_T$. Finally, $\mathcal{T}_k^B$ cannot intersect $\mathcal{W}_T$. If $u \in \mathcal{T}_k^B \cap \mathcal{W}_T$, then $u \geq \tau$ and $u < \hat{\tau}$. Since every $t \in \mathcal{R}_k^R$ occurs after all periods in $\mathcal{T}_k^B$, we have $u < t < \hat{\tau}$. Because $\mathcal{W}_T$ is an interval, $t \in \mathcal{W}_T$, contradicting $t \in \mathcal{T}_T$. Therefore, $\mathcal{T}_k^B \subseteq \mathcal{T}_T$, which proves the claim.

If $\mathcal{T}_k^B \neq \emptyset$, then define $\mathcal{H}^{k,B} = \mathcal{H}^{\max\{u+1: u \in \mathcal{T}_k^B\}}$. For every $k \geq 0$, define $\Delta_u(F^1) = \Delta(S^u, F^1)$ for $u \in \mathcal{T}_k^B$. The event $\{\mathcal{R}_k^R \neq \emptyset\}$ is $\mathcal{H}^{k,B}$ -measurable because it only records whether the cycle reaches the monitoring periods associated with $\mathcal{T}_k^B$; it does not depend on the particular index selected by U_k . Since U_k is drawn uniformly from $\mathcal{T}_k^B$ conditionally on $\mathcal{H}^{k,B}$, one has:

$$\mathbb{E}_{\mathbb{P}_\tau} [\mathbf{1}\{\mathcal{R}_k^R \neq \emptyset\} \Delta(\tilde{S}_k, F^1) \mid \mathcal{H}^{k,B}] = \mathbf{1}\{\mathcal{R}_k^R \neq \emptyset\} \frac{1}{m_T} \sum_{u \in \mathcal{T}_k^B} \Delta_u(F^1).$$

Moreover, $\tilde{S}_k$ is replayed only on $\mathcal{T}_k^J \cup \mathcal{T}_{k+1}^I$, whose total length is at most $2q_T$. Therefore, for every $k \geq 0$, using Step 1, the following inequality is satisfied:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_\tau} [\mathbf{1}\{\mathcal{R}_k^R \neq \emptyset\} |\mathcal{R}_k^R| \Delta(\tilde{S}_k, F^1) \mid \mathcal{H}^{k,B}] &\leq 2q_T \mathbb{E}_{\mathbb{P}_\tau} [\mathbf{1}\{\mathcal{R}_k^R \neq \emptyset\} \Delta(\tilde{S}_k, F^1) \mid \mathcal{H}^{k,B}] \\ &= \frac{2q_T}{m_T} \sum_{u \in \mathcal{T}_k^B} \Delta_u(F^1) = \frac{1}{4} \sum_{u \in \mathcal{T}_k^B} \Delta_u(F^1). \end{aligned}$$

Taking expectations, summing over k , and using the disjointness of the sets $(\mathcal{R}_k^R : k \geq 0)$ gives:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_\tau} \left[\sum_{t \in \mathcal{T}_T^R} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] &\leq \mathbb{E}_{\mathbb{P}_\tau} \left[\sum_{t \in \mathcal{T}_T^R} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \right] \\ &= \sum_{k \geq 0} \mathbb{E}_{\mathbb{P}_\tau} [\mathbf{1}\{\mathcal{R}_k^R \neq \emptyset\} |\mathcal{R}_k^R| \Delta(\tilde{S}_k, F^1)] \end{aligned}$$

$$\begin{aligned}
&\leq \frac{1}{4} \sum_{k \geq 0} \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{u \in \mathcal{T}_k^B} \Delta_u(F^1) \right] \\
&\leq \frac{1}{4} \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \right].
\end{aligned}$$

Let $n_T^B = |\mathcal{T}_T^B|$, and let $\sigma_1 < \dots < \sigma_{n_T^B}$ be the ordered elements of $\mathcal{T}_T^B$. Since monitoring periods do not update $\mathcal{A}$, the sequence of assortments $(S^{\pi, \sigma_j} : 1 \leq j \leq n_T^B)$ is exactly the sequence $(S^{\mathcal{A}, \sigma_j} : 1 \leq j \leq n_T^B)$ generated by a single run of $\mathcal{A}$ on its own internal time scale, which differs from the calendar time of π because of the monitoring phases. On $\mathcal{E}_{\text{good}}$, all observations in $\mathcal{T}_T^B$ are generated under the same static preferences. Hence, we obtain the following inequalities:

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] &= \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{j=1}^{n_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, \sigma_j}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \\
&\stackrel{(a)}{\leq} \mathbb{E}_{\mathbb{P}^\mathcal{A}} \left[\sum_{j=1}^T (r(S^*(F^1), F^1) - r(S^{\mathcal{A}, j}, F^1)) \right] \\
&\leq \mathcal{R}^\mathcal{A}(\mathcal{F}_S(\gamma_T), T),
\end{aligned}$$

where (a) follows because, on $\mathcal{E}_{\text{good}}$ and during periods in $\mathcal{T}_T^B$, the strategy π simply follows $\mathcal{A}$. In that case, the only randomization involved is the one induced by the subroutine $\mathcal{A}$. Moreover,

$$\begin{aligned}
\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \right] &= \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \\
&\quad + \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}^c\} \right] \\
&\leq \mathcal{R}^\mathcal{A}(\mathcal{F}_S(\gamma_T), T) + T \|w\|_\infty \mathbb{P}[\mathcal{E}_{\text{good}}^c].
\end{aligned}$$

Combining this bound with the testing-period bound gives:

$$\begin{aligned}
&\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T} (r(S^*(F^t), F^t) - r(S^{\pi, t}, F^t)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \\
&= \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] + \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^R} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \\
&\leq \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] + \frac{1}{4} \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t \in \mathcal{T}_T^B} (r(S^*(F^1), F^1) - r(S^{\pi, t}, F^1)) \right] \\
&\leq \frac{5}{4} \mathcal{R}^\mathcal{A}(\mathcal{F}_S(\gamma_T), T) + \frac{T \|w\|_\infty}{4} \mathbb{P}[\mathcal{E}_{\text{good}}^c].
\end{aligned}$$

Step 7 (After detection). If $\hat{\tau} \leq T$, the strategy restarts $\mathcal{A}$ at period $\hat{\tau} + 1$. On $\mathcal{E}_{\text{good}}$, all periods after $\hat{\tau}$ are generated under the static preferences F^τ . Therefore, one has:

$$\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=\hat{\tau}+1}^T (r(S^*(F^\tau), F^\tau) - r(S^{\pi, t}, F^\tau)) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \leq \mathcal{R}^\mathcal{A}(\mathcal{F}_S(\gamma_T), T).$$

If $\hat{\tau} = T + 1$, then the sum on the left-hand side is empty, and the inequality is trivially satisfied.

Step 8 (Conclusion). Combining Steps 5-7, the following inequality is satisfied:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\sum_{t=1}^T (r(S^*(F^t), F^t) - \sum_{t=1}^T r(S^{\pi,t}, F^t)) \mathbf{1}_{\{\mathcal{E}_{\text{good}}\}} \right] &\leq \frac{9}{4} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), T) + 21q_T \|w\|_\infty + \frac{T\|w\|_\infty}{4} \mathbb{P}[\mathcal{E}_{\text{good}}^c] \\ &\leq 3\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), T) + 21q_T \|w\|_\infty + \frac{T\|w\|_\infty}{4} \mathbb{P}[\mathcal{E}_{\text{good}}^c]. \end{aligned}$$

On $\mathcal{E}_{\text{good}}^c$, the regret is at most $T\|w\|_\infty$. Therefore, by using Step 4, we obtain the following inequality:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\sum_{t=1}^T r(S^*(F^t), F^t) - \sum_{t=1}^T r(S^{\pi,t}, F^t) \right] &\leq 3\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), T) + 21q_T \|w\|_\infty + 9T\|w\|_\infty \mathbb{P}[\mathcal{E}_{\text{good}}^c]/4 \\ &\leq 3\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), T) + 21q_T \|w\|_\infty + 5\|w\|_\infty T^{-1}. \end{aligned}$$

For $T \geq T_0$, one has $\log(T) \geq 1$, $\log \log(T) \geq 1$, and $q_T = \lceil \log(T) \rceil \lceil \log \log(T) \rceil \leq 4 \log(T) \log \log(T)$.

Moreover, $5\|w\|_\infty T^{-1} \leq 5\|w\|_\infty$. Defining $c_1 \equiv 89\|w\|_\infty$, we obtain, for all $T \geq T_0$,

$$J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) \leq c_1 \log(T) \log \log(T) + 3\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), T).$$

Taking the supremum over $F^{(\mathbb{N})} \in \mathcal{F}_D(\gamma_T, \varepsilon)$ completes the proof. $\blacksquare$

E.C.2.3 Proof of Proposition 2

We fix $T \geq 9$ and a subsegment length $\Delta \in [T-2]$. Let $\tilde{T} := \lfloor (T-2)/\Delta \rfloor + 1$ and $\ell_j := 2 + (j-1)\Delta$ for $j \in [\tilde{T}]$. Then, $\ell_{\tilde{T}} \leq T$, and each subsegment $\{\ell_j, \dots, \ell_{j+1} - 1\}$ for $j \in [\tilde{T} - 1]$, contains exactly Δ periods. Any periods after $\ell_{\tilde{T}} - 1$ are left outside these subsegments. We assume that F^1 and F^τ are such that the path $F^{(\mathbb{N})} \in \mathcal{F}_U(\gamma_T, \kappa, \phi)$ they induce satisfies $\mathcal{K}(F^1, F^\tau, S^*(F^1)) = 0$ with τ to be specified. Such paths necessarily exist by the definition of $\mathcal{F}_U(\gamma_T, \kappa, \phi)$. For any strategy $\pi \in \mathcal{P}$, let $\mathbb{P}_{\ell_j}^\pi$ be the distribution induced by π when the change occurs at $\tau = \ell_j$; refer to Electronic Companion E.C.4.

The analysis proceeds through two essential lemmas that address distinct strategy classes. First, Lemma 5 analyzes policies that do not sufficiently explore alternative assortments within at least one sub-segment. Then, Lemma 6 considers strategies that are always guaranteed to sufficiently explore new assortments. We quantify exploration intensity through the KL divergence between successive scenarios $\mathbb{P}_{\ell_{j+1}}^\pi$ and $\mathbb{P}_{\ell_j}^\pi$. At a high level, this divergence measures whether shifting the change point from one sub-segment to the next leaves a visible trace in the retailer's sales history.

When the divergence is small, the two sales histories remain statistically close, and the retailer cannot reliably distinguish the two change points. When the divergence is large, the strategy must have generated enough information about the change, which requires selecting assortments that are informative but potentially costly. The proof formalizes this information and regret trade-off using probabilistic arguments adapted from Besbes and Zeevi (2011) and Tsybakov (2003). Formally:

Lemma 5. *Assume that there exist an admissible strategy $\pi \in \mathcal{P}$ and a constant $\beta > 0$ that satisfy:*

$$\min \left\{ \mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) : 1 \leq j \leq \tilde{T} - 1 \right\} \leq \beta.$$

Then, there exists a finite constant $C \equiv C(\gamma_T, \beta) > 0$ such that:

$$\max \left\{ \mathbb{E}_{\mathbb{P}_\tau^\pi} [J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T)] : 2 \leq \tau \leq T + 1 \right\} \geq C\Delta.$$

Proof. We denote by F^1 the preferences before the change occurs and F^τ those after the change occurs, with the actual change time $2 \leq \tau \leq T+1$ to be specified later. Specifically, for all $t \in \mathbb{N}$, one has $F^t \equiv F^1$ if $t < \tau$ and $F^t \equiv F^\tau$, otherwise. Also, we assume that there exist an admissible strategy $\pi \in \mathcal{P}$ and a constant $\beta > 0$ such that:

$$\min \left\{ \mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) : 1 \leq j \leq \tilde{T} - 1 \right\} \leq \beta.$$

Next, let $i_0 \in [\tilde{T} - 1]$ be such that $\mathcal{K}(\mathbb{P}_{\ell_{i_0+1}}^\pi, \mathbb{P}_{\ell_{i_0}}^\pi) \leq \beta$. We consider the following two hypotheses:

$$H_0 : \tau = \ell_{i_0+1},$$

$$H_1 : \tau = \ell_{i_0}.$$

Under $\mathbb{P}_{\ell_{i_0+1}}^\pi$ from the null H_0 , no such shift occurs within $\{\ell_{i_0}, \dots, \ell_{i_0+1} - 1\}$. Conversely, under the probability measure $\mathbb{P}_{\ell_{i_0}}^\pi$ from the alternative H_1 , the preferences undergo a shift at ℓ_{i_0} , where they change from F^1 to F^τ . Then, fix an arbitrary admissible ($\mathcal{H}^{\ell_{i_0+1}-}$ -measurable) decision rule:

$$\varphi : (\mathcal{S} \times \{0, 1\})^{\mathcal{N} \cup \{0\}}^{\ell_{i_0+1}-1} \rightarrow \{0, 1\},$$

where $\varphi = 0$ indicates that “no change” occurs before $\ell_{i_0+1} - 1$, which essentially implies that $\tau \notin \{\ell_{i_0}, \dots, \ell_{i_0+1} - 1\}$, whereas $\varphi = 1$ indicates that a change has occurred precisely at ℓ_{i_0} . Thus, φ maps the set of all possible histories from customers 1 to $\ell_{i_0+1} - 1$ to $\{0, 1\}$.

By using Theorem 2.2 of Tsybakov 2003, the following inequality is satisfied:

$$\inf_{\varphi} \max \left\{ \mathbb{P}_{\ell_{i_0}}^\pi [\varphi \neq 1], \mathbb{P}_{\ell_{i_0+1}}^\pi [\varphi \neq 0] \right\} \geq \max \left\{ \frac{1}{4} \exp(-\beta), \frac{1 - \sqrt{\beta/2}}{2} \right\}.$$

In other words, the following inequality holds:

$$\inf_{\varphi} \max \left\{ \mathbb{P}_{\ell_{i_0}}^\pi [\varphi = 0], \mathbb{P}_{\ell_{i_0+1}}^\pi [\varphi = 1] \right\} \geq \max \left\{ \frac{1}{4} \exp(-\beta), \frac{1 - \sqrt{\beta/2}}{2} \right\}.$$

In addition, we define the following constant:

$$\tilde{C} := \frac{\gamma_T}{4} \max \left\{ \frac{1}{4} \exp(-\beta), \frac{1 - \sqrt{\beta/2}}{2} \right\},$$

where $\gamma_T > 0$ by assumption.

Then, suppose, for the sake of contradiction, that the following inequality holds:

$$\max \left\{ \mathbb{E}_{\mathbb{P}_{\ell_k}^\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] : k \in \{i_0, i_0 + 1\} \right\} \leq \tilde{C}\Delta. \quad (\text{E.C.2.9})$$

Next, consider the following decision rule:

$$\varphi(\pi) = \begin{cases} 0 & \text{if } \sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \leq \gamma_T \Delta / 2, \\ 1 & \text{if } \sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) > \gamma_T \Delta / 2, \end{cases}$$

where the decision rule φ depends implicitly on the observed realization of the purchasing decisions, because the strategy π depends on the history $(\mathcal{H}^t)_{t=1}^{\ell_{i_0+1}-1}$. We write $\varphi(\pi)$ only to keep the notation concise. We complete the proof in three steps. We first bound the probability of Type I error, then bound the probability of Type II error, and finally combine the two bounds.

Step 1 (Type I error). We first establish an upper bound on the probability of Type I errors,

the probability of incorrectly rejecting the null hypothesis by indicating a change when none has occurred before sub-segment $i_0 + 1$. The probability of the Type I error can be expressed as:

$$\begin{aligned}\mathbb{P}_{\ell_{i_0+1}}^\pi[\varphi = 1] &= \mathbb{P}_{\ell_{i_0+1}}^\pi \left[\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) > \gamma_T \Delta / 2 \right] \\ &\stackrel{(a)}{\leq} \frac{2}{\gamma_T \Delta} \mathbb{E}_{\mathbb{P}_{\ell_{i_0+1}}^\pi} \left[\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \right] \\ &\stackrel{(b)}{\leq} \frac{2}{\gamma_T \Delta} \mathbb{E}_{\mathbb{P}_{\ell_{i_0+1}}^\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] \stackrel{(c)}{\leq} \frac{2\tilde{C}}{\gamma_T} = \frac{1}{2} \max \left\{ \frac{1}{4} \exp(-\beta), \frac{1 - \sqrt{\beta/2}}{2} \right\}.\end{aligned}$$

The derivations above employ three key steps: (a) applies Markov's inequality (Jacod and Protter 2004), (b) follows from the optimality of $S^*(F^1)$ and the definition of $\mathcal{J}^\pi(F^{(\mathbb{N})}, T)$ as established in Lemma 13, and (c) uses the bound from equation (E.C.2.9).

Step 2 (Type II error). We establish an upper bound for the probability of Type II error, defined as the probability that our decision rule fails to detect a change when one has actually occurred. To proceed, let us assume that $\varphi = 0$. Under these conditions, the following inequality is satisfied:

$$\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \leq \gamma_T \Delta / 2.$$

In particular, the following inequality also naturally holds:

$$\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \mathbf{1} \left\{ \|S^*(F^1) - S^{\pi,t}\|_1 \geq 1 \right\} \leq \gamma_T \Delta / 2,$$

and we obtain the following inequality:

$$\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} \mathbf{1} \left\{ \|S^*(F^1) - S^{\pi,t}\|_1 \geq 1 \right\} \leq \frac{\gamma_T \Delta}{2\gamma_T} = \Delta / 2.$$

Therefore, we obtain the following sequence of inequalities:

$$\begin{aligned}\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^\tau), F^\tau) - r(S^{\pi,t}, F^\tau)) &\geq \gamma_T \sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} \mathbf{1} \left\{ \|S^*(F^1) - S^{\pi,t}\|_1 \leq 0 \right\} \\ &= \gamma_T \sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (1 - \mathbf{1} \left\{ \|S^*(F^1) - S^{\pi,t}\|_1 \geq 1 \right\}) \\ &= \gamma_T \left(\Delta - \sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} \mathbf{1} \left\{ \|S^*(F^1) - S^{\pi,t}\|_1 \geq 1 \right\} \right) \\ &= \gamma_T (\Delta - \Delta / 2) = \gamma_T \Delta / 2.\end{aligned}$$

As a consequence, we derive the following upper bounds for the probability of Type II errors for φ :

$$\begin{aligned}\mathbb{P}_{\ell_{i_0}}^\pi[\varphi = 0] &\leq \mathbb{P}_{\ell_{i_0}}^\pi \left[\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^\tau), F^\tau) - r(S^{\pi,t}, F^\tau)) \geq \frac{\gamma_T \Delta}{2} \right] \\ &\stackrel{(a)}{\leq} \frac{2}{\gamma_T \Delta} \mathbb{E}_{\mathbb{P}_{\ell_{i_0}}^\pi} \left[\sum_{t=\ell_{i_0}}^{\ell_{i_0+1}-1} (r(S^*(F^\tau), F^\tau) - r(S^{\pi,t}, F^\tau)) \right]\end{aligned}$$

$$\stackrel{(b)}{\leq} \frac{2}{\gamma_T \Delta} \mathbb{E}_{\mathbb{P}_{\ell_{i_0}}^\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] \stackrel{(c)}{\leq} \frac{2}{\gamma_T \Delta} \tilde{C} \Delta = \frac{1}{2} \max \left\{ \frac{1}{4} \exp(-\beta), \frac{1 - \sqrt{\beta/2}}{2} \right\},$$

where (a) follows from Markov's inequality (Jacod and Protter 2004), while (b) follows from the optimality of $S^*(F^\tau)$ and Lemma 13. Finally the last inequality (c) follows from (E.C.2.9).

Step 3 (Conclusion). Consequently, based on the results from both Step 1 and Step 2, we conclude that the minimax testing error is bounded above by:

$$\inf_{\varphi} \max \left\{ \mathbb{P}_{\ell_{i_0}}^\pi [\varphi = 0], \mathbb{P}_{\ell_{i_0+1}}^\pi [\varphi = 1] \right\} \leq \frac{1}{2} \max \left\{ \frac{1}{4} \exp(-\beta), \frac{1 - \sqrt{\beta/2}}{2} \right\},$$

which is a contradiction. Hence, the following inequality must hold:

$$\max \left\{ \mathbb{E}_{\mathbb{P}_{\ell_k}^\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] : k \in \{i_0, i_0 + 1\} \right\} > \tilde{C} \Delta.$$

Therefore, we conclude that:

$$\max \left\{ \mathbb{E}_{\mathbb{P}_\tau^\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] : 2 \leq \tau \leq T + 1 \right\} > C(\gamma_T, \beta) \Delta,$$

where $C(\gamma_T, \beta) = \frac{\gamma_T}{4} \max \left\{ \frac{1}{4} \exp(-\beta), \frac{1 - \sqrt{\beta/2}}{2} \right\}$, where β is as defined in the proposition. $\blacksquare$

We proceed to establish a lower bound on the attainable regret for admissible policies that exhibit adequate exploration within each sub-segment. Formally:

Lemma 6. Assume that there exist an admissible strategy $\pi \in \mathcal{P}$ and a constant $\beta > 0$ that satisfy:

$$\min \left\{ \mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) : 1 \leq j \leq \tilde{T} - 1 \right\} > \beta.$$

Then, there exists a finite constant $C \equiv C(\gamma_T, \phi, \beta) > 0$ such that:

$$\max \left\{ \mathbb{E}_{\mathbb{P}_\tau^\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] : 2 \leq \tau \leq T + 1 \right\} \geq C(\tilde{T} - 1).$$

Proof. We denote by F^1 the preferences before the change occurs and F^τ those after the change occurs, with the actual change time $2 \leq \tau \leq T + 1$ to be specified later. Specifically, for all $t \in \mathbb{N}$, one has $F^t \equiv F^1$ if $t < \tau$ and $F^t \equiv F^\tau$, otherwise. Suppose also that there exist an admissible strategy $\pi \equiv (\psi_t(\cdot) : t \in [T]) \in \mathcal{P}$ and a constant $\beta > 0$ such that:

$$\min \left\{ \mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) : 1 \leq j \leq \tilde{T} - 1 \right\} > \beta.$$

In this case, the strategy is guaranteed to explore sufficiently within each time segment. Hence, the environment can choose to change the preferences at the last candidate change time, i.e., $\tau = \ell_{\tilde{T}} \leq T$. To proceed, we work with the random variable $\mathcal{J}^\pi(F^{(\mathbb{N})}, T)$ instead of its expected value; refer to Lemma 13 for further details. Therefore, the following sequence of inequalities holds:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{\ell_{\tilde{T}}}^\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] &= \mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\sum_{t=1}^{\ell_{\tilde{T}}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \right] \\ &\quad + \mathbb{E}_{\mathbb{P}_{\ell_{\tilde{T}}}^\pi} \left[\sum_{t=\ell_{\tilde{T}}}^T (r(S^*(F^\tau), F^\tau) - r(S^{\pi,t}, F^\tau)) \right] \\ &\stackrel{(a)}{\geq} \mathbb{E}_{\mathbb{P}_{\ell_{\tilde{T}}}^\pi} \left[\sum_{t=\ell_1}^{\ell_{\tilde{T}}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \right] \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^{\tilde{T}-1} \mathbb{E}_{\mathbb{P}_{\ell_{\tilde{T}}^\pi}^\pi \left[\sum_{t=\ell_i}^{\ell_{i+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \right] \\
&\stackrel{(b)}{=} \sum_{i=1}^{\tilde{T}-1} \mathbb{E}_{\mathbb{P}_{\ell_{i+1}}^\pi} \left[\sum_{t=\ell_i}^{\ell_{i+1}-1} (r(S^*(F^1), F^1) - r(S^{\pi,t}, F^1)) \right] \\
&\stackrel{(c)}{\geq} \gamma_T \sum_{i=1}^{\tilde{T}-1} \mathbb{E}_{\mathbb{P}_{\ell_{i+1}}^\pi} \left[\sum_{t=\ell_i}^{\ell_{i+1}-1} \mathbf{1}\{\|S^*(F^1) - S^{\pi,t}\|_1 > 0\} \right] \\
&\stackrel{(d)}{\geq} \gamma_T \sum_{i=1}^{\tilde{T}-1} \frac{\mathcal{K}(\mathbb{P}_{\ell_{i+1}}^\pi, \mathbb{P}_{\ell_i}^\pi)}{\mathcal{K}(F^1, F^\tau)} \geq \frac{\gamma_T \beta (\tilde{T} - 1)}{\mathcal{K}(F^1, F^\tau)},
\end{aligned}$$

where (a) follows from the optimality of $S^*(F^\tau)$. For any $i \in [\tilde{T} - 1]$, the distribution of the purchasing decision of customer $t \in \{\ell_i, \dots, \ell_{i+1} - 1\}$ is independent of the time at which the change occurs, provided it takes place after (or at) ℓ_{i+1} , which justifies (b). The inequality (c) follows from the definition of γ_T . Finally, (d) follows from Lemma 17 (whose conditions are satisfied as $\mathcal{K}(p(S^*(F^1), F^1), p(S^*(F^1), F^\tau)) = 0$) and $\mathcal{K}(F^1, F^\tau) := \max\{\mathcal{K}(p(S, F^1), p(S, F^\tau)) : S \in \mathcal{S}\}$.

Next, recall that by the construction of the risk profile $\mathcal{F}_U(\gamma_T, \kappa, \phi)$, one has $\mathcal{K}(F^1, F^\tau) := \max\{\mathcal{K}(p(S, F^1), p(S, F^\tau)) : S \in \mathcal{S}\} < 2\phi/\varsigma$. Finally, we obtain the following inequality on the difference between the expected revenue obtained by the clairvoyant retailer and by the strategy π :

$$\max \left\{ \mathbb{E}_{\mathbb{P}_\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] : 2 \leq \tau \leq T + 1 \right\} > C(\tilde{T} - 1),$$

where $C \equiv C(\varsigma, \gamma_T, \phi, \beta) := \gamma_T \beta \varsigma / (2\phi)$. This inequality concludes the proof. $\blacksquare$

We now present the proof of Proposition 2. The proof follows from the observation that for any given $\beta > 0$ and any given $\pi \in \mathcal{P}$, the conditions of either Lemma 5 or Lemma 6 must be satisfied. The proposition's conclusion therefore follows directly from the application of the relevant lemma.

Proof of Proposition 2. Fix $F^{(\mathbb{N})} \in \mathcal{F}_U(\gamma_T, \kappa, \phi)$ such that $\mathcal{K}(F^1, F^\tau, S^*(F^1)) = 0$. For $T \geq 9$, we define $\Delta = \lceil T^{1/2} \rceil$. Thus, we have $\Delta \geq T^{1/2}$ and $\tilde{T} - 1 \geq \frac{1}{6}\sqrt{T}$. Next, we fix $\beta = 1$. For any strategy $\pi \in \mathcal{P}$, we apply either Lemma 5 or Lemma 6, depending on whether the strategy explores sufficiently or not. Hence, defining $c(\varsigma, \gamma_T, \phi) = \min\{C(\gamma_T, 1), \frac{1}{6}C(\varsigma, \gamma_T, \phi, 1)\}$, the following inequality holds:

$$\max \left\{ \mathbb{E}_{\mathbb{P}_\pi} [J^*(F^{(\mathbb{N})}, T) - \mathcal{J}^\pi(F^{(\mathbb{N})}, T)] : 2 \leq \tau \leq T + 1 \right\} \geq \min \{C(\gamma_T, 1)\Delta, C(\varsigma, \gamma_T, \phi, 1)(\tilde{T} - 1)\} \\ \geq c(\varsigma, \gamma_T, \phi)\sqrt{T},$$

Taking the supremum over $\mathcal{F}_U(\gamma_T, \kappa, \phi)$ concludes the proof. $\blacksquare$

E.C.2.4 Proof of the claim in Remark 3

For a sentinel family $\mathcal{E} \subseteq \mathcal{S}$ and a separation level $\eta > 0$, define:

$$\tilde{\mathcal{F}}_U \equiv \tilde{\mathcal{F}}_U(\mathcal{E}, \eta) := \{F^{(\mathbb{N})} \in \mathcal{F}_U : \exists \tau \geq 2 \text{ such that } \max\{\mathcal{K}^\tau(S) : S \in \mathcal{E}\} > \eta\}.$$

The class $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$ contains the paths in $\mathcal{F}_U$ whose change is visible from the sentinel family. For

each path, the maximum selects the assortment in $\mathcal{E}$ that separates the purchasing distributions before and after the change the most. Requiring this maximum to exceed η ensures that the monitoring strategy has at least one assortment through which the change can be detected.

Proposition 6. *Assume that $\mathcal{E} \subseteq \mathcal{S}$ is a product cover. For all $\kappa > 0$, there exists $\eta > 0$ such that $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta) \cap \mathcal{F}_{\text{MNL}} = \mathcal{F}_U(\gamma_T, \kappa, \phi) \cap \mathcal{F}_{\text{MNL}}$.*

Proof. The inclusion $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta) \cap \mathcal{F}_{\text{MNL}} \subseteq \mathcal{F}_U(\gamma_T, \kappa, \phi) \cap \mathcal{F}_{\text{MNL}}$ is immediate from the definition of the set $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$. We prove the reverse inclusion. Recall that $\mathcal{F}_{\text{MNL}}$ is the set of paths of customers' MNL preferences represented by their attraction parameters, whose induced purchasing probabilities lie in $[\varsigma, 1 - \varsigma]^{|S|+1}$ for all feasible assortments $S \in \mathcal{S}$. Hereafter, we work directly with these vectors of attraction parameters. For the proof, we denote by $\mathcal{V}_{\text{MNL}}$ the set of MNL attraction parameters $\nu \in \mathbb{R}^{\mathcal{N}}$ whose induced purchasing probabilities satisfy the condition above. The set $\mathcal{V}_{\text{MNL}}$ is a compact subset of $\mathbb{R}^{\mathcal{N}}$. For every $\nu \in \mathcal{V}_{\text{MNL}}$ and $S \in \mathcal{S}$, the MNL purchasing probabilities satisfy:

$$p_i(S, \nu) = \frac{\nu_i}{1 + \sum_{j \in S} \nu_j} \quad \text{for } i \in S, \quad \text{and} \quad p_0(S, \nu) = \frac{1}{1 + \sum_{j \in S} \nu_j}.$$

Hence, for every $S \in \mathcal{S}$ and every $i \in S$, one has $p_i(S, \nu)/p_0(S, \nu) = \nu_i$.

For $\nu, \nu' \in \mathcal{V}_{\text{MNL}}$, define the mapping $\Phi(\nu, \nu') = \max\{\mathcal{K}(p(S, \nu'), p(S, \nu)) : S \in \mathcal{S}\}$ and the mapping $\Psi_{\mathcal{E}}(\nu, \nu') = \max\{\mathcal{K}(p(E, \nu'), p(E, \nu)) : E \in \mathcal{E}\}$. Both maps are continuous because $p_i(S, \cdot)$ are continuous on $\mathcal{V}_{\text{MNL}}$ for all $i \in S$ and all $S \in \mathcal{S}$, because both sets $\mathcal{S}$ and $\mathcal{E}$ are finite, and because the KL divergence for categorical distributions $\mathcal{K}(\cdot, \cdot)$ is continuous on $[\varsigma, 1 - \varsigma]^{k+1} \times [\varsigma, 1 - \varsigma]^{k+1}$ for all $k \in [K]$. Define $\eta(\kappa) = \inf\{\Psi_{\mathcal{E}}(\nu, \nu') : \nu, \nu' \in \mathcal{V}_{\text{MNL}}, \Phi(\nu, \nu') \geq \kappa\}$. The set over which the infimum is taken is not empty without loss of generality (provided that ς is small enough). Then, the infimum is taken over a compact set as it is a closed subset of the compact set $\mathcal{V}_{\text{MNL}} \times \mathcal{V}_{\text{MNL}}$, where closedness follows from the continuity of Φ . We now show that this infimum is strictly positive.

Suppose, toward a contradiction, that $\eta(\kappa) = 0$. Since the infimum is attained, there exist $\nu, \nu' \in \mathcal{V}_{\text{MNL}}$ such that $\Phi(\nu, \nu') \geq \kappa$ and $\Psi_{\mathcal{E}}(\nu, \nu') = 0$. The equality $\Psi_{\mathcal{E}}(\nu, \nu') = 0$ implies that, for every $E \in \mathcal{E}$, the two purchasing distributions $p(E, \nu')$ and $p(E, \nu)$ are equal. Fix any product $i \in \mathcal{N}$. Since $\mathcal{E}$ is a product cover, there exists $E_i \in \mathcal{E}$ such that $i \in E_i$. The equality of the two purchasing distributions on E_i gives $p_i(E_i, \nu')/p_0(E_i, \nu') = p_i(E_i, \nu)/p_0(E_i, \nu)$. Using the MNL ratio identity, we obtain $\nu'_i = \nu_i$. Since i was arbitrary, $\nu' = \nu$. Thus $p(S, \nu') = p(S, \nu)$ for every $S \in \mathcal{S}$, so $\Phi(\nu, \nu') = 0$, contradicting in turn $\Phi(\nu, \nu') \geq \kappa > 0$. Therefore, one has $\eta(\kappa) > 0$.

Set $\eta = \eta(\kappa)/2$. Fix $F^{(\mathbb{N})} \in \mathcal{F}_U(\gamma_T, \kappa, \phi) \cap \mathcal{F}_{\text{MNL}}$, and let τ be the change time. We denote by $\nu^t \equiv \nu(F^t) \in \mathcal{V}_{\text{MNL}}$ the attraction parameters associated with preferences F^t for $t \in [T]$. Since $F^{(\mathbb{N})} \in \mathcal{F}_U(\gamma_T, \kappa, \phi)$, one has $\max\{\mathcal{K}^\tau(S) : S \in \mathcal{S}\} \geq \kappa$. Consequently, by the definition of $\eta \equiv \eta(\kappa)/2$ and since $\mathcal{K}^\tau(S) \equiv \mathcal{K}(p(S, \nu^\tau), p(S, \nu^1))$, one also has $\max\{\mathcal{K}^\tau(E) : E \in \mathcal{E}\} > \eta$. Hence, $F^{(\mathbb{N})} \in \tilde{\mathcal{F}}_U(\mathcal{E}, \eta) \cap \mathcal{F}_{\text{MNL}}$. This result proves the reverse inclusion, completing in turn the proof. ■

E.C.2.5 Proof of Theorem 2

The two-sample test described in Electronic Companion E.C.2.2 is used in Algorithm 2 to detect changes in customers' preferences. We refer to the proof of Theorem 1 for the details of the testing procedure and use the same notation and construction throughout the remainder of the proof.

Under the assumptions of Theorem 2, the sentinel family satisfies $\eta(\mathcal{E}) > 0$. Therefore, for any $\eta \in (0, \eta(\mathcal{E}))$, the definition of $\tilde{\mathcal{F}}_U$ before Proposition 6 gives $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta) = \mathcal{F}_U(\gamma_T, \kappa, \phi)$. We state the regret bound below over $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$ in order to keep explicit the dependence on the parameter η .

Proof. Let $\pi \equiv \pi(\mathcal{A}, \mathcal{E}, T)$ be the active monitoring strategy from Algorithm 2. Fix an arbitrary path of customers' preferences $F^{(\mathbb{N})} \in \tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$. Let τ denote its change time, so that $F^1 = \dots = F^{\tau-1}$ and $F^\tau = \dots = F^T$ when $1 < \tau \leq T$, with $\tau = T + 1$ when $F^t \equiv F^1$ for all $t \in [T]$. When $\tau \leq T$, the definition of $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$ gives $\max\{\mathcal{K}^\tau(S) : S \in \mathcal{E}\} \geq \eta$. For the remainder of the proof, write $M = |\mathcal{E}|$, $q_T = \lceil \log(T) \log \log(T) \rceil$, $D_T = M q_T$, and $H_T = \lceil \sqrt{T D_T} \rceil$.

Let $c_\varsigma > 0$ be the finite constant from Lemma 9. Define $T_1(M) = \lceil \max\{e^9, 16M^2\} \rceil$. Define $T_2(\eta, K, \varsigma)$ as the smallest integer $\tilde{T} \geq 2$ such that, for every $T \geq \tilde{T}$, the following three inequalities hold. First, $q_T \geq 8c_\varsigma(K+1)^2\eta^{-1}(4\log(T) + \log(4(K+1)))$. Second, $4(K+1)T^{-5} \leq T^{-4}$. Third, $q_T\eta/(16c_\varsigma) > 8(K+1)\log(2q_T) + 8\log(T+1)$. Such an integer exists because $q_T/\log(T) \rightarrow +\infty$ as $T \rightarrow \infty$. Finally, fix $T_0 := \max\{T_1(M), T_2(\eta, K, \varsigma)\}$.

For every $T \geq T_1(M)$, the elementary relations $q_T \geq 1$, $q_T \leq T$, and $D_T \leq H_T \leq T$ hold. Indeed, if $x = \log(T)$, then $T \geq e^9$ gives $x \geq 9$. For $x \geq 9$, one has $\log(x) \leq x/4$ and $x^2/4 \leq e^{x/2}$. Thus $\log(T) \log \log(T) \leq \sqrt{T}$. Since $T \geq e^9$, this product is at least one, and $q_T \leq 2\sqrt{T}$. Moreover, $T \geq 16M^2$ implies $D_T = M q_T \leq T/2$. Hence, $D_T \leq \sqrt{T D_T} \leq H_T$. Also, $D_T \leq T/2$ implies $\sqrt{T D_T} < T$, and since T is an integer, $H_T \leq T$. Finally, $q_T \leq 2\sqrt{T} \leq T$, and $q_T \geq 1$ follows from $T \geq e^9$.

For every assortment $S \in \mathcal{E}$ and customers' preferences F , we write $p(S, F) = (p_i(S, F) : i \in S \cup \{0\})$ for the corresponding vector of purchasing probabilities. For every $t \in [T]$, define the one-period regret by $\Delta^{\pi, t} = r(S^*(F^t), F^t) - r(S^{\pi, t}, F^t)$. Since each customer purchases at most one product, both one-period revenues are bounded by $\|w\|_\infty$. Hence, $0 \leq \Delta^{\pi, t} \leq \|w\|_\infty$ for every $t \in [T]$.

Step 1 (Clean comparisons). Assume that $\tau \leq T - D_T$. Since $F^{(\mathbb{N})} \in \tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$, there exists $S^\tau \in \mathcal{E}$ such that $\mathcal{K}^\tau(S^\tau) \geq \eta$. We first show that the two-sample test rejects the no-change hypothesis with probability at least $1 - T^{-4}$ whenever it compares two sales histories of length q_T collected under S^τ , with the reference history generated under $F^{\tau-1}$ and the test history generated under F^τ . We call such a comparison *clean*. In a clean comparison, the assortment is fixed, so the only difference between the distributions generating the two sales histories comes from the change in customers' preferences.

Let $d_\tau = |S^\tau \cup \{0\}|$. Since every feasible assortment contains at most K products, $d_\tau \leq K + 1$. By the definition of $\mathcal{K}^\tau(S^\tau)$, we have $\mathcal{K}(p(S^\tau, F^\tau), p(S^\tau, F^{\tau-1})) = \mathcal{K}^\tau(S^\tau) \geq \eta$. By

Lemma 9, $\mathcal{K}(p(S^\tau, F^\tau), p(S^\tau, F^{\tau-1})) \leq c_\varsigma (\sum_{i \in S^\tau \cup \{0\}} |p_i(S^\tau, F^\tau) - p_i(S^\tau, F^{\tau-1})|)^2$. Hence, with $\delta_\tau = \sum_{i \in S^\tau \cup \{0\}} |p_i(S^\tau, F^\tau) - p_i(S^\tau, F^{\tau-1})|$, we obtain $\delta_\tau \geq \sqrt{\eta/c_\varsigma}$. To proceed, define $a = \frac{1}{4}\sqrt{\eta/c_\varsigma}$.

The reference history $\mathcal{J}(S^\tau)$ has length q_T and is generated under customers' preferences $F^{\tau-1}$, whereas the test history $\mathcal{I}(S^\tau)$ has length q_T and is generated under customers' preferences F^τ . Let $\hat{p}(S^\tau, F^{\tau-1})$ denote the empirical purchasing distribution constructed from $\mathcal{J}(S^\tau)$, and let $\hat{p}(S^\tau, F^\tau)$ denote the empirical purchasing distribution constructed from $\mathcal{I}(S^\tau)$. Define the event:

$$\mathcal{E}_T = \left\{ \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^{\tau-1}) - p_i(S^\tau, F^{\tau-1})| \leq a \right\} \cap \left\{ \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - p_i(S^\tau, F^\tau)| \leq a \right\}.$$

We first bound the probability of $\mathcal{E}_T^c$. If $\sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^{\tau-1}) - p_i(S^\tau, F^{\tau-1})| > a$, then at least one option $i \in S^\tau \cup \{0\}$ satisfies $|\hat{p}_i(S^\tau, F^{\tau-1}) - p_i(S^\tau, F^{\tau-1})| > a/d_\tau$. Hence, by Hoeffding's inequality (Boucheron et al. 2013) and a union bound over the d_τ products, we obtain:

$$\begin{aligned} \mathbb{P}_\tau^\pi \left[\sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^{\tau-1}) - p_i(S^\tau, F^{\tau-1})| > a \right] &\leq \sum_{i \in S^\tau \cup \{0\}} \mathbb{P}_\tau^\pi [|\hat{p}_i(S^\tau, F^{\tau-1}) - p_i(S^\tau, F^{\tau-1})| > a/d_\tau] \\ &\leq 2d_\tau \exp(-2q_T a^2/d_\tau^2). \end{aligned}$$

The same argument for the empirical distribution $\hat{p}(S^\tau, F^\tau)$ gives:

$$\mathbb{P}_\tau^\pi \left[\sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - p_i(S^\tau, F^\tau)| > a \right] \leq 2d_\tau \exp(-2q_T a^2/d_\tau^2).$$

Combining the two inequalities, using $d_\tau \leq K+1$, and using $a^2 = \eta/(16c_\varsigma)$, we obtain:

$$\mathbb{P}_\tau^\pi[\mathcal{E}_T^c] \leq 4d_\tau \exp(-2q_T a^2/d_\tau^2) \leq 4(K+1) \exp(-q_T \eta/(8c_\varsigma(K+1)^2)).$$

Therefore, for every $T \geq T_0$, the previous inequality gives $\mathbb{P}_\tau^\pi[\mathcal{E}_T^c] \leq T^{-4}$.

We now prove that the likelihood-ratio statistic is bounded below by the threshold used in the statistical test of the strategy π on $\mathcal{E}_T$. On $\mathcal{E}_T$, the triangle inequality gives the following inequality:

$$\begin{aligned} \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - \hat{p}_i(S^\tau, F^{\tau-1})| &\geq \sum_{i \in S^\tau \cup \{0\}} |p_i(S^\tau, F^\tau) - p_i(S^\tau, F^{\tau-1})| \\ &\quad - \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - p_i(S^\tau, F^\tau)| \\ &\quad - \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^{\tau-1}) - p_i(S^\tau, F^{\tau-1})| \geq \delta_\tau - 2a \geq \frac{1}{2}\sqrt{\eta/c_\varsigma}. \end{aligned}$$

Let $\hat{p}^0$ be the pooled empirical distribution. Since both histories have length q_T , one has $\hat{p}^0 = (\hat{p}(S^\tau, F^{\tau-1}) + \hat{p}(S^\tau, F^\tau))/2$. Hence, the following equalities are satisfied:

$$\sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^{\tau-1}) - \hat{p}_i^0| = \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - \hat{p}_i^0| = \frac{1}{2} \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - \hat{p}_i(S^\tau, F^{\tau-1})|.$$

By Pinsker's inequality (Tsybakov 2003), we obtain:

$$\begin{aligned} \mathcal{K}(\hat{p}(S^\tau, F^{\tau-1}), \hat{p}^0) &\geq \frac{1}{2} \left(\sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^{\tau-1}) - \hat{p}_i^0| \right)^2, \\ \mathcal{K}(\hat{p}(S^\tau, F^\tau), \hat{p}^0) &\geq \frac{1}{2} \left(\sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - \hat{p}_i^0| \right)^2. \end{aligned}$$

Therefore, on $\mathcal{E}_T$, the following sequence of inequalities holds:

$$\begin{aligned}
G(S^\tau, \mathcal{J}(S^\tau), \mathcal{I}(S^\tau)) &= q_T \mathcal{K}(\hat{p}(S^\tau, F^{\tau-1}), \hat{p}^0) + q_T \mathcal{K}(\hat{p}(S^\tau, F^\tau), \hat{p}^0) \\
&\geq q_T \frac{1}{2} \left(\frac{1}{2} \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - \hat{p}_i(S^\tau, F^{\tau-1})| \right)^2 \\
&\quad + q_T \frac{1}{2} \left(\frac{1}{2} \sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - \hat{p}_i(S^\tau, F^{\tau-1})| \right)^2 \\
&= \frac{q_T}{4} \left(\sum_{i \in S^\tau \cup \{0\}} |\hat{p}_i(S^\tau, F^\tau) - \hat{p}_i(S^\tau, F^{\tau-1})| \right)^2 \geq \frac{q_T \eta}{16c_\zeta}.
\end{aligned}$$

In a clean comparison with q_T reference observations and q_T test observations, the test uses the threshold $\Gamma(q_T, q_T, T) = 8(K+1)\log(2q_T) + 8\log(T+1)$. By the definition of T_0 , for $T \geq T_0$, one has:

$$\frac{q_T \eta}{16c_\zeta} > 8(K+1)\log(2q_T) + 8\log(T+1).$$

Hence, for every $T \geq T_0$, on the event $\mathcal{E}_T$, one has $G(S^\tau, \mathcal{J}(S^\tau), \mathcal{I}(S^\tau)) > \Gamma(q_T, q_T, T)$. Consequently:

$$\mathbb{P}_\tau^\pi[G(S^\tau, \mathcal{J}(S^\tau), \mathcal{I}(S^\tau)) \leq \Gamma(q_T, q_T, T)] \leq \mathbb{P}_\tau^\pi[\mathcal{E}_T^c] \leq T^{-4}.$$

Step 2 (Testing periods). Fix an arbitrary ordering $\mathcal{E} = \{E_1, \dots, E_M\}$. For $S \in \mathcal{E}$, let $\mathcal{T}_0(S) \subseteq [T]$ be the set of periods used to collect the initial reference history $\mathcal{J}(S)$. For each cycle $m \geq 1$, let $\mathcal{T}_m^{\mathcal{A}} \subseteq [T]$ be the set of periods during which the strategy runs $\mathcal{A}$, and let $\mathcal{T}_m(S) \subseteq [T]$ be the set of periods used to collect the test history for S . Define the set of testing periods by:

$$\mathcal{T}^\mathcal{E} = \bigcup_{S \in \mathcal{E}} \mathcal{T}_0(S) \bigcup \left(\bigcup_{m \geq 1} \bigcup_{S \in \mathcal{E}} \mathcal{T}_m(S) \right).$$

Let N_T be the number of cycles started before the horizon. Since every complete cycle starts with an interval in which $\mathcal{A}$ is run for H_T periods, one has $N_T \leq \lceil T/H_T \rceil$. The initial reference histories require at most D_T periods, and each complete monitoring interval in which assortments from $\mathcal{E}$ are offered has length at most D_T . Hence, $|\mathcal{T}^\mathcal{E}| \leq D_T + D_T \lceil T/H_T \rceil \leq 2D_T + TD_T/H_T$. Therefore, the following inequality is always satisfied:

$$\sum_{t \in \mathcal{T}^\mathcal{E}} \Delta^{\pi, t} \leq \|w\|_\infty (2D_T + TD_T/H_T).$$

Step 3 (False alarms). We now control rejections that occur when the two sales histories entering the test are generated under the same static preferences. For $S \in \mathcal{E}$, recall from Step 2 that $\mathcal{T}_0(S)$ denotes the set of periods used to collect the initial reference history $\mathcal{J}(S)$.

For $m \geq 1$, $\mathcal{T}_m(S)$ is the set of periods used to collect the test history for S in cycle m . We denote the corresponding sales history by $\mathcal{I}_m(S)$. Thus, $\mathcal{I}_m(S)$ is the batch called $\mathcal{I}(S)$ inside the algorithm during cycle m . Define $n_0(S) = |\mathcal{J}(S)|$ and $n_m(S) = |\mathcal{I}_m(S)|$. A comparison for S in cycle m is performed only when $n_0(S)n_m(S) > 0$. In that case, the strategy computes $G(S, \mathcal{J}(S), \mathcal{I}_m(S))$ and compares it with $\Gamma(n_0(S), n_m(S), T)$. Accordingly, define the rejection event, namely the event on which the strategy declares a change in cycle m , by:

$$\mathcal{R}_{m,S} = \{n_0(S)n_m(S) > 0 \text{ and } G(S, \mathcal{J}(S), \mathcal{I}_m(S)) > \Gamma(n_0(S), n_m(S), T)\}.$$

Next, we define the event that the test compares two sales histories generated under the same preferences. Let $\mathcal{N}_{m,S}$ be the event that $n_0(S)n_m(S) > 0$ and that all periods in $\mathcal{T}_0(S) \cup \mathcal{T}_m(S)$ lie either before τ or from τ onward. Equivalently, define:

$$\mathcal{N}_{m,S} = \{n_0(S)n_m(S) > 0\} \cap \{\{\max\{u : u \in \mathcal{T}_0(S) \cup \mathcal{T}_m(S)\} < \tau\} \cup \{\min\{u : u \in \mathcal{T}_0(S) \cup \mathcal{T}_m(S)\} \geq \tau\}\}.$$

On $\mathcal{N}_{m,S}$, every observation entering the comparison $(\mathcal{J}(S), \mathcal{I}_m(S))$ is generated by repeatedly offering assortment S under the same preferences. Therefore, the two sales histories consist of categorical observations on $S \cup \{0\}$ drawn from a common purchasing distribution.

We now apply the two-sample categorical bound from Lemma 10. Fix any two positive integers a and b , and consider two samples of lengths a and b from a common categorical distribution on $S \cup \{0\}$. Since $|S \cup \{0\}| \leq K + 1$, the bound in the lemma (for the appropriate measure $\mathbb{P}$) gives:

$$\mathbb{P}[G(S, \mathcal{J}(S), \mathcal{I}(S)) > x] \leq (a + b)^{2(K+1)} \exp(-x).$$

Taking $x = \Gamma(a, b, T)$, and using the definition of the threshold, gives:

$$(a + b)^{2(K+1)} \exp(-\Gamma(a, b, T)) \leq (a + b)^{-6(K+1)} (T + 1)^{-8} \leq (T + 1)^{-8}.$$

Hence, conditional on any fixed values of $n_0(S)$ and $n_m(S)$, the probability that the test rejects the equality hypothesis when both sales histories are generated under the same preferences is at most $(T + 1)^{-8}$. Since $n_0(S)$ and $n_m(S)$ can only take the values 0 and q_T , for all $T \geq T_0$, the last bound implies:

$$\mathbb{P}_\tau^\pi[\mathcal{N}_{m,S} \cap \mathcal{R}_{m,S}] \leq (T + 1)^{-8} \leq (T + 1)^{-6}.$$

Define the event that at least one test comparing two sales histories generated under the same preferences incorrectly declares a change before the end of the horizon by $\mathcal{E}_{\text{false}} = \bigcup_{m \geq 1} \bigcup_{S \in \mathcal{E}} (\mathcal{N}_{m,S} \cap \mathcal{R}_{m,S})$. There are at most T cycles and $M = |\mathcal{E}|$ monitoring assortments. A union bound gives:

$$\mathbb{P}_\tau^\pi[\mathcal{E}_{\text{false}}] \leq \sum_{m \geq 1} \sum_{S \in \mathcal{E}} \mathbb{P}_\tau^\pi[\mathcal{N}_{m,S} \cap \mathcal{R}_{m,S}] \leq MT(T + 1)^{-6} \leq MT^{-3}.$$

Step 4 (Controlling the probability of error). For cycle $m \geq 1$ and $S \in \mathcal{E}$, recall that $\mathcal{T}_m(S)$ is the set of periods used to collect the test batch $\mathcal{I}_m(S)$ in cycle m , while $\mathcal{T}_0(S)$ is the set of periods used to collect the initial reference batch $\mathcal{J}(S)$. Define the set of all periods in which the retailer offers an assortment from $\mathcal{E}$ by $\mathcal{T}^\mathcal{E} = \bigcup_{S \in \mathcal{E}} \mathcal{T}_0(S) \cup \bigcup_{m \geq 1} \bigcup_{S \in \mathcal{E}} \mathcal{T}_m(S)$. Let $\hat{\tau}$ be the first period at which the strategy declares DETECTED = True, with the convention that $\hat{\tau} = T + 1$ if no change is declared before the horizon. When $\tau \leq T$, fix $S^\tau \in \mathcal{E}$ such that $\mathcal{K}^\tau(S^\tau) \geq \eta$.

For every cycle $m \geq 1$, Step 3 implies that $\mathcal{R}_{m,S^\tau}$ is the event that the test comparing $\mathcal{J}(S^\tau)$ and $\mathcal{I}_m(S^\tau)$ declares DETECTED = True. We also define the event that this comparison is clean by:

$$\mathcal{C}_m^\tau = \{|\mathcal{I}_m(S^\tau)| = q_T\} \cap \{\{\max\{u : u \in \mathcal{T}_0(S^\tau)\} < \tau\} \cap \{\min\{u : u \in \mathcal{T}_m(S^\tau)\} \geq \tau\}\}.$$

Thus, on $\mathcal{C}_m^\tau$, the reference batch for S^τ is collected under $F^{\tau-1}$, while the m -th cycle test batch for S^τ is collected under F^τ . Step 1 therefore applies to this clean comparison. Next, we define the event:

$$\mathcal{E}_{\text{pow}} = \bigcap_{m \geq 1} [(\mathcal{C}_m^\tau)^c \cup \mathcal{R}_{m,S^\tau}].$$

Thus, on $\mathcal{E}_{\text{pow}}$, every clean comparison for the sentinel assortment S^τ identifies a change. By Step 1, for every $m \geq 1$, one has $\mathbb{P}_\tau^\pi[\mathcal{C}_m^\tau \cap \mathcal{R}_{m,S^\tau}^c] \leq T^{-4}$. A union bound over $m \geq 1$ gives $\mathbb{P}_\tau^\pi[\mathcal{E}_{\text{pow}}^c] \leq T^{-3}$.

To proceed, define $\mathcal{E}_{\text{good}} = \mathcal{E}_{\text{false}}^c \cap \mathcal{E}_{\text{pow}}$. On this event, the testing procedure behaves as required: it makes no false detection and detects the change whenever a detectable change occurs before the horizon. Combining Step 3 with the preceding inequality, we obtain:

$$\mathbb{P}_\tau^\pi[\mathcal{E}_{\text{good}}^c] \leq (M+1)T^{-3}. \quad (\text{E.C.2.13})$$

Step 5 (Regret from $\mathcal{A}$ during the detection delay). We now bound the regret accumulated after the change and before detection from the statistical test, during periods in which the strategy π still runs $\mathcal{A}$. Let $\mathcal{W}_T^{\mathcal{A}}$ denote the set of such periods, namely the periods t with $t \geq \tau$, $t \leq \hat{\tau}$, and t belonging to a block where the strategy runs $\mathcal{A}$. Formally:

$$\mathcal{W}_T^{\mathcal{A}} = \begin{cases} \emptyset, & \tau \leq D_T \text{ or } \tau = T+1, \\ \{\tau, \tau+1, \dots, \min\{\hat{\tau}, T\}\} \cap ([T] \setminus \mathcal{T}^{\mathcal{E}}), & D_T < \tau \leq T. \end{cases}$$

Periods in $\mathcal{T}^{\mathcal{E}}$ are excluded because their regret contribution is already bounded in Step 2. We prove that $|\mathcal{W}_T^{\mathcal{A}}| \leq H_T$ on $\mathcal{E}_{\text{good}}$. If $\tau = T+1$, then $\mathcal{W}_T^{\mathcal{A}} = \emptyset$ by definition. If the change occurs while the initial reference histories are being collected, then $\mathcal{W}_T^{\mathcal{A}} = \emptyset$ again. Indeed, the strategy has not yet started running $\mathcal{A}$, and the periods spent collecting the initial reference batches belong to $\mathcal{T}^{\mathcal{E}}$.

It remains to consider the case in which $\tau \leq T$ and all initial reference batches have been collected before τ . Then, $\mathcal{J}(S^\tau)$ is a reference batch generated under $F^{\tau-1}$. Moreover, on $\mathcal{E}_{\text{false}}^c$, the strategy cannot declare DETECTED = True before τ , because every test completed before τ involves two batches generated under the same preferences $F^{\tau-1}$. Let m_τ be the unique cycle such that τ belongs either to $\mathcal{T}_{m_\tau}^{\mathcal{A}}$ or to $\bigcup_{S \in \mathcal{E}} \mathcal{T}_{m_\tau}(S)$.

Suppose first that $\tau \in \mathcal{T}_{m_\tau}^{\mathcal{A}}$. After the remaining periods of the interval, the strategy enters the monitoring part of the same cycle. The number of periods in $\mathcal{T}_{m_\tau}^{\mathcal{A}}$ after τ is at most H_T . If the horizon ends before the strategy completes $\mathcal{I}_{m_\tau}(S^\tau)$, then there are no further periods to count, and $|\mathcal{W}_T^{\mathcal{A}}| \leq H_T$. If $\mathcal{I}_{m_\tau}(S^\tau)$ is completed before the horizon, then all its observations are collected after τ . Therefore, $\mathcal{C}_{m_\tau}^\tau$ occurs. On $\mathcal{E}_{\text{pow}}$, the statistical test rejects the no-change hypothesis. Hence, no later period can contribute to $\mathcal{W}_T^{\mathcal{A}}$, and again $|\mathcal{W}_T^{\mathcal{A}}| \leq H_T$.

Suppose that the change occurs while the strategy collects test histories in cycle m_τ , that is, $\tau \in \bigcup_{S \in \mathcal{E}} \mathcal{T}_{m_\tau}(S)$. Let $S^\tau \in \mathcal{E}$ be an assortment satisfying $\mathcal{K}^\tau(S^\tau) \geq \eta$, whose existence follows from the definition of $\tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$. We distinguish the three possible positions of τ relative to $\mathcal{T}_{m_\tau}(S^\tau)$. Formally:

(Case 1) If $\min\{u : u \in \mathcal{T}_{m_\tau}(S^\tau)\} \geq \tau$ and $|\mathcal{T}_{m_\tau}(S^\tau)| = qT$, then $\mathcal{C}_{m_\tau}^\tau$ occurs. Hence, on $\mathcal{E}_{\text{pow}}$, the comparison between $\mathcal{J}(S^\tau)$ and $\mathcal{I}_{m_\tau}(S^\tau)$ rejects the no-change hypothesis in cycle m_τ . Since no period in which the strategy runs $\mathcal{A}$ occurs between τ and this rejection, one has $\mathcal{W}_T^{\mathcal{A}} = \emptyset$.

(Case 2) If $\min\{u : u \in \mathcal{T}_{m_\tau}(S^\tau)\} < \tau \leq \max\{u : u \in \mathcal{T}_{m_\tau}(S^\tau)\}$, then the comparison in cycle

m_τ may contain observations from both sides of the change. If it rejects, then $\mathcal{W}_T^A = \emptyset$. Otherwise, the next periods in which the strategy runs $\mathcal{A}$ are contained in $\mathcal{T}_{m_\tau+1}^A$, whose length is at most H_T . If $\mathcal{T}_{m_\tau+1}(S^\tau)$ is not completed before T , then $|\mathcal{W}_T^A| \leq H_T$. If $|\mathcal{T}_{m_\tau+1}(S^\tau)| = qT$, then every period in $\mathcal{T}_{m_\tau+1}(S^\tau)$ is at least τ , so $\mathcal{C}_{m_\tau+1}^\tau$ occurs. On $\mathcal{E}_{\text{pow}}$, the strategy rejects in cycle $m_\tau + 1$. Therefore, no period after $\mathcal{T}_{m_\tau+1}(S^\tau)$ contributes to $\mathcal{W}_T^A$. Thus, in all cases, $|\mathcal{W}_T^A| \leq H_T$.

(Case 3) If $\max\{u : u \in \mathcal{T}_{m_\tau}(S^\tau)\} < \tau$, then the comparison for S^τ in cycle m_τ is generated before the change and does not reject. If some comparison in cycle m_τ rejects, then $\mathcal{W}_T^A = \emptyset$. Otherwise, the next periods in which the strategy runs $\mathcal{A}$ are contained in $\mathcal{T}_{m_\tau+1}^A$, with $|\mathcal{T}_{m_\tau+1}^A| \leq H_T$. If $\mathcal{T}_{m_\tau+1}(S^\tau)$ is not completed before T , then $\mathcal{W}_T^A \subseteq \mathcal{T}_{m_\tau+1}^A$, and hence $|\mathcal{W}_T^A| \leq H_T$. If $|\mathcal{T}_{m_\tau+1}(S^\tau)| = qT$, then all periods in $\mathcal{T}_{m_\tau+1}(S^\tau)$ are at least τ , so $\mathcal{C}_{m_\tau+1}^\tau$ occurs. On $\mathcal{E}_{\text{pow}}$, the strategy rejects in cycle $m_\tau + 1$. Hence, no later period in which the strategy runs $\mathcal{A}$ contributes to $\mathcal{W}_T^A$, and $|\mathcal{W}_T^A| \leq H_T$.

We have shown that, on $\mathcal{E}_{\text{good}}$, the cardinality of $\mathcal{W}_T^A$ is at most H_T . Since $\Delta^{\pi,t} \leq \|w\|_\infty$ for every $t \in [T]$, it follows that, on the event $\mathcal{E}_{\text{good}}$, the following inequality is satisfied:

$$\sum_{t \in \mathcal{W}_T^A} \Delta^{\pi,t} \leq H_T \|w\|_\infty.$$

Step 6 (Regret on the good event). We bound the regret on $\mathcal{E}_{\text{good}}$. We separate the regret into three terms: the regret accumulated while the strategy offers assortments from $\mathcal{E}$, namely on $\mathcal{T}^\mathcal{E}$; the regret accumulated after the change and before detection while the strategy still runs $\mathcal{A}$, namely on $\mathcal{W}_T^A$; and the regret accumulated during periods in which the strategy runs $\mathcal{A}$ under static preferences.

Recall that $\mathcal{T}^\mathcal{E}$ is the set of all monitoring periods and that $\mathcal{W}_T^A$ is the set of periods after the change, before detection, outside testing intervals. Define the remaining periods by $\mathcal{V}_T^A = [T] \setminus (\mathcal{T}^\mathcal{E} \cup \mathcal{W}_T^A)$. Thus, $\mathcal{V}_T^A$ contains the periods that are not charged to testing intervals and not charged to delayed detection. On $\mathcal{E}_{\text{good}}$, we have the following decomposition:

$$\sum_{t=1}^T \Delta^{\pi,t} = \sum_{t \in \mathcal{T}^\mathcal{E}} \Delta^{\pi,t} + \sum_{t \in \mathcal{W}_T^A} \Delta^{\pi,t} + \sum_{t \in \mathcal{V}_T^A} \Delta^{\pi,t}.$$

The first term is bounded by Step 2. Namely,

$$\sum_{t \in \mathcal{T}^\mathcal{E}} \Delta^{\pi,t} \leq \|w\|_\infty (2D_T + \frac{TD_T}{H_T}).$$

The second term is bounded by Step 5. On $\mathcal{E}_{\text{good}}$,

$$\sum_{t \in \mathcal{W}_T^A} \Delta^{\pi,t} \leq H_T \|w\|_\infty.$$

It remains to control the regret over $\mathcal{V}_T^A$.

We first describe precisely how $\mathcal{V}_T^A$ is generated on $\mathcal{E}_{\text{good}}$. There are only three possible cases.

(Case 1) If $\tau = T + 1$, then no change occurs before the horizon. On $\mathcal{E}_{\text{false}}^c$, the strategy never declares a change before T . In that case, all periods in $\mathcal{V}_T^A$ belong to a single run of $\mathcal{A}$ under the static preferences F^1 . The monitoring periods only create pauses in calendar time: they are not

passed to $\mathcal{A}$, and therefore do not affect the internal history of $\mathcal{A}$.

(Case 2) If $\tau \leq T$ and the change occurs while the initial reference batches are being collected, then $\mathcal{W}_T^A = \emptyset$ by definition. After the initial reference batches have been collected, every period in which the strategy runs $\mathcal{A}$ is generated under F^τ . If the strategy never declares a change later, then $\mathcal{V}_T^A$ is contained in a single run of $\mathcal{A}$ under F^τ . If the strategy declares a change later, then $\mathcal{V}_T^A$ is the union of two runs under F^τ : the run of $\mathcal{A}$ before detection and the refreshed run of $\mathcal{A}$ after detection.

(Case 3) Finally, suppose that $\tau \leq T$ and that all initial reference batches have been collected before τ . On $\mathcal{E}_{\text{false}}^c$, the strategy cannot declare a change before τ , because every test completed before τ compares two histories generated under $F^{\tau-1}$. Hence, the periods of $\mathcal{V}_T^A$ before τ form a single run of $\mathcal{A}$ under $F^{\tau-1}$. The periods after τ and before detection have been removed and placed in $\mathcal{W}_T^A$. After detection, the strategy restarts $\mathcal{A}$, and the remaining periods of $\mathcal{V}_T^A$ form a single run of $\mathcal{A}$ under F^τ . Thus, also in this case, $\mathcal{V}_T^A$ is covered by at most two runs of $\mathcal{A}$, each of length at most T .

We now translate the preceding case analysis into a bound. On $\mathcal{E}_{\text{good}}$, there exist two disjoint random sets of periods $\mathcal{V}_{T,1}^A$ and $\mathcal{V}_{T,2}^A$, possibly empty, such that $\mathcal{V}_T^A = \mathcal{V}_{T,1}^A \cup \mathcal{V}_{T,2}^A$. Each nonempty set corresponds to one run of $\mathcal{A}$ under static preferences belonging to $\mathcal{F}_S(\gamma_T)$. In the no-change case, we take $\mathcal{V}_{T,2}^A = \emptyset$. If the change occurs during the collection of the initial reference batches, then $\mathcal{A}$ has not yet begun to run, so every subsequent run is generated under F^τ and we may have $\mathcal{V}_{T,1}^A \neq \emptyset$ and $\mathcal{V}_{T,2}^A = \emptyset$. If all initial reference batches have been collected before τ , then $\mathcal{V}_{T,1}^A$ is generated under $F^{\tau-1}$, whereas $\mathcal{V}_{T,2}^A$, when nonempty, is generated under F^τ after the restart. In all cases, each run has length at most T .

For $j \in \{1, 2\}$, let $L_j = |\mathcal{V}_{T,j}^A|$. During the j -th run, $\mathcal{A}$ is restarted and then uses only sales generated under static preferences. Therefore, for $j \in \{1, 2\}$, one has:

$$\mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\sum_{t \in \mathcal{V}_{T,j}^A} \Delta^{\pi,t} \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \leq \mathcal{R}^A(\mathcal{F}_S(\gamma_T), T).$$

Summing over $j \in \{1, 2\}$, we obtain the following inequality:

$$\mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\sum_{t \in \mathcal{V}_T^A} \Delta^{\pi,t} \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \leq 2\mathcal{R}^A(\mathcal{F}_S(\gamma_T), T).$$

Combining the bound on the runs of $\mathcal{A}$ with the bounds on the testing periods and on $\mathcal{W}_T^A$ (recall Step 5), we obtain the following inequality:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\sum_{t=1}^T \Delta^{\pi,t} \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] &= \mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\left(\sum_{t \in \mathcal{T}^\mathcal{E}} \Delta^{\pi,t} + \sum_{t \in \mathcal{W}_T^A} \Delta^{\pi,t} + \sum_{t \in \mathcal{V}_T^A} \Delta^{\pi,t} \right) \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \\ &\leq \|w\|_\infty (2D_T + \frac{TD_T}{H_T}) + H_T \|w\|_\infty + 2\mathcal{R}^A(\mathcal{F}_S(\gamma_T), T). \end{aligned}$$

Equivalently, we obtain the following inequality:

$$\mathbb{E}_{\mathbb{P}_\tau^\pi} \left[\sum_{t=1}^T \Delta^{\pi,t} \mathbf{1}\{\mathcal{E}_{\text{good}}\} \right] \leq 2\mathcal{R}^A(\mathcal{F}_S(\gamma_T), T) + \|w\|_\infty (H_T + 2D_T + \frac{TD_T}{H_T}). \quad (\text{E.C.2.14})$$

Step 7 (Bad event and conclusion). On $\mathcal{E}_{\text{good}}^c$, we use the crude bound $\sum_{t=1}^T \Delta^{\pi,t} \leq T\|w\|_\infty$. Together with (E.C.2.13), this inequality gives:

$$\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=1}^T \Delta^{\pi,t} \mathbf{1}_{\{\mathcal{E}_{\text{good}}^c\}} \right] \leq (M+1)T^{-2}\|w\|_\infty. \quad (\text{E.C.2.15})$$

Combining (E.C.2.14) and (E.C.2.15), we obtain the following inequality on the regret:

$$\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=1}^T \Delta^{\pi,t} \right] \leq 2\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), T) + \|w\|_\infty \left(H_T + 2D_T + \frac{TD_T}{H_T} + (M+1)T^{-2} \right).$$

Since $H_T \leq \sqrt{TD_T} + 1$, $TD_T/H_T \leq \sqrt{TD_T}$, $D_T \leq \sqrt{TD_T}$, and $(M+1)T^{-2} \leq \sqrt{TD_T}$ for every $T \geq T_0$, the term in parentheses is at most $6\sqrt{TD_T}$. Also, $q_T \leq 2\log(T)\log\log(T)$, and thus $\sqrt{TD_T} \leq \sqrt{2TM\log(T)\log\log(T)}$. Consequently, the following inequality is satisfied:

$$J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) = \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=1}^T \Delta^{\pi,t} \right] \leq 2\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), T) + 9\|w\|_\infty \sqrt{TM\log(T)\log\log(T)}.$$

Taking the supremum over all paths $F^{(\mathbb{N})} \in \tilde{\mathcal{F}}_U(\mathcal{E}, \eta)$ completes the proof. $\blacksquare$

E.C.2.6 Proof of Proposition 3

Proof. We prove the lower bound on achievable regret by constructing a challenging path of preferences. Throughout the proof, we work with paths of MNL preferences fully characterized by their attraction parameters. The outside attraction is normalized to one. For MNL preferences F , we denote by $v_i(F)$ the attraction of product i . Given a feasible assortment $S \in \mathcal{S}$, the purchasing probabilities are $p_i(S, F) = v_i(F) \cdot (1 + \sum_{j \in S} v_j(F))^{-1}$ for all $i \in S$ and $p_0(S, F) = (1 + \sum_{j \in S} v_j(F))^{-1}$. We use $w_i = 1$ for all $i \in [N]$. Since the lower bound already holds in this special case, allowing arbitrary positive revenues only changes the constant. The construction proceeds with 9 steps.

Step 1 (Construction of a sequence of numbers). For a block length $\Delta \geq 2$, define $\varphi_0 = \varphi_\Delta = 0$, and for $s = 1, \dots, \Delta$, define $\varphi_s = 4(s/\Delta)(1 - s/\Delta)$. We first record three elementary bounds on this sequence. Since $\varphi_s = 4s(\Delta - s)/\Delta^2$, we have:

$$\begin{aligned} \sum_{s=1}^{\Delta} \varphi_s &= \frac{4}{\Delta^2} \sum_{s=1}^{\Delta} s(\Delta - s) \\ &= \frac{4}{\Delta^2} \left(\Delta \sum_{s=1}^{\Delta} s - \sum_{s=1}^{\Delta} s^2 \right) = \frac{4}{\Delta^2} \left(\frac{\Delta^2(\Delta + 1)}{2} - \frac{\Delta(\Delta + 1)(2\Delta + 1)}{6} \right) = \frac{2}{3} \left(\Delta - \frac{1}{\Delta} \right) \geq \frac{\Delta}{2}, \end{aligned}$$

where the last inequality follows from $\Delta \geq 2$. Moreover, since $s(\Delta - s) \leq \Delta^2/4$ for all $s = 1, \dots, \Delta$,

$$\sum_{s=1}^{\Delta} \varphi_s^2 = \frac{16}{\Delta^4} \sum_{s=1}^{\Delta} s^2(\Delta - s)^2 \leq \frac{16}{\Delta^4} \sum_{s=1}^{\Delta} \frac{\Delta^4}{16} = \Delta.$$

Finally, for all $s = 1, \dots, \Delta$, one has:

$$\varphi_s - \varphi_{s-1} = 4 \frac{s}{\Delta} \left(1 - \frac{s}{\Delta} \right) - 4 \frac{s-1}{\Delta} \left(1 - \frac{s-1}{\Delta} \right) = \frac{4(\Delta - 2s + 1)}{\Delta^2}.$$

Therefore, we obtain the following inequality:

$$\sum_{s=1}^{\Delta} (\varphi_s - \varphi_{s-1})^2 = \frac{16}{\Delta^4} \sum_{s=1}^{\Delta} (\Delta - 2s + 1)^2 = \frac{16}{\Delta^4} \cdot \frac{\Delta(\Delta^2 - 1)}{3} \leq \frac{16}{3\Delta}.$$

Thus, to summarize, for every $\Delta \geq 2$, we obtain:

$$\sum_{s=1}^{\Delta} \varphi_s \geq \frac{\Delta}{2}, \quad \sum_{s=1}^{\Delta} \varphi_s^2 \leq \Delta, \quad \sum_{s=1}^{\Delta} (\varphi_s - \varphi_{s-1})^2 \leq \frac{16}{3\Delta}. \quad (\text{E.C.2.16})$$

Step 2 (MNL instance construction). Let $C := \{1, \dots, K-1\}$ denote the set of *core* products. Products K and $K+1$ are *candidate* products. Products $K+2, \dots, N$ are called *dummy* products. We fix $\xi = 1/8$. For every period t , we fix $v_i(F^t) = 1$ for all $i \in C$ and $v_i(F^t) = \xi$ for all $i \in \{K+2, \dots, N\}$. The two candidate attractions are defined as $v_K(F^t) = 1 + g_t$ and $v_{K+1}(F^t) = 1 - g_t$, where $g_t \in [-1/4, 1/4]$.

Since all revenues are equal to one, the expected revenue of an assortment $S \in \mathcal{S}$ is given by:

$$r(S, F^t) = \frac{\sum_{i \in S} v_i(F^t)}{1 + \sum_{i \in S} v_i(F^t)}.$$

Hence, the revenue is increasing in the total attraction of the assortment. In particular, among assortments of cardinality at most K , the optimal assortment consists of the K products with largest attractions. Thus, if $g_t > 0$, then $S^*(F^t) = C \cup \{K\}$, whereas if $g_t < 0$, then $S^*(F^t) = C \cup \{K+1\}$.

For $g \in [-1/4, 1/4]$, let $F(g)$ denote the MNL preferences in the preceding construction with $v_K(F(g)) = 1 + g$ and $v_{K+1}(F(g)) = 1 - g$, all other attractions being fixed as above. We next derive a Lipschitz property for the KL divergence taken with respect to the purchasing distributions induced by $F(g)$. Fix $S \in \mathcal{S}$ and $i \in S \cup \{0\}$. Since $|S| \leq K$, $g \in [-1/4, 1/4]$, and all product attractions in the construction lie in $[1/8, 5/4]$, $D_S(g) := 1 + \sum_{j \in S} v_j(F(g))$ satisfies $1 \leq D_S(g) \leq 1 + 5K/4$. Moreover, every product in the support $S \cup \{0\}$ has a purchasing probability at least $(8 + 10K)^{-1}$. Indeed, the outside option has probability $1/D_S(g)$, while every product in S has attraction at least $1/8$.

We now bound the change in $p_i(S, F(g))$ when g changes. For $i \in S$, one has $p_i(S, F(g)) = v_i(F(g))/D_S(g)$. The derivative of $v_i(F(g))$ with respect to g belongs to $\{-1, 0, 1\}$, and the derivative of $D_S(g)$ belongs to $\{-1, 0, 1\}$. Therefore, for every $i \in S$, the following inequalities are satisfied:

$$\left| \frac{d}{dg} p_i(S, F(g)) \right| = \left| \frac{v'_i(F(g))D_S(g) - v_i(F(g))D'_S(g)}{D_S(g)^2} \right| \leq \frac{|v'_i(F(g))|}{D_S(g)} + \frac{v_i(F(g))|D'_S(g)|}{D_S(g)^2} \leq 1 + \frac{5}{4} = \frac{9}{4}.$$

For the outside option, we have $p_0(S, F(g)) = 1/D_S(g)$ and hence:

$$\left| \frac{d}{dg} p_0(S, F(g)) \right| = \frac{|D'_S(g)|}{D_S(g)^2} \leq 1 \leq \frac{9}{4}.$$

Thus, for every $g, g' \in [-1/4, 1/4]$, we obtain the following inequality:

$$\sum_{i \in S \cup \{0\}} |p_i(S, F(g)) - p_i(S, F(g'))| \leq \frac{9}{4}(K+1)|g - g'|.$$

Recall that by Lemma 9, the following inequality:

$$\sum_i p_i \log \left(\frac{p_i}{q_i} \right) \leq \sum_i \frac{(p_i - q_i)^2}{q_i} \leq \left(\min_i q_i \right)^{-1} \left(\sum_i |p_i - q_i| \right)^2,$$

is valid whenever the two distributions have the same finite support and $q_i > 0$ on that support. We use this inequality over $(p_i(S, F(g)) : i \in S \cup \{0\})$, to obtain, for all $g, g' \in [-1/4, 1/4]$ and all $S \in \mathcal{S}$,

$$\sum_{i \in S \cup \{0\}} p_i(S, F(g)) \log \left(\frac{p_i(S, F(g))}{p_i(S, F(g'))} \right) \leq (8 + 10K)(9(K+1)/4)^2 (g - g')^2.$$

Define $L_K := (8 + 10K)(9(K + 1)/4)^2$. Then, the previous inequality can be written as:

$$\sum_{i \in S \cup \{0\}} p_i(S, F(g)) \log \left(\frac{p_i(S, F(g))}{p_i(S, F(g'))} \right) \leq L_K (g - g')^2.$$

In particular, taking $g = \varepsilon\varphi_s$ and $g' = -\varepsilon\varphi_s$ gives:

$$\sum_{i \in S \cup \{0\}} p_i(S, F(\varepsilon\varphi_s)) \log \left(\frac{p_i(S, F(\varepsilon\varphi_s))}{p_i(S, F(-\varepsilon\varphi_s))} \right) \leq 4L_K \varepsilon^2 \varphi_s^2. \quad (\text{E.C.2.17})$$

Finally, define $\ell_R := (1 + 2K)^{-2}$. This constant is a uniform lower bound on the derivative of the revenue map $x \mapsto x/(1+x)$ over all total-attraction values that can be taken in our construction. Indeed, each feasible assortment contains at most K products, and the perturbations keep every attraction at most $1 + 1/4$. Hence, every feasible assortment has total attraction at most $K + 1/4$. Since $1 + K + 1/4 \leq 2K$ for $K \geq 2$, the derivative of the revenue map is at least ℓ_R throughout the relevant range $[0, 2K]$. This bound is used below to translate gaps in total attraction into gaps in the corresponding expected revenues. We fix $a := \min\{1/4, 1/\sqrt{32L_K}\}$ so that $a \leq 1/4$ and $4L_K a^2 \leq 1/8$.

Step 3 (Randomized construction). Since $M_T \leq T/8$, one has $\lfloor (T/M_T)^{1/3} \rfloor \geq 2$. Then, we define $\Delta := \lfloor (T/M_T)^{1/3} \rfloor$, $B = \lfloor T/\Delta \rfloor$ and $\varepsilon := a\Delta^{-1/2}$. Since $\Delta \geq 2$, $a \leq 1/4$ (recall Step 2), and $\varphi_s \leq 1$ (recall Step 1), we obtain $\varepsilon\varphi_s \leq 1/4$ for all $s \in \{1, \dots, \Delta\}$. Thus, all perturbations $\pm\varepsilon\varphi_s$ remain in the interval $[-1/4, 1/4]$. For each block $b \in [B]$, let $\theta_b \in \{-1, 1\}$. For $t = (b-1)\Delta + s$, with $s \in [\Delta]$, we define $g_t := \theta_b \varepsilon\varphi_s$. For periods $t > B\Delta$, fix $g_t = 0$. Following this construction together with Step 2, every sign vector $\theta = (\theta_1, \dots, \theta_B)$ defines a path of MNL preferences. Let $\mathcal{F}_{\text{MNL}}(M_T)$ be the class of all such paths. Each path is characterized by the attraction parameters from Step 2, with their values taken as defined above: those of the core and candidate products are set to one, while the dummy products retain their attraction after period $B\Delta + 1$.

The class $\mathcal{F}_{\text{MNL}}(M_T)$ constructed above has only finitely many paths. To prove the lower bound, we place a uniform distribution on these finite paths. Formally, let $\Theta_1, \dots, \Theta_B$ be independent random variables satisfying $\mathbb{P}[\Theta_b = 1] = \mathbb{P}[\Theta_b = -1] = 1/2$ for all $b \in [B]$. For each deterministic sign vector $\theta = (\theta_1, \dots, \theta_B) \in \{-1, 1\}^B$, let $F^{(\mathbb{N})}(\theta) \in \mathcal{F}_{\text{MNL}}(M_T)$ denote the path of preferences constructed before by setting $\Theta_b = \theta_b$ for every $b \in [B]$. Thus, under the randomized construction, the path faced by the retailer is a realization of the random vector $F^{(\mathbb{N})}(\Theta)$, where each realization $F^{(\mathbb{N})}(\theta)$ for $\theta \in \{-1, 1\}^B$ has probability 2^{-B} .

For a realization $F^{(\mathbb{N})}(\theta)$, define the dynamic regret of strategy π achieved on that path by:

$$\mathcal{R}^\pi(F^{(\mathbb{N})}(\theta), T) := J^*(F^{(\mathbb{N})}(\theta), T) - J^\pi(F^{(\mathbb{N})}(\theta), T).$$

The expected regret under the randomized construction is therefore:

$$\mathbb{E}_{\mathbb{P}^{\pi, \Theta}} \left[\mathcal{R}^\pi(F^{(\mathbb{N})}(\Theta), T) \right] := \frac{1}{2^B} \sum_{\theta \in \{-1, 1\}^B} \mathcal{R}^\pi(F^{(\mathbb{N})}(\theta), T).$$

This randomized construction is only a proof device. If the expected regret under the random sign vector satisfies the desired lower bound, then some realization of Θ must satisfy the same lower bound.

Step 4 (Magnitude of the constructed class). Fix any path $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}(M_T)$. By Step 2, for every feasible assortment $S \in \mathcal{S}$ and every period t , one has $\mathcal{K}^t(S) \leq L_K(g_t - g_{t-1})^2$, where we have $g_0 = 0$ by default. Inside a block $b \in [B]$, the path satisfies $g_{(b-1)\Delta+s} = \theta_b \varepsilon \varphi_s$, and thus:

$$g_{(b-1)\Delta+s} - g_{(b-1)\Delta+s-1} = \theta_b \varepsilon (\varphi_s - \varphi_{s-1}).$$

Therefore, using Step 1, the contribution of block b to the magnitude is bounded by:

$$\sum_{s=1}^{\Delta} \max \{ \mathcal{K}^{(b-1)\Delta+s}(S) : S \in \mathcal{S} \} \leq L_K \varepsilon^2 \sum_{s=1}^{\Delta} (\varphi_s - \varphi_{s-1})^2 \leq \frac{16}{3} L_K \frac{\varepsilon^2}{\Delta} = \frac{16}{3} L_K \frac{a^2}{\Delta^2},$$

since $\varepsilon := a\Delta^{-1/2}$ by Step 3.

The preceding bound sums all within-block increments, including the first increment in the first block (i.e., $g_1 - g_0$). The magnitude, however, only counts changes between consecutive periods of the horizon, starting at $t = 2$. Thus, the first term in the first block is not part of the magnitude. Keeping it only makes the upper bound larger. There is also no extra boundary term between two consecutive blocks because each block ends at $\varphi_{\Delta} = 0$ and the next block starts from $\varphi_0 = 0$. Hence,

$$\sum_{t=2}^T \max \{ \mathcal{K}^t(S) : S \in \mathcal{S} \} \leq B \frac{16}{3} L_K \frac{a^2}{\Delta^2} \leq \frac{16}{3} L_K a^2 \frac{T}{\Delta^3},$$

by definition of B ; recall Step 3. Since $\Delta = \lfloor (T/M_T)^{1/3} \rfloor \geq 2$, we have $\Delta \geq \frac{1}{2}(T/M_T)^{1/3}$, and therefore $T/\Delta^3 \leq 8M_T$. Consequently, every path $F^{(\mathbb{N})}$ in the constructed class $\mathcal{F}_{\text{MNL}}(M_T)$ satisfies:

$$\mathcal{M}(F^{(\mathbb{N})}, T) = \sum_{t=2}^T \max \{ \mathcal{K}^t(S) : S \in \mathcal{S} \} \leq \frac{128}{3} L_K a^2 M_T \leq 4M_T/3. \quad (\text{E.C.2.18})$$

Step 5 (Information in one block). Fix an arbitrary admissible strategy $\pi \in \mathcal{P}$. The strategy may be randomized, and all probabilities and expectations are taken over the strategy randomization and the purchase outcomes. Fix a block $b \in [B]$. Let $m_b := \min\{\Delta, T - (b-1)\Delta\}$ denote the number of periods in block b , and define $t_{b,s} := (b-1)\Delta + s$ for $s \in [m_b]$. Let $\mathcal{H}^{\pi,t}$ denote the history observed before period t , including the assortments selected by the strategy and the realized purchases.

In what follows, all probability distributions and expectations are conditional on the pre-block history $\mathcal{H}^{\pi,t_{b,0}}$ and on the signs $(\theta_{b'} : b' \neq b)$. After this conditioning, the behavior outside block b is fixed, and the remaining randomness comes from the block sign θ_b , the strategy's within-block randomization, and the purchase outcomes generated in block b . This conditioning is well defined because θ_b does not affect purchasing distributions before $t_{b,0}$ and it is independent of the other signs. Hence, the conditional distribution of the pre-block history $\mathcal{H}^{\pi,t_{b,0}}$ is the same under $\theta_b = 1$ and under $\theta_b = -1$. In particular, any difference between the two conditional distributions on block b can only arise from purchase outcomes, and from the strategy decisions based on those outcomes, inside block b .

Formally, for $\sigma \in \{-1, 1\}$, let $\mathbb{P}_{b,\sigma}^{\pi}$ denote the conditional probability distribution of the within-block history $\mathcal{H}^{\pi,b} := ((S^{\pi,t_{b,1}}, i^{t_{b,1}}), \dots, (S^{\pi,t_{b,m_b}}, i^{t_{b,m_b}}))$ given $\mathcal{H}^{\pi,t_{b,0}}$, $(\theta_{b'} : b' \neq b)$, and $\theta_b = \sigma$. We define $\mathbb{P}_{b,+}^{\pi}$ and $\mathbb{P}_{b,-}^{\pi}$ for the cases $\sigma = 1$ and $\sigma = -1$, respectively.

We compute the KL divergence between these two conditional distributions. For any realized his-

tory within a block, the conditional probability of observing an assortment S in period $t_{b,s}$, given the previous within-block history and the fixed pre-block history, is determined by the strategy π under both $\mathbb{P}_{b,+}^\pi$ and $\mathbb{P}_{b,-}^\pi$. This observation remains true when π is randomized. Hence, in the likelihood ratio between $\mathbb{P}_{b,+}^\pi$ and $\mathbb{P}_{b,-}^\pi$, the assortment-selection probabilities cancel. Only the conditional purchasing probabilities remain. We essentially use similar ideas to those used in the proof of Lemma 15.

Applying the chain rule for KL divergence (and using similar arguments to those used in the proof of Lemma 15 that we do not repeat for conciseness) to the sequence of within-block observations gives:

$$\mathcal{K}(\mathbb{P}_{b,+}^\pi, \mathbb{P}_{b,-}^\pi) = \sum_{s=1}^{\Delta} \mathbb{E}_{\mathbb{P}_{b,+}^\pi} [\mathcal{K}((p_i(S_{t_{b,s}}^\pi, F(\varepsilon\varphi_s)) : i \in S_{t_{b,s}}^\pi \cup \{0\}), (p_i(S_{t_{b,s}}^\pi, F(-\varepsilon\varphi_s)) : i \in S_{t_{b,s}}^\pi \cup \{0\}))].$$

The bound from Step 2 is uniform over all feasible assortments. Therefore, for every realized assortment from $S^{\pi, t_{b,s}}$, the following inequality is satisfied:

$$\mathcal{K}((p_i(S_{t_{b,s}}^\pi, F(\varepsilon\varphi_s)) : i \in S_{t_{b,s}}^\pi \cup \{0\}), (p_i(S_{t_{b,s}}^\pi, F(-\varepsilon\varphi_s)) : i \in S_{t_{b,s}}^\pi \cup \{0\})) \leq 4L_K \varepsilon^2 \varphi_s^2.$$

Consequently, using Step 1 and Step 3, we obtain the following inequalities:

$$\mathcal{K}(\mathbb{P}_{b,+}^\pi, \mathbb{P}_{b,-}^\pi) \leq \sum_{s=1}^{\Delta} 4L_K \varepsilon^2 \varphi_s^2 \leq 4L_K \varepsilon^2 \Delta = 4L_K a^2.$$

By the definition of a in Step 2, $4L_K a^2 \leq 1/8$. Thus, one has $\mathcal{K}(\mathbb{P}_{b,+}^\pi, \mathbb{P}_{b,-}^\pi) \leq 1/8$. Pinsker's inequality (Tsybakov 2003) gives $D_{\text{TV}}(\mathbb{P}_{b,+}^\pi, \mathbb{P}_{b,-}^\pi) \leq 1/4$, where D_{TV} denotes the total variation distance.

Step 6 (Revenue loss from choosing the wrong candidate). For a feasible assortment $S \in \mathcal{S}$, we define the indicator function $d(S) := \mathbf{1}\{K \in S\} - \mathbf{1}\{K \notin S\}$. If $\theta_b = 1$, then the correct candidate product for the optimal assortment is K . If $\theta_b = -1$, then the correct candidate product for the optimal assortment is $K + 1$. Recall our discussion at the beginning of Step 2 for the difference between core, candidate and dummy products. We claim that, for $t = (b - 1)\Delta + s$, every feasible assortment S , and every sign $\theta_b \in \{-1, 1\}$, the following inequality is satisfied:

$$r(S^*(F^t), F^t) - r(S, F^t) \geq \ell_R \varepsilon \varphi_s \mathbf{1}\{d(S) \neq \theta_b\}. \quad (\text{E.C.2.19})$$

Indeed, suppose that $\theta_b = 1$. Then $g_t = \varepsilon\varphi_s$, and the revenue-maximizing assortment is $C \cup \{K\}$, with total attraction $K + \varepsilon\varphi_s$. If $d(S) = -1$, then $K \notin S$. Among feasible assortments excluding K , the largest total attraction is attained by $C \cup \{K + 1\}$ and equals $K - \varepsilon\varphi_s$. Hence, any feasible assortment with $d(S) = -1$ has total attraction at least $\varepsilon\varphi_s$ below the maximum. Now suppose that $\theta_b = -1$. Then $g_t = -\varepsilon\varphi_s$, and the revenue-maximizing assortment is $C \cup \{K + 1\}$, whose total attraction is again $K + \varepsilon\varphi_s$. If $d(S) = 1$, then $K \in S$. If $K + 1 \notin S$, the total attraction of S is at most $K - \varepsilon\varphi_s$. If S contains both K and $K + 1$, then $|S| \leq K$ implies that S excludes at least one core product, so its total attraction is at most K . In either case, the total attraction is at least $\varepsilon\varphi_s$ below the maximum.

Finally, with unit revenues, the revenue of an assortment with total attraction x is $x/(1 + x)$. By Step 2, the derivative of this function is at least ℓ_R on all total-attraction values that arise in the construction. Hence, the total-attraction gap of at least $\varepsilon\varphi_s$ gives the revenue gap in (E.C.2.19).

Step 7 (Hypothesis test induced by the strategy). Fix a block $b \in [B]$ and $s \in [\Delta]$, and recall that $t_{b,s} = (b-1)\Delta + s$. We use the two distributions introduced in Step 5. More precisely, before comparing the two hypotheses, we fix a realization of the history observed before the beginning of block b , and we fix the values of all signs outside block b as we did in Step 5. The only remaining unknown sign is θ_b . We consider the hypothesis of the form:

$$\begin{cases} H_0 : & \theta_b = 1, \\ H_1 : & \theta_b = -1. \end{cases}$$

Under H_0 , the preferences inside block b are $F^{t_{b,u}} = F(\varepsilon\varphi_u)$ for all $u \in [\Delta]$ whereas under H_1 , the preferences inside block b are $F^{t_{b,u}} = F(-\varepsilon\varphi_u)$ for all $u \in [\Delta]$. The data generated inside block b are the assortment and purchase observations $X_b^\pi = ((S^{\pi,t_{b,1}}, i^{t_{b,1}}), \dots, (S^{\pi,t_{b,\Delta}}, i^{t_{b,\Delta}}))$. If H_0 is true, then X_b^π is distributed according to $\mathbb{P}_{b,+}^\pi$. If H_1 is true, then X_b^π is distributed according to $\mathbb{P}_{b,-}^\pi$.

We now define the decision rule induced by the strategy π at period $t_{b,s}$. Formally:

$$\phi_{b,s}^\pi(X_b^\pi) = \mathbf{1}\{K \notin S^{\pi,t_{b,s}}\}.$$

By convention, $\phi_{b,s}^\pi = 0$ means that H_0 is not rejected, while $\phi_{b,s}^\pi = 1$ means that H_0 is rejected in favor of H_1 . This decision rule uses only the assortment chosen by the strategy π at period $t_{b,s}$ rather than all the data from the block.

Equivalently, define the event $A_{b,s} = \{K \in S^{\pi,t_{b,s}}\}$. Then, by construction, one has $\{\phi_{b,s}^\pi(X_b^\pi) = 0\} = A_{b,s}$ and $\{\phi_{b,s}^\pi(X_b^\pi) = 1\} = A_{b,s}^c$. Under H_0 , product K is the candidate product for the optimal assortment in period $t_{b,s}$. Hence, the decision rule makes an error under H_0 on the event $\{\phi_{b,s}^\pi(X_b^\pi) \neq 0\} = A_{b,s}^c$. Under H_1 , product $K+1$ is the correct candidate product in period $t_{b,s}$. Hence, the decision rule makes an error under H_1 on the event $\{\phi_{b,s}^\pi(X_b^\pi) \neq 1\} = A_{b,s}$. Therefore, one has:

$$\mathbb{P}_{b,+}^\pi[\phi_{b,s}^\pi(X_b^\pi) \neq 0] + \mathbb{P}_{b,-}^\pi[\phi_{b,s}^\pi(X_b^\pi) \neq 1] = \mathbb{P}_{b,+}^\pi[A_{b,s}^c] + \mathbb{P}_{b,-}^\pi[A_{b,s}].$$

We now lower bound this sum. By the definition of the total variation distance (Tsybakov 2003), for every decision rule ϕ from the observations X_b^π into $\{0, 1\}$, one has:

$$\mathbb{P}_{b,+}^\pi[\phi(X_b^\pi) \neq 0] + \mathbb{P}_{b,-}^\pi[\phi(X_b^\pi) \neq 1] \geq 1 - D_{\text{TV}}(\mathbb{P}_{b,+}^\pi, \mathbb{P}_{b,-}^\pi).$$

Applying it to the particular rule $\phi = \phi_{b,s}^\pi$, and using Step 5, gives $\mathbb{P}_{b,+}^\pi[A_{b,s}^c] + \mathbb{P}_{b,-}^\pi[A_{b,s}] \geq 3/4$.

We now translate this testing error into regret. Under H_0 , the preferences in period $t_{b,s}$ are $F(\varepsilon\varphi_s)$. In that case, on the event $A_{b,s}^c$, the strategy does not include the correct candidate product K . Therefore, by using the arguments in Step 6, the following inequality holds:

$$r(S^*(F(\varepsilon\varphi_s)), F(\varepsilon\varphi_s)) - r(S^{\pi,t_{b,s}}, F(\varepsilon\varphi_s)) \geq \ell_R \varepsilon \varphi_s \mathbf{1}\{A_{b,s}^c\}.$$

Taking expectation under $\mathbb{P}_{b,+}^\pi$, we obtain:

$$\mathbb{E}_{\mathbb{P}_{b,+}^\pi} \left[r(S^*(F(\varepsilon\varphi_s)), F(\varepsilon\varphi_s)) - r(S^{\pi,t_{b,s}}, F(\varepsilon\varphi_s)) \right] \geq \ell_R \varepsilon \varphi_s \mathbb{P}_{b,+}^\pi[A_{b,s}^c].$$

Similarly, under H_1 , the preferences in period $t_{b,s}$ are $F(-\varepsilon\varphi_s)$, and the event $A_{b,s}$ means that the

strategy includes product K , while the correct candidate product is $K + 1$. Thus, again by Step 6,

$$\mathbb{E}_{\mathbb{P}_{b,-}^\pi} \left[r(S^*(F(-\varepsilon\varphi_s)), F(-\varepsilon\varphi_s)) - r(S^{\pi,t_{b,s}}, F(-\varepsilon\varphi_s)) \right] \geq \ell_R \varepsilon \varphi_s \mathbb{P}_{b,-}^\pi[A_{b,s}].$$

By averaging both expectations, we obtain the following inequalities:

$$\begin{aligned} & \frac{1}{2} \mathbb{E}_{\mathbb{P}_{b,+}^\pi} \left[r(S^*(F(\varepsilon\varphi_s)), F(\varepsilon\varphi_s)) - r(S^{\pi,t_{b,s}}, F(\varepsilon\varphi_s)) \right] \\ & + \frac{1}{2} \mathbb{E}_{\mathbb{P}_{b,-}^\pi} \left[r(S^*(F(-\varepsilon\varphi_s)), F(-\varepsilon\varphi_s)) - r(S^{\pi,t_{b,s}}, F(-\varepsilon\varphi_s)) \right] \\ & \geq \frac{1}{2} \ell_R \varepsilon \varphi_s \left(\mathbb{P}_{b,+}^\pi[A_{b,s}^c] + \mathbb{P}_{b,-}^\pi[A_{b,s}] \right) \geq \frac{3}{8} \ell_R \varepsilon \varphi_s. \end{aligned}$$

Summing the previous inequality over $s = 1, \dots, \Delta$, we obtain the following inequality:

$$\begin{aligned} & \frac{1}{2} \sum_{s=1}^{\Delta} \mathbb{E}_{\mathbb{P}_{b,+}^\pi} \left[r(S^*(F(\varepsilon\varphi_s)), F(\varepsilon\varphi_s)) - r(S^{\pi,t_{b,s}}, F(\varepsilon\varphi_s)) \right] \\ & + \frac{1}{2} \sum_{s=1}^{\Delta} \mathbb{E}_{\mathbb{P}_{b,-}^\pi} \left[r(S^*(F(-\varepsilon\varphi_s)), F(-\varepsilon\varphi_s)) - r(S^{\pi,t_{b,s}}, F(-\varepsilon\varphi_s)) \right] \geq \sum_{s=1}^{\Delta} \frac{3}{8} \ell_R \varepsilon \varphi_s = \frac{3}{8} \ell_R \varepsilon \sum_{s=1}^{\Delta} \varphi_s. \end{aligned}$$

The left-hand side is the expected regret (in the sense defined in Step 3) generated by block b , conditional on the fixed pre-block history and on the fixed signs outside block b , and averaged over the two possible values of θ_b . Since Θ_b is independent of the other signs and satisfies $\mathbb{P}[\Theta_b = 1] = \mathbb{P}[\Theta_b = -1] = 1/2$, the two conditional regrets enter with equal weight. Since $\varepsilon = a\Delta^{-1/2}$ by Step 3 and $\sum_{s=1}^{\Delta} \varphi_s \geq \Delta/2$ by Step 1, we further obtain $\frac{3}{8} \ell_R \varepsilon \sum_{s=1}^{\Delta} \varphi_s \geq \frac{3}{8} \ell_R a \Delta^{-1/2} \frac{\Delta}{2} = \frac{3}{16} \ell_R a \Delta^{1/2}$.

Step 8 (Regret over the horizon). We now turn the blockwise lower bound into a worst-case lower bound over deterministic paths. The random signs are used only as a proof device. For each sign vector $\theta = (\theta_1, \dots, \theta_B) \in \{-1, 1\}^B$, let $F^{(\mathbb{N})}(\theta) \in \mathcal{F}_{\text{MNL}}(M_T)$ be the path of customers' preferences constructed in Step 3. For $b \in [B]$, define the regret accumulated over block b by:

$$\mathcal{R}_b^\pi(F^{(\mathbb{N})}(\theta)) := \sum_{s=1}^{\Delta} \left[r(S^*(F^{t_{b,s}}(\theta)), F^{t_{b,s}}(\theta)) - r(S_{t_{b,s}}^\pi, F^{t_{b,s}}(\theta)) \right],$$

where $t_{b,s} = (b-1)\Delta + s$. The remaining periods after $B\Delta$, if any, can be ignored because their contribution to the regret is non-negative. Hence, the following inequality is satisfied:

$$\mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathcal{R}^\pi(F^{(\mathbb{N})}(\theta), T) \right] \geq \mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\sum_{b=1}^B \mathcal{R}_b^\pi(F^{(\mathbb{N})}(\theta)) \right] = \sum_{b=1}^B \mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathcal{R}_b^\pi(F^{(\mathbb{N})}(\theta)) \right].$$

For each block b , let $\mathcal{G}_b^\pi := \sigma(\mathcal{H}^{\pi,t_{b,0}}, \theta_{b'} : b' \neq b)$ be the information consisting of the pre-block history and the signs outside block b . By law of total expectations (Jacod and Protter 2004), we obtain the following equality:

$$\mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathcal{R}_b^\pi(F^{(\mathbb{N})}(\theta)) \right] = \mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathbb{E} \left[\mathcal{R}_b^\pi(F^{(\mathbb{N})}(\theta)) \mid \mathcal{G}_b^\pi \right] \right].$$

Step 7 gives, for every realization of $\mathcal{G}_b^\pi$, one has:

$$\mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathcal{R}_b^\pi(F^{(\mathbb{N})}(\theta)) \mid \mathcal{G}_b^\pi \right] \geq \frac{3}{16} \ell_R a \sqrt{\Delta}.$$

Therefore, the following inequality is satisfied:

$$\mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathcal{R}^\pi(F^{(\mathbb{N})}(\theta), T) \right] \geq \sum_{b=1}^B \mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathbb{E}_{\mathbb{P}^\pi, \theta} \left[\mathcal{R}_b^\pi(F^{(\mathbb{N})}(\theta)) \mid \mathcal{G}_b^\pi \right] \right] \geq \sum_{b=1}^B \frac{3}{16} \ell_R a \sqrt{\Delta} = \frac{3}{16} \ell_R a B \sqrt{\Delta}.$$

In the present case, $\Delta = \lfloor (T/M_T)^{1/3} \rfloor \geq 2$; recall Step 3. Moreover, $\Delta \leq (T/M_T)^{1/3}$ and therefore $T/\Delta \geq T^{2/3} M_T^{1/3} \geq 4$ as $T \geq 8$ and $M_T \geq 1$. Thus, $B = \lfloor T/\Delta \rfloor \geq T/(2\Delta)$. Using the previous inequality, we obtain the following inequality:

$$\mathbb{E}_{\mathbb{P}^{\pi, \Theta}} \left[\mathcal{R}^{\pi}(F^{(\mathbb{N})}(\Theta), T) \right] \geq \frac{3}{32} \ell_R a \frac{T}{\sqrt{\Delta}}.$$

Since $\mathbb{E}_{\mathbb{P}^{\pi, \Theta}} \left[\mathcal{R}^{\pi}(F^{(\mathbb{N})}(\Theta), T) \right]$ is the expectation of finitely many non-negative numbers indexed by $\theta \in \{-1, 1\}^B$, at least one sign vector $\theta^{\pi} \in \{-1, 1\}^B$ satisfies the following inequality:

$$J^*(F^{(\mathbb{N})}(\theta^{\pi}), T) - J^{\pi}(F^{(\mathbb{N})}(\theta^{\pi}), T) \geq \frac{3}{32} \ell_R a \frac{T}{\sqrt{\Delta}}.$$

Because $F^{(\mathbb{N})}(\theta^{\pi}) \in \mathcal{F}_{\text{MNL}}(M_T)$, we conclude that:

$$\mathcal{R}^{\pi}(\mathcal{F}_{\text{MNL}}(M_T), T) \geq \frac{3}{32} \ell_R a \frac{T}{\sqrt{\Delta}}.$$

Finally, since $\Delta \leq (T/M_T)^{1/3}$, we have $T/\sqrt{\Delta} \geq T^{5/6} M_T^{1/6}$. Therefore, it follows that:

$$\mathcal{R}^{\pi}(\mathcal{F}_{\text{MNL}}(M_T), T) \geq \frac{3}{32} \ell_R a T^{5/6} M_T^{1/6}.$$

Step 9 (Conclusion). First, (E.C.2.18) gives $\mathcal{M}(F^{(\mathbb{N})}, T) \leq 4M_T/3$ for all $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}(M_T)$. Second, Step 8 implies that every strategy $\pi \in \mathcal{P}$ satisfies $\mathcal{R}^{\pi}(\mathcal{F}_{\text{MNL}}(M_T), T) \geq c_1 T^{5/6} M_T^{1/6}$ for $c_1 := \frac{3}{32} \ell_R a$. Since $\pi \in \mathcal{P}$ is arbitrary, taking the infimum over all strategies $\pi \in \mathcal{P}$ gives:

$$\mathcal{R}^*(\mathcal{F}_{\text{MNL}}(M_T), T) \geq c_1 T^{5/6} M_T^{1/6}.$$

The constant c_1 depends only on K , because L_K , a , and ℓ_R depend only on K . ■

E.C.2.7 Proof of Theorem 3

Proof. Let $\pi \equiv \pi(\mathcal{A}, T, \Delta)$ be the periodic restarting strategy from Algorithm 3. Fix an arbitrary path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$. Let $B := \lceil T/\Delta \rceil$ be the number of restarts. For each $j \in [B]$, we define the interval $I_j := \{(j-1)\Delta + 1, \dots, \min\{j\Delta, T\}\} \subseteq [T]$. Then, $(I_j : j \in [B])$ forms a partition of the horizon $[T]$, and $|I_j| \leq \Delta$ for all $j \in [B]$.

By definition, the periodic restarting strategy $\pi(\mathcal{A}, T, \Delta)$ starts a new copy of $\mathcal{A}$ at the beginning of each interval I_j . Hence, by Corollary 5, there exists $\tilde{c} \equiv \tilde{c}(\varsigma)$ such that for every $j \in [B]$, one has:

$$\sum_{t \in I_j} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\pi}} \left[\sum_{t \in I_j} r(S^{\pi(\mathcal{A}, T, \Delta), t}, F^t) \right] \leq \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + 2\tilde{c} \|w\|_{\infty} |I_j|^2 \sqrt{\mathcal{M}_{I_j}}.$$

Summing this inequality over all intervals gives the following inequality:

$$J^*(F^{(\mathbb{N})}, T) - J^{\pi(\mathcal{A}, T, \Delta)}(F^{(\mathbb{N})}, T) \leq \sum_{j=1}^B \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + 2\tilde{c} \|w\|_{\infty} \sum_{j=1}^B |I_j|^2 \sqrt{\mathcal{M}_{I_j}}.$$

To proceed, we control the first term in the right-hand side above. Since $|I_j| \leq \Delta$, we obtain:

$$\sum_{j=1}^B \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) \leq B \cdot \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) = \lceil T/\Delta \rceil \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta).$$

Next, we control the second term. Since $|I_j| \leq \Delta$, one has:

$$\sum_{j=1}^B |I_j|^2 \sqrt{\mathcal{M}_{I_j}} \leq \Delta^2 \sum_{j=1}^B \sqrt{\mathcal{M}_{I_j}} \leq \Delta^2 \sqrt{B \sum_{j=1}^B \mathcal{M}_{I_j}}.$$

where the last inequality follows from applying Cauchy–Schwarz inequality. Moreover, since the intervals form a partition of $[T]$, the sum of the within-intervals magnitudes is bounded by the total magnitude of the path (which takes boundary into account), i.e., $\sum_{j=1}^B \mathcal{M}_{I_j} \leq \mathcal{M}(F^{(\mathbb{N})}, T)$.

Putting these bounds together, we obtain the following inequality:

$$\sum_{j=1}^B |I_j|^2 \sqrt{\mathcal{M}_{I_j}} \leq \Delta^2 \sqrt{B \mathcal{M}(F^{(\mathbb{N})}, T)} \leq \Delta^2 \sqrt{2(T/\Delta) \mathcal{M}(F^{(\mathbb{N})}, T)} = \Delta^{3/2} \sqrt{2T \mathcal{M}(F^{(\mathbb{N})}, T)},$$

where the second inequality follows from $B = \lceil T/\Delta \rceil \leq 2T/\Delta$, since $\Delta \in [T]$. Consequently,

$$J^*(F^{(\mathbb{N})}, T) - J^{\pi(\mathcal{A}, T, \Delta)}(F^{(\mathbb{N})}, T) \leq \lceil T/\Delta \rceil \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + 2\tilde{c}\|w\|_{\infty} \Delta^{3/2} \sqrt{2T \mathcal{M}(F^{(\mathbb{N})}, T)}.$$

Since $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$, by the definition of $\mathcal{F}_G(\gamma_T, M_T)$, we have $\mathcal{M}(F^{(\mathbb{N})}, T) \leq M_T$. Thus,

$$J^*(F^{(\mathbb{N})}, T) - J^{\pi(\mathcal{A}, T, \Delta)}(F^{(\mathbb{N})}, T) \leq \lceil T/\Delta \rceil \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + 2\tilde{c}\|w\|_{\infty} \Delta^{3/2} \sqrt{2TM_T}.$$

Taking the supremum over $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$ and setting $c \equiv c(\varsigma) = 2\tilde{c}$ proves the result. $\blacksquare$

E.C.2.8 Proof of Corollary 1

Proof. Let $\pi(\mathcal{A}, T, \Delta)$ be the periodic restarting strategy from Algorithm 3. We define $\gamma_{\mathcal{A}} := (2C_{\mathcal{A}}/(3C_{\varsigma}))^{1/2}$ and $C_{\varsigma} := c\|w\|_{\infty}\sqrt{2}$ with $c \equiv c(\varsigma)$ as in Theorem 3. By Theorem 3 and the assumption on the policy $\mathcal{A}$, for every $\Delta \in [T]$, one has:

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq \lceil T/\Delta \rceil C_{\mathcal{A}}\sqrt{\Delta} + C_{\varsigma}\Delta^{3/2}\sqrt{TM_T}.$$

Since $\Delta \leq T$, we have $\lceil T/\Delta \rceil \leq T/\Delta + 1 \leq 2T/\Delta$. Therefore, the following inequality holds:

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq 2C_{\mathcal{A}}T\Delta^{-1/2} + C_{\varsigma}\Delta^{3/2}\sqrt{TM_T}. \quad (\text{E.C.2.20})$$

We now optimize the right-hand side over a continuous restart length $x > 0$. To proceed, we define the function $\phi : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ by $\phi(x) := 2C_{\mathcal{A}}Tx^{-1/2} + C_{\varsigma}x^{3/2}\sqrt{TM_T}$. Taking the first derivative, we obtain $\phi'(x) = -C_{\mathcal{A}}Tx^{-3/2} + \frac{3}{2}C_{\varsigma}\sqrt{TM_T}x^{1/2}$. The first-order condition $\phi'(x) = 0$ is equivalent to $C_{\mathcal{A}}Tx^{-3/2} = \frac{3}{2}C_{\varsigma}\sqrt{TM_T}x^{1/2}$. Multiplying both sides by $x^{3/2}$ gives:

$$x^2 = \frac{2C_{\mathcal{A}}T}{3C_{\varsigma}\sqrt{TM_T}} = \frac{2C_{\mathcal{A}}}{3C_{\varsigma}} \sqrt{\frac{T}{M_T}}.$$

Since $\phi''(x) > 0$, the minimum of $\phi(\cdot)$ is achieved at:

$$x^* = \left(\frac{2C_{\mathcal{A}}}{3C_{\varsigma}}\right)^{1/2} \left(\frac{T}{M_T}\right)^{1/4} = \gamma_{\mathcal{A}} \left(\frac{T}{M_T}\right)^{1/4}.$$

Next, we fix $\Delta^* = \lceil x^* \rceil$. Since $x^* \geq 1$ (provided that T is large enough), it follows that $\Delta^* \leq x^* + 1 \leq 2x^*$. Moreover, by assumption, $\Delta^* \in [T]$, so we may apply (E.C.2.20) with $\Delta = \Delta^*$. Since $\Delta^* \geq x^*$, we have $2C_{\mathcal{A}}T(\Delta^*)^{-1/2} \leq 2C_{\mathcal{A}}T(x^*)^{-1/2}$. Since $\Delta^* \leq 2x^*$, we also have $C_{\varsigma}(\Delta^*)^{3/2}\sqrt{TM_T} \leq 2^{3/2}C_{\varsigma}(x^*)^{3/2}\sqrt{TM_T}$. At the minimizer x^* , the first-order condition implies $2C_{\mathcal{A}}T(x^*)^{-1/2} = 3C_{\varsigma}(x^*)^{3/2}\sqrt{TM_T}$. All together, it follows that the following inequality is satisfied:

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta^*)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq \left(3 + 2^{3/2}\right) C_{\varsigma}(x^*)^{3/2}\sqrt{TM_T}.$$

Substituting the expression for x^* , we obtain:

$$C_\varsigma(x^*)^{3/2}\sqrt{TM_T} = C_\varsigma \left[\left(\frac{2C_{\mathcal{A}}}{3C_\varsigma} \right)^{1/2} \left(\frac{T}{M_T} \right)^{1/4} \right]^{3/2} T^{1/2} M_T^{1/2} = \left(\frac{2}{3} \right)^{3/4} C_{\mathcal{A}}^{3/4} C_\varsigma^{1/4} T^{7/8} M_T^{1/8}.$$

Hence, the following inequality is satisfied:

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta^*)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq (3 + 2^{3/2}) (2/3)^{3/4} C_{\mathcal{A}}^{3/4} C_\varsigma^{1/4} T^{7/8} M_T^{1/8},$$

and the claim follows. $\blacksquare$

E.C.2.9 Proof of Theorem 4

Proof. Let $\pi(\mathcal{A}, T, \Delta)$ be the periodic restarting strategy from Algorithm 3. Fix an arbitrary path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$. Let $B := \lceil T/\Delta \rceil$ be the number of restarts for $\Delta \in [T]$. For each $j \in [B]$, we define the interval $I_j := \{(j-1)\Delta + 1, \dots, \min\{j\Delta, T\}\} \subseteq [T]$. Then, $(I_j : j \in [B])$ forms a partition of $[T]$, and $|I_j| \leq \Delta$ for all $j \in [B]$. Fix some $j \in [B]$. For an interval $I = \{a, \dots, b\} \subseteq [T]$ and a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}$, define:

$$\mathcal{V}_I \equiv \mathcal{V}_I(F^{(\mathbb{N})}) := \sum_{t=a+1}^b \max \{ |r(S, F^t) - r(S, F^{t-1})| : S \in \mathcal{S} \}.$$

The quantity $\mathcal{V}_I(F^{(\mathbb{N})})$ measures how much the revenue of a fixed assortment can change inside I .

We decompose the regret on the j -th interval I_j into the regret of the fixed assortment that maximizes cumulative expected revenue over I_j relative to the clairvoyant retailer and the regret of the strategy relative to that assortment. Specifically,

$$\begin{aligned} \sum_{t \in I_j} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^\pi} \left[\sum_{t \in I_j} r(S^{\pi(\mathcal{A}, T, \Delta), t}, F^t) \right] &= \left(\sum_{t \in I_j} r(S^*(F^t), F^t) - \max \left\{ \sum_{t \in I_j} r(S, F^t) : S \in \mathcal{S} \right\} \right) \\ &\quad + \left(\max \left\{ \sum_{t \in I_j} r(S, F^t) : S \in \mathcal{S} \right\} - \mathbb{E}_{\mathbb{P}^\pi} \left[\sum_{t \in I_j} r(S^{\pi(\mathcal{A}, T, \Delta), t}, F^t) \right] \right). \end{aligned}$$

The first bracket is bounded by Lemma 12. The second bracket is bounded by Definition 1, because $\pi(\mathcal{A}, T, \Delta)$ runs a new copy of $\mathcal{A}$ on I_j . Therefore, we obtain the following inequality:

$$\sum_{t \in I_j} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^\pi} \left[\sum_{t \in I_j} r(S^{\pi(\mathcal{A}, T, \Delta), t}, F^t) \right] \leq c_{\mathcal{A}} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + (2 + c_{\mathcal{A}}) |I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})}).$$

Summing over $j \in [B]$ gives the following inequality:

$$J^*(F^{(\mathbb{N})}, T) - J^{\pi(\mathcal{A}, T, \Delta)}(F^{(\mathbb{N})}, T) \leq c_{\mathcal{A}} \sum_{j=1}^B \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + (2 + c_{\mathcal{A}}) \sum_{j=1}^B |I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})}).$$

Since $|I_j| \leq \Delta$, and using the definition of regret $\mathcal{R}^{\mathcal{A}}$, we obtain the following inequality:

$$\sum_{j=1}^B \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) \leq \lceil T/\Delta \rceil \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta).$$

We now control the variation term. By the definition of $\mathcal{V}_I(F^{(\mathbb{N})})$, for every block I_j ,

$$\mathcal{V}_{I_j}(F^{(\mathbb{N})}) = \sum_{t \in I_j : t > a_j} \max \{ |r(S, F^t) - r(S, F^{t-1})| : S \in \mathcal{S} \},$$

where a_j is the first period of the interval I_j . For every $S \in \mathcal{S}$ and every $t \geq 2$,

$$|r(S, F^t) - r(S, F^{t-1})| = \left| \sum_{i \in S} w_i [p_i(S, F^t) - p_i(S, F^{t-1})] \right| \leq \|w\|_\infty \sum_{i \in S \cup \{0\}} |p_i(S, F^t) - p_i(S, F^{t-1})|.$$

By applying Pinsker's inequality (Tsybakov 2003), we obtain an inequality of the form:

$$\sum_{i \in S \cup \{0\}} |p_i(S, F^t) - p_i(S, F^{t-1})| \leq \sqrt{2\mathcal{K}^t(S)}.$$

Therefore, the following inequality is also satisfied:

$$\max \{|r(S, F^t) - r(S, F^{t-1})| : S \in \mathcal{S}\} \leq \sqrt{2}\|w\|_\infty \sqrt{\max \{\mathcal{K}^t(S) : S \in \mathcal{S}\}}.$$

It follows that the following inequality is satisfied:

$$\sum_{j=1}^B \mathcal{V}_{I_j}(F^{(\mathbb{N})}) \leq \sqrt{2}\|w\|_\infty \sum_{t=2}^T \sqrt{\max \{\mathcal{K}^t(S) : S \in \mathcal{S}\}}. \quad (\text{E.C.2.21})$$

Applying Cauchy–Schwarz inequality gives:

$$\sum_{t=2}^T \sqrt{\max \{\mathcal{K}^t(S) : S \in \mathcal{S}\}} \leq (T \sum_{t=2}^T \max \{\mathcal{K}^t(S) : S \in \mathcal{S}\})^{1/2} = \sqrt{T\mathcal{M}(F^{(\mathbb{N})}, T)}. \quad (\text{E.C.2.22})$$

Since $|I_j| \leq \Delta$, (E.C.2.21) and (E.C.2.22) imply together:

$$\sum_{j=1}^B |I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})}) \leq \Delta \sum_{j=1}^B \mathcal{V}_{I_j}(F^{(\mathbb{N})}) \leq \sqrt{2}\|w\|_\infty \Delta \sqrt{T\mathcal{M}(F^{(\mathbb{N})}, T)}.$$

Putting these bounds together, we obtain a bound of the form:

$$J^*(F^{(\mathbb{N})}, T) - J^{\pi(\mathcal{A}, T, \Delta)}(F^{(\mathbb{N})}, T) \leq c_{\mathcal{A}} \lceil T/\Delta \rceil \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + (2 + c_{\mathcal{A}}) \sqrt{2}\|w\|_\infty \Delta \sqrt{T\mathcal{M}(F^{(\mathbb{N})}, T)}.$$

Since $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$, we have $\mathcal{M}(F^{(\mathbb{N})}, T) \leq M_T$. Taking the supremum over $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$ proves the theorem. $\blacksquare$

E.C.2.10 Proof of Corollary 2

Proof. To begin, recall that T and M_T are chosen so that $\Delta \in [T]$. This condition implicitly requires $M_T \geq C_{\mathcal{A}}^2/T^2$. If the condition is not satisfied, then $\Delta = T$, and the resulting regret rate is essentially the same as in the case for static preferences. Applying Theorem 4 for every $\Delta \in [T]$, we obtain:

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq c_{\mathcal{A}} \lceil T/\Delta \rceil \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + (2 + c_{\mathcal{A}}) \sqrt{2}\|w\|_\infty \Delta \sqrt{TM_T}.$$

Using the assumption $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), n) \leq C_{\mathcal{A}} \sqrt{n}$, evaluated at $n = \Delta$, gives:

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq (2 + c_{\mathcal{A}}) C_{\mathcal{A}} \lceil T/\Delta \rceil \sqrt{\Delta} + (2 + c_{\mathcal{A}}) \sqrt{2}\|w\|_\infty \Delta \sqrt{TM_T}.$$

Since $\Delta \in [T]$, we have $\lceil T/\Delta \rceil \leq 2T/\Delta$. Hence,

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq 2(2 + c_{\mathcal{A}}) C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} + (2 + c_{\mathcal{A}}) \sqrt{2}\|w\|_\infty \Delta \sqrt{TM_T}. \quad (\text{E.C.2.23})$$

Let $x := (C_{\mathcal{A}}^2 T/M_T)^{1/3}$ and $\Delta = \lceil x \rceil$. Since $M_T \leq T$ and $C_{\mathcal{A}} \geq 1$ without loss of generality, one has $x \geq 1$. Thus, $x \leq \Delta \leq 2x$. We bound the two terms in (E.C.2.23) separately. First, since $\Delta \geq x$,

$$\begin{aligned} 2(2 + c_{\mathcal{A}}) C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} &\leq 2(2 + c_{\mathcal{A}}) C_{\mathcal{A}} \frac{T}{\sqrt{x}} \\ &= 2(2 + c_{\mathcal{A}}) C_{\mathcal{A}} T \left(\frac{M_T}{C_{\mathcal{A}}^2 T} \right)^{1/6} = 2(2 + c_{\mathcal{A}}) C_{\mathcal{A}}^{2/3} T^{5/6} M_T^{1/6}. \end{aligned}$$

Second, because $\Delta \leq 2x$,

$$(2 + c_{\mathcal{A}}) \sqrt{2}\|w\|_\infty \Delta \sqrt{TM_T} \leq 2(2 + c_{\mathcal{A}}) \sqrt{2}\|w\|_\infty x \sqrt{TM_T}$$

$$\begin{aligned}
&= 2(2 + c_{\mathcal{A}})\sqrt{2}\|w\|_{\infty} \left(\frac{C_{\mathcal{A}}^2 T}{M_T} \right)^{1/3} T^{1/2} M_T^{1/2} \\
&= 2(2 + c_{\mathcal{A}})\sqrt{2}\|w\|_{\infty} C_{\mathcal{A}}^{2/3} T^{5/6} M_T^{1/6}.
\end{aligned}$$

Combining the last two inequalities gives:

$$\mathcal{R}^{\pi(\mathcal{A}, T, \Delta)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq \left(2(2 + c_{\mathcal{A}}) + 2(2 + c_{\mathcal{A}})\sqrt{2}\|w\|_{\infty} \right) C_{\mathcal{A}}^{2/3} T^{5/6} M_T^{1/6}.$$

Thus, the result follows with $c = 2(2 + c_{\mathcal{A}}) + 2(2 + c_{\mathcal{A}})\sqrt{2}\|w\|_{\infty}$. $\blacksquare$

E.C.2.11 Proof of the claim in Remark 4

Proof. First, we denote by $\mathcal{A}_{\mathbf{F}}$ the policy of Foussoul et al. (2023). Then, fix a path $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}}$, an interval $I = \{a, \dots, b\} \subseteq [T]$, and let $n := |I|$. We start a new copy of $\mathcal{A}_{\mathbf{F}}$ on I (at time a). Relabel the interval as $[n]$, and define, for all $t \in [n]$, $\bar{F}^t := F^{a+t-1}$ and $\bar{S}^{\mathcal{A}_{\mathbf{F}}, t} := S^{\mathcal{A}_{\mathbf{F}}, a+t-1}$. Let $\nu^t := \nu(\bar{F}^t)$ denote the MNL attraction parameters characterizing $F^{(\mathbb{N})}$ at relabeled period t . Following Foussoul et al. (2023), define, for all $t \in \{1, \dots, n-1\}$, $\Delta_{\mathbf{F}}(t) := 26\|\nu^t - \nu^{t+1}\|_{L_{\infty}^{2K}}$, where the norm $\|\cdot\|_{L_{\infty}^{2K}}$ is defined by $\|x\|_{L_{\infty}^{2K}} := \sup \{ \sum_{i \in S} |x_i| : S \subseteq \mathcal{N}, |S| \leq 2K \}$.

For an interval $I = \{a, \dots, b\} \subseteq [T]$ and a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}$, define:

$$\mathcal{V}_I \equiv \mathcal{V}_I(F^{(\mathbb{N})}) := \sum_{t=a+1}^b \max \{ |r(S, F^t) - r(S, F^{t-1})| : S \in \mathcal{S} \}.$$

The quantity $\mathcal{V}_I(F^{(\mathbb{N})})$ measures how much the revenue of a fixed assortment can change inside I . Also, define the cumulative variation over the relabeled interval (as originally defined in Foussoul et al. (2023)) by $B_n := \sum_{t=1}^{n-1} \Delta_{\mathbf{F}}(t)$. Throughout this proof, we denote by $\mathcal{F}_S(\gamma_T) \subseteq \mathcal{F}_{\text{MNL}}$ the set of static MNL preferences with optimality gap γ_T satisfying $\gamma_T < c/\sqrt{T}$ for some appropriate $c > 0$.

Step 1 (Relating variation in preferences and in revenue). We relate the cumulative variation B_n to the revenue variation $\mathcal{V}_I(F^{(\mathbb{N})})$. For any singleton assortment $\{i\} \in \mathcal{S}$, the corresponding expected revenue is given by $r(\{i\}, F) = w_i \nu_i(F) / (1 + \nu_i(F))$. The derivative of the map $x \mapsto w_i x / (1 + x)$ is $w_i / (1 + x)^2$. Since $w_i \geq \underline{w} \equiv \min\{w_i : i \in \mathcal{N}\}$ and $\nu_i(F^t) \leq \bar{\nu}_{\zeta}$ for all $t \in [T]$ since $F^{(\mathbb{N})} \in \mathcal{F}_{\text{MNL}} \subseteq \mathcal{F}$ (for some $\bar{\nu}_{\zeta}$ induced by $\mathcal{F}$), the following inequality holds for all $x \in [0, \bar{\nu}_{\zeta}]$:

$$\frac{w_i}{(1+x)^2} \geq \frac{\underline{w}}{(1+\bar{\nu}_{\zeta})^2}.$$

Hence, for every $t = 1, \dots, n-1$ and every product $i \in \mathcal{N}$, we obtain the following inequality:

$$|\nu_i^{t+1} - \nu_i^t| \leq \frac{(1+\bar{\nu}_{\zeta})^2}{\underline{w}} |r(\{i\}, \bar{F}^{t+1}) - r(\{i\}, \bar{F}^t)| \leq \frac{(1+\bar{\nu}_{\zeta})^2}{\underline{w}} \max \{ |r(S, \bar{F}^{t+1}) - r(S, \bar{F}^t)| : S \in \mathcal{S} \}.$$

Therefore, the following inequality holds:

$$\begin{aligned}
\Delta_{\mathbf{F}}(t) &= 26 \sup \left\{ \sum_{i \in S} |\nu_i^{t+1} - \nu_i^t| : S \subseteq \mathcal{N}, |S| \leq 2K \right\} \\
&\leq 52K \frac{(1+\bar{\nu}_{\zeta})^2}{\underline{w}} \max \{ |r(S, \bar{F}^{t+1}) - r(S, \bar{F}^t)| : S \in \mathcal{S} \}.
\end{aligned} \tag{E.C.2.24}$$

Summing over $t = 1, \dots, n-1$, and returning to the original indexing, gives simultaneously $B_n \leq \tilde{c} \mathcal{V}_I(F^{(\mathbb{N})})$ for $\tilde{c} \equiv \tilde{c}(\zeta, w, K) := 52K(1+\bar{\nu}_{\zeta})^2/\underline{w}$.

Step 2 (Near stationary definition used in Foussoul et al. (2023)). We now use the analysis of Foussoul et al. (2023). Their Theorem 1 and its proof imply that their Algorithm 1 outputs revenue certificates $\hat{R}_t = r(S^{\mathcal{A}_F, t}, \hat{\nu}(t))$ for all $t \in [n]$, where $\hat{\nu}(t)$ is an estimator used in their policy. Moreover, there exists an event $\mathcal{E}_n$, with probability at least $1 - \delta_n$ for some δ_n , on which the following two inequalities hold. First, for every $t \in [n]$, the following inequality is satisfied:

$$\hat{R}_t \geq \min\{r(S^*(\bar{F}^\tau), \bar{F}^\tau) : \tau \in [t]\} - \sum_{u=1}^{t-1} \Delta_F(u). \quad (\text{E.C.2.25})$$

Second, the following inequality is satisfied:

$$\sum_{t=1}^n (\hat{R}_t - R_t^{\mathcal{A}_F}) \leq n\rho(n) + nB_n, \quad (\text{E.C.2.26})$$

where $R_t^{\mathcal{A}_F}$ is the realized revenue collected by $\mathcal{A}_F$ in relabeled period t , and:

$$\rho(n) = (149 \log Nn)^{3/2} \sqrt{\frac{N}{n}} + (55 \log Nn)^3 \frac{N}{n} + \sqrt{\frac{2 \log n}{n}}.$$

The event probability in their proof satisfies $\delta_n = 6/n$. We next convert these certificates into our near-stationary robustness inequality. Define the clairvoyant retailer's expected revenue at period t by $r_t^* := r(S^*(\bar{F}^t), \bar{F}^t)$ for all $t \in [n]$. For every $u = 1, \dots, n-1$, by the definition of $\mathcal{V}_I(F^{\mathbb{N}})$ and by the fact that $\Delta_F(u)$ is the variation measure in the sense of Foussoul et al. (2023), one has $|r_{u+1}^* - r_u^*| \leq \|w\|_\infty \Delta_F(u)$. Indeed, if $r_{u+1}^* \geq r_u^*$, then:

$$\begin{aligned} r_{u+1}^* - r_u^* &= r(S^*(\bar{F}^{u+1}), \bar{F}^{u+1}) - r(S^*(\bar{F}^u), \bar{F}^u) \leq r(S^*(\bar{F}^{u+1}), \bar{F}^{u+1}) - r(S^*(\bar{F}^{u+1}), \bar{F}^u) \\ &\leq \Delta_F(u) \|w\|_\infty, \end{aligned}$$

where $\|w\|_\infty \equiv \max\{w_i : i \in \mathcal{N}\}$ and the reverse case is analogous. Therefore, for every $\tau \leq t$,

$$r_t^* \leq r_\tau^* + \|w\|_\infty \sum_{u=\tau}^{t-1} \Delta_F(u) \leq r_\tau^* + \|w\|_\infty \sum_{u=1}^{t-1} \Delta_F(u).$$

Taking the minimum over $\tau \in [t]$, and using (E.C.2.25), gives, on $\mathcal{E}_n$,

$$r_t^* \leq \hat{R}_t + (1 + \|w\|_\infty) \sum_{u=1}^{t-1} \Delta_F(u), \quad (\text{E.C.2.27})$$

for all $t \in [n]$.

Since the best fixed assortment over the interval is dominated by the period-by-period oracle,

$$\max \left\{ \sum_{t=1}^n r(S, \bar{F}^t) : S \in \mathcal{S} \right\} \leq \sum_{t=1}^n r_t^*.$$

Combining this inequality with (E.C.2.27), we obtain, on $\mathcal{E}_n$,

$$\max \left\{ \sum_{t=1}^n r(S, \bar{F}^t) : S \in \mathcal{S} \right\} \leq \sum_{t=1}^n \hat{R}_t + (1 + \|w\|_\infty) \sum_{t=1}^n \sum_{u=1}^{t-1} \Delta_F(u) \leq \sum_{t=1}^n \hat{R}_t + (1 + \|w\|_\infty) nB_n.$$

Subtracting the cumulative realized reward, and then applying (E.C.2.26), gives, on $\mathcal{E}_n$,

$$\begin{aligned} \max \left\{ \sum_{t=1}^n r(S, \bar{F}^t) : S \in \mathcal{S} \right\} - \sum_{t=1}^n R_t^{\mathcal{A}_F} &\leq \sum_{t=1}^n (\hat{R}_t - R_t^{\mathcal{A}_F}) + (1 + \|w\|_\infty) nB_n \\ &\leq n\rho(n) + (2 + \|w\|_\infty) nB_n. \end{aligned} \quad (\text{E.C.2.28})$$

We now pass from realized rewards to expected revenues. On $\mathcal{E}_n^c$, the regret over n periods is

at most $n\|w\|_\infty$. Thus, taking expectations in (E.C.2.28) gives the following inequality:

$$\max \left\{ \sum_{t=1}^n r(S, \bar{F}^t) : S \in \mathcal{S} \right\} - \mathbb{E}_{\mathbb{P}^{\mathcal{A}_F}} \left[\sum_{t=1}^n R_t^{\mathcal{A}_F} \right] \leq n\rho(n) + (2 + \|w\|_\infty)nB_n + \delta_n n\|w\|_\infty.$$

By the law of total expectations (Jacod and Protter 2004), $\mathbb{E}[R_t^{\mathcal{A}_F}] = \mathbb{E}[r(\bar{S}^{\mathcal{A}_F, t}, \bar{F}^t)]$ because the conditional expected reward in period t , given the history before period t , is $r(\bar{S}^{\mathcal{A}_F, t}, \bar{F}^t)$. Therefore,

$$\max \left\{ \sum_{t=1}^n r(S, \bar{F}^t) : S \in \mathcal{S} \right\} - \mathbb{E}_{\mathbb{P}^{\mathcal{A}_F}} \left[\sum_{t=1}^n r(\bar{S}^{\mathcal{A}_F, t}, \bar{F}^t) \right] \leq n\rho(n) + (2 + \|w\|_\infty)nB_n + \delta_n n\|w\|_\infty.$$

Using (E.C.2.24), and returning to the original indexing of I , we obtain:

$$\max \left\{ \sum_{t \in I} r(S, F^t) : S \in \mathcal{S} \right\} - \mathbb{E}_{\mathbb{P}^{\mathcal{A}_F}} \left[\sum_{t \in I} r(S^{\mathcal{A}_F, t}, F^t) \right] \leq n\rho(n) + (2 + \|w\|_\infty)\tilde{c}n\mathcal{V}_I(F^{(\mathbb{N})}) + \delta_n n\|w\|_\infty.$$

Step 3 (Near-stationary robustness). It remains to absorb the certificate terms from Step 2 into the regret term for static preferences. Given $n := |I|$, define $L_T := \log(NT)$ and $m_n := \min\{n, \sqrt{Nn}\}$. Since $n \leq T$, we have $\log(Nn) \leq L_T$. Therefore, one has $n\rho(n) \leq 149^{3/2}L_T^{3/2}\sqrt{Nn} + 55^3L_T^3N + \sqrt{2n\log n}$. We now compare each term with m_n . First, $\sqrt{Nn} \leq \sqrt{N}m_n$. Indeed, if $n \leq N$, then $m_n = n$ and $\sqrt{Nn} \leq \sqrt{N}n$, while if $n > N$, then $m_n = \sqrt{Nn}$. Second, since $m_n \geq 1$, $N \leq Nm_n$. Third, since $\sqrt{n} \leq m_n$, we have $\sqrt{2n\log n} \leq (\sqrt{2\log n})m_n$. Moreover, $L_T \geq \log 2$, and hence $L_T^{3/2} \leq (\log 2)^{-3/2}L_T^3$ and $\sqrt{2\log n} \leq \sqrt{2}(\log 2)^{-5/2}L_T^3$. Combining the preceding bounds gives $n\rho(n) \leq c_0L_T^3m_n$, where $c_0 := 149^{3/2}(\log 2)^{-3/2}\sqrt{N} + 55^3N + \sqrt{2}(\log 2)^{-5/2}$.

The high-probability event in the analysis of Foussoul et al. (2023) has failure probability at most $6/n$ when the algorithm is run with horizon n . Thus, $\delta_n n\|w\|_\infty \leq 6\|w\|_\infty$. Since $m_n \geq 1$ and $L_T \geq \log 2$, one has $\delta_n n\|w\|_\infty \leq 6\|w\|_\infty(\log 2)^{-3}L_T^3m_n$. Hence, $n\rho(n) + \delta_n n\|w\|_\infty \leq c_1L_T^3m_n$ with the constant $c_1 := c_0 + 6\|w\|_\infty(\log 2)^{-3}$.

The quantity $\mathcal{R}^{\mathcal{A}_F}(\mathcal{F}_S(\gamma_T), n)$ is a worst-case regret over static MNL preferences with $\gamma_T < c/\sqrt{T}$ for some $c > 0$. By the lower-bound construction for the MNL bandit in (Chen and Wang 2018) for $K \leq N/4$, there exists a constant $c_2 > 0$ such that every admissible policy, and in particular $\mathcal{A}_F$, satisfies $\mathcal{R}^{\mathcal{A}_F}(\mathcal{F}_S(\gamma_T), n) \geq c_2m_n$. Consequently,

$$n\rho(n) + \delta_n n\|w\|_\infty \leq c_1L_T^3m_n \leq \frac{c_1}{c_2}L_T^3\mathcal{R}^{\mathcal{A}_F}(\mathcal{F}_S(\gamma_T), n).$$

Returning to Step 2 and defining $c_3 := (2 + \|w\|_\infty)\tilde{c}$, we obtain the following inequality:

$$\max \left\{ \sum_{t \in I} r(S, F^t) : S \in \mathcal{S} \right\} - \mathbb{E} \left[\sum_{t \in I} r(S^{\mathcal{A}_F, t}, F^t) \right] \leq \frac{c_1}{c_2} \log(NT)^3 \mathcal{R}^{\mathcal{A}_F}(\mathcal{F}_S(\gamma_T), n) + c_3 n \mathcal{V}_I(F^{(\mathbb{N})}).$$

Since $n = |I|$, Definition 1 follows with $c_{\mathcal{A}_F} := \max\{\frac{c_1}{c_2} \log(NT)^3, c_3\}$. ■

E.C.3 Proofs for Section 6

We first state and prove an adversarial bandit lemma used in the regret analysis of Algorithm 4. The lemma is derived from the analysis of Neu (2015). We include both its statement and its proof for completeness. To proceed, we fix a finite set of actions $\mathcal{D}$ and a number of rounds K . At every round $k \in [K]$, an adversary chooses a loss vector $\ell^k = (\ell^k(\Delta) : \Delta \in \mathcal{D}) \in [0, 1]^{\mathcal{D}}$. The vector ℓ^k may depend on all previous choices of the learner. The learner chooses $\Delta^k \in \mathcal{D}$, observes only the realized loss $\ell^k(\Delta^k)$, and does not observe $\ell^k(\Delta)$ for $\Delta \neq \Delta^k$. The implicit-exploration aggregation rule maintains weights $v^k(\Delta)$ for $\Delta \in \mathcal{D}$. It initializes $v^1(\Delta) = 1$ for every $\Delta \in \mathcal{D}$. At round k , it sets $q^k(\Delta) = v^k(\Delta)(\sum_{\delta \in \mathcal{D}} v^k(\delta))^{-1}$ for every $\Delta \in \mathcal{D}$, and draws $\Delta^k \sim q^k$. For a fixed learning rate $\eta \in (0, 1]$, let $\beta = \eta/2$. After observing $\ell^k(\Delta^k)$, the learner forms, for all $\Delta \in \mathcal{D}$,

$$\tilde{\ell}^k(\Delta) = \frac{\ell^k(\Delta^k) \mathbf{1}\{\Delta = \Delta^k\}}{q^k(\Delta) + \beta}, \quad \Delta \in \mathcal{D}.$$

It updates $v^{k+1}(\Delta) = v^k(\Delta) \exp\{-\eta \tilde{\ell}^k(\Delta)\}$ for every $\Delta \in \mathcal{D}$. We denote the resulting policy by π^N .

Lemma 7 (Neu (2015)). *Assume that $|\mathcal{D}| \geq 2$. For the policy π^N described above with $\eta := \min\{1, \sqrt{\log |\mathcal{D}| / (K|\mathcal{D}|)}\}$, there exists $c > 0$ such that, for every $\Delta \in \mathcal{D}$,*

$$\mathbb{E}_{\pi^N} \left[\sum_{k=1}^K \ell^k(\Delta^{\pi^N, k}) - \sum_{k=1}^K \ell^k(\Delta) \right] \leq 3(K|\mathcal{D}| \log |\mathcal{D}|)^{1/2} + 3 \log |\mathcal{D}|.$$

Proof. Fix $\eta \in (0, 1]$ and $\Delta \in \mathcal{D}$, and define $\beta := \eta/2$. Let $\mathbb{E}_{\pi^N}[\cdot]$ denote the expectation with respect to the randomization of the policy π^N . For $k \geq 1$, we define the sigma-algebra generated by the history observed before period k as $\mathcal{H}^{k-1} := \sigma(\Delta^{\pi^N, 1}, \ell^1(\Delta^{\pi^N, 1}), \dots, \Delta^{\pi^N, k-1}, \ell^{k-1}(\Delta^{\pi^N, k-1}))$. Let $\mathcal{G}^k := \sigma(\mathcal{H}^{k-1}, \ell^k)$ be the information obtained after the adversary has chosen the loss vector ℓ^k , but before the learner draws $\Delta^{\pi^N, k}$. The distribution q^k is $\mathcal{H}^{k-1}$ -measurable, and hence $\mathcal{G}^k$ -measurable.

Without loss of generality, we assume that the randomization of π^N is implemented so that $\mathbb{P}^{\pi^N}[\Delta^{\pi^N, k} = \delta \mid \mathcal{G}^k] = q^k(\delta)$ for $\delta \in \mathcal{D}$. Indeed, if the policy is specified only through the probability vector q^k , then we can realize the same policy on an enlarged probability space as follows. Fix an ordering $\mathcal{D} = \{\delta_1, \dots, \delta_d\}$. Let U_k be a random variable uniformly distributed on $[0, 1]$, independent of $\mathcal{G}^k$. Define $Q_0^k := 0$ and $Q_j^k := \sum_{r=1}^j q^k(\delta_r)$ for $j \in [d]$. The action drawn at period k is then defined by $\Delta^{\pi^N, k} = \delta_j$ if and only if $Q_{j-1}^k \leq U_k < Q_j^k$. Since q^k is $\mathcal{H}^{k-1}$ -measurable, the random variables Q_{j-1}^k and Q_j^k are both $\mathcal{G}^k$ -measurable. Therefore, for every $j \in [d]$,

$$\mathbb{P}^{\pi^N}[\Delta^{\pi^N, k} = \delta_j \mid \mathcal{G}^k] = \mathbb{E}_{\pi^N}[\mathbf{1}\{Q_{j-1}^k \leq U_k < Q_j^k\} \mid \mathcal{G}^k] = Q_j^k - Q_{j-1}^k = q^k(\delta_j).$$

The estimator formed in the policy is $\tilde{\ell}^k(\delta) := \ell^k(\delta) \mathbf{1}\{\Delta^{\pi^N, k} = \delta\} (q^k(\delta) + \beta)^{-1}$ for all $\delta \in \mathcal{D}$. Therefore, for every $\delta \in \mathcal{D}$, the following identities are satisfied:

$$\mathbb{E}_{\pi^N}[\tilde{\ell}^k(\delta) \mid \mathcal{G}^k] = \mathbb{E}_{\pi^N} \left[\frac{\ell^k(\delta) \mathbf{1}\{\Delta^{\pi^N, k} = \delta\}}{q^k(\delta) + \beta} \mid \mathcal{G}^k \right] = \frac{\ell^k(\delta)}{q^k(\delta) + \beta} \mathbb{P}^{\pi^N}[\Delta^{\pi^N, k} = \delta \mid \mathcal{G}^k] \leq \ell^k(\delta).$$

Applying the last inequality to Δ , and then taking expectations, gives $\mathbb{E}_{\mathbb{P}^{\pi^N}}[\tilde{\ell}^k(\Delta)] \leq \mathbb{E}_{\mathbb{P}^{\pi^N}}[\ell^k(\Delta)]$.

Moreover, the following inequality is also satisfied since $\ell^k(\delta) \in [0, 1]$ for all $\delta \in \mathcal{D}$:

$$\mathbb{E}_{\mathbb{P}^{\pi^N}} \left[\sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta)^2 \mid \mathcal{G}^k \right] = \sum_{\delta \in \mathcal{D}} q^k(\delta) \mathbb{E}_{\mathbb{P}^{\pi^N}} \left[\frac{\ell^k(\delta)^2 \mathbf{1}\{\Delta^{\pi^N, k} = \delta\}}{(q^k(\delta) + \beta)^2} \mid \mathcal{G}^k \right] = \sum_{\delta \in \mathcal{D}} \frac{q^k(\delta)^2}{(q^k(\delta) + \beta)^2} \ell^k(\delta)^2 \leq |\mathcal{D}|.$$

By applying the law of total expectations (Jacod and Protter 2004), we obtain the following inequality:

$$\mathbb{E}_{\mathbb{P}^{\pi^N}} \left[\sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta)^2 \right] \leq |\mathcal{D}|. \quad (\text{E.C.3.1})$$

Let $Z_k = \sum_{\delta \in \mathcal{D}} v^k(\delta)$. Using $\exp(-x) \leq 1 - x + x^2$ for $x \geq 0$, and using $\eta \tilde{\ell}^k(\delta) \geq 0$, we obtain:

$$\frac{Z_{k+1}}{Z_k} = \sum_{\delta \in \mathcal{D}} q^k(\delta) \exp\{-\eta \tilde{\ell}^k(\delta)\} \leq 1 - \eta \sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta) + \eta^2 \sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta)^2.$$

Hence, using $\log(1+x) \leq x$, we obtain the following inequality:

$$\log Z_{K+1} - \log Z_1 \leq -\eta \sum_{k=1}^K \sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta) + \eta^2 \sum_{k=1}^K \sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta)^2.$$

On the other hand, Z_{K+1} is the sum of all final weights, and therefore dominates the final weight of Δ . Since $v^1(\Delta) = 1$, the multiplicative update gives $v^{K+1}(\Delta) = \exp\{-\eta \sum_{k=1}^K \tilde{\ell}^k(\Delta)\}$. Hence, $Z_{K+1} = \sum_{\delta \in \mathcal{D}} v^{K+1}(\delta) \geq v^{K+1}(\Delta)$, and taking logarithms gives:

$$\log Z_{K+1} \geq -\eta \sum_{k=1}^K \tilde{\ell}^k(\Delta).$$

Combining the two inequalities, using $Z_1 = |\mathcal{D}|$ and rearranging, we obtain:

$$\sum_{k=1}^K \sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta) - \sum_{k=1}^K \tilde{\ell}^k(\Delta) \leq \frac{\log |\mathcal{D}|}{\eta} + \eta \sum_{k=1}^K \sum_{\delta \in \mathcal{D}} q^k(\delta) \tilde{\ell}^k(\delta)^2. \quad (\text{E.C.3.2})$$

We now translate (E.C.3.2) into a statement about actual losses. First,

$$\begin{aligned} \mathbb{E}_{\mathbb{P}^{\pi^N}} [\ell^k(\Delta^{\pi^N, k}) \mid \mathcal{G}^k] &= \sum_{\delta \in \mathcal{D}} q^k(\delta) \ell^k(\delta) \\ &= \sum_{\delta \in \mathcal{D}} q^k(\delta) \mathbb{E}_{\mathbb{P}^{\pi^N}} [\tilde{\ell}^k(\delta) \mid \mathcal{G}^k] + \sum_{\delta \in \mathcal{D}} \frac{\beta q^k(\delta)}{q^k(\delta) + \beta} \ell^k(\delta) \leq \sum_{\delta \in \mathcal{D}} q^k(\delta) \mathbb{E}_{\mathbb{P}^{\pi^N}} [\tilde{\ell}^k(\delta) \mid \mathcal{G}^k] + \beta |\mathcal{D}|. \end{aligned}$$

Then, $\sum_{k=1}^K \mathbb{E}_{\mathbb{P}^{\pi^N}} [\tilde{\ell}^k(\Delta) \mid \mathcal{G}^k] \leq \sum_{k=1}^K \ell^k(\Delta)$. Taking expectations in (E.C.3.2) and then applying the tower property with respect to $\mathcal{G}^k$ (Jacod and Protter 2004) and using the two preceding inequalities with (E.C.3.1), we obtain the following inequality:

$$\sum_{k=1}^K \mathbb{E}_{\mathbb{P}^{\pi^N}} [\ell^k(\Delta^{\pi^N, k}) \mid \mathcal{G}^k] - \sum_{k=1}^K \ell^k(\Delta) \leq \frac{\log |\mathcal{D}|}{\eta} + \eta K |\mathcal{D}| + \beta K |\mathcal{D}|.$$

Since $\beta = \eta/2$, the claim follows by substituting the value for η selected in the lemma.

Indeed, if $\sqrt{\log(|\mathcal{D}|)/(K|\mathcal{D}|)} > 1$, then $\eta = 1$ and we obtain:

$$\sum_{k=1}^K \mathbb{E}_{\mathbb{P}^{\pi^N}} [\ell^k(\Delta^{\pi^N, k}) \mid \mathcal{G}^k] - \sum_{k=1}^K \ell^k(\Delta) \leq 3 \log |\mathcal{D}|.$$

On the other hand, if $\sqrt{\log(|\mathcal{D}|)/(K|\mathcal{D}|)} \leq 1$, then one has:

$$\sum_{k=1}^K \mathbb{E}_{\mathbb{P}^{\pi^N}} [\ell^k(\Delta^{\pi^N, k}) \mid \mathcal{G}^k] - \sum_{k=1}^K \ell^k(\Delta) \leq \frac{5}{2} \sqrt{K |\mathcal{D}| \log |\mathcal{D}|}.$$

Fixing $c \equiv 3$ and using the law of total expectations (Jacod and Protter 2004) concludes the proof. $\blacksquare$

For $b_T = \lceil T^{2/3} \rceil$, fix $K_T := \lfloor T/b_T \rfloor$ and $T_0 \equiv K_T b_T$ so that $T - T_0 < b_T$. In Algorithm 4, the strategy treats each interval $B_k := \{(k-1)b_T + 1, \dots, kb_T\}$ as a round. Given B^k , for $1 \leq k \leq \lfloor T/b_T \rfloor$, we associate with each $\Delta \in \mathcal{D}_T := \{2^\ell : 0 \leq \ell \leq \lfloor \log_2 b_T \rfloor\}$ the loss:

$$\ell^k(\Delta) := 1 - \frac{1}{\|w\|_\infty b_T} \sum_{t \in B^k} w_{i^{\Delta,t}} \mathbf{1}\{i^{\Delta,t} \neq 0\}.$$

Here, $i^{\Delta,t}$ is the purchase outcome that would be observed at period t if the retailer ran the periodic restarting strategy $\pi(\mathcal{A}, b_T, \Delta)$ from Algorithm 3 on the block B^k . Thus, $\ell^k = (\ell^k(\Delta) : \Delta \in \mathcal{D}_T) \in [0, 1]^{\mathcal{D}_T}$ is the interval-level loss vector faced by Algorithm 4. At round k , Algorithm 4 selects $\Delta^{\pi,k}$, runs the corresponding restarted policy on B^k , and observes only the realized loss $\ell^k(\Delta^{\pi,k})$. This realized loss is denoted by the variable r_k in Algorithm 4, namely $r_k := \ell^k(\Delta^{\pi,k})$.

For every $\Delta \in \mathcal{D}_T$, let π^Δ be the strategy that, on every interval $B^k = \{(k-1)b_T + 1, \dots, kb_T\}$ for $k \in [K_T]$, runs the periodic restarting strategy $\pi(\mathcal{A}, b_T, \Delta)$. Let $i^{\Delta,t}$ be the purchase that would be observed at time $t \leq T_0$ under this counterfactual strategy. Define the oracle revenue by $J^*(F^{(\mathbb{N})}, T_0) := \sum_{t=1}^{T_0} r(S^*(F^t), F^t)$ and define the regret of the fixed restarting frequency Δ over all intervals by:

$$\mathcal{R}^\Delta(F^{(\mathbb{N})}, T) := J^*(F^{(\mathbb{N})}, T_0) - \mathbb{E}_{\mathbb{P}^{\pi^\Delta}} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta,t}} \mathbf{1}\{i^{\Delta,t} \neq 0\} \right].$$

Lemma 8. *Given $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$ and $\Delta \in \mathcal{D}_T$, one has:*

$$J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) \leq \mathcal{R}^\Delta(F^{(\mathbb{N})}, T) + \|w\|_\infty b_T (3\sqrt{K_T |\mathcal{D}_T| \log |\mathcal{D}_T|} + 3 \log |\mathcal{D}_T| + 1).$$

Proof. Let $\pi \equiv \pi(\mathcal{A}, T)$ be the learning to restart strategy from Algorithm 4. Fix a path $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$. Let $T_0 := K_T b_T$ and $A_T := |\mathcal{D}_T|$. Since $b_T = \lceil T^{2/3} \rceil$ and $K_T = \lfloor T/b_T \rfloor$, one has $T - T_0 < b_T$. To apply Lemma 7, we view the interval losses on an enlarged probability space. Conditional on the history before interval B^k , and before drawing $\Delta^{\pi,k}$, we independently sample, for every $\Delta \in \mathcal{D}_T$, the utility vectors of all customers in B^k and the internal randomization used by the strategy $\pi(\mathcal{A}, b_T, \Delta)$, independently of the past. These copies define the full loss vector $\ell^k = (\ell^k(\Delta) : \Delta \in \mathcal{D}_T)$. After this vector is defined, the strategy draws $\Delta^{\pi,k} \sim q^k$ and observes only $\ell^k(\Delta^{\pi,k})$.

Let $\tilde{\mathbb{P}}^\pi$ be the enlarged distribution just constructed. The actual distribution associated with the learning to restart strategy is $\mathbb{P}^\pi$. The enlarged one is used to make all counterfactual losses $(\ell^k(\Delta) : \Delta \in \mathcal{D}_T)$ available before the draw of $\Delta^{\pi,k}$. Moreover, as in Lemma 7, conditional on the history before B^k , the draw $\Delta^{\pi,k} \sim q^k$ can be made independently of the randomness generating all counterfactual losses. Hence,

$$\mathbb{E}_{\tilde{\mathbb{P}}^\pi} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta^{\pi,k},t}} \mathbf{1}\{i^{\Delta^{\pi,k},t} \neq 0\} \right] = \mathbb{E}_{\mathbb{P}^\pi} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta^{\pi,k},t}} \mathbf{1}\{i^{\Delta^{\pi,k},t} \neq 0\} \right].$$

Similarly, for every fixed $\Delta \in \mathcal{D}_T$, the counterfactual trajectory $(i^{\Delta,t} : 1 \leq t \leq T_0)$ constructed on the enlarged space has the same distribution as the trajectory generated by the strategy π^Δ , which

runs $\pi(\mathcal{A}, b_T, \Delta)$ independently on each interval. Hence,

$$\mathbb{E}_{\tilde{\mathbb{P}}^\pi} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta,t}} \mathbf{1}\{i^{\Delta,t} \neq 0\} \right] = \mathbb{E}_{\mathbb{P}^{\pi^\Delta}} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta,t}} \mathbf{1}\{i^{\Delta,t} \neq 0\} \right].$$

Applying Lemma 7 with $\mathcal{D} = \mathcal{D}_T$ and $K = K_T$ gives, for every $\Delta \in \mathcal{D}_T$, the following inequality:

$$\mathbb{E}_{\tilde{\mathbb{P}}^\pi} \left[\sum_{k=1}^{K_T} \ell^k(\Delta^{\pi,k}) - \sum_{k=1}^{K_T} \ell^k(\Delta) \right] \leq 3\sqrt{K_T A_T \log A_T} + 3 \log A_T =: R_T. \quad (\text{E.C.3.3})$$

Multiplying by $\|w\|_\infty b_T$ in (E.C.3.3), we first obtain:

$$\mathbb{E}_{\tilde{\mathbb{P}}^\pi} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta,t}} \mathbf{1}\{i^{\Delta,t} \neq 0\} \right] - \mathbb{E}_{\tilde{\mathbb{P}}^\pi} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta^{\pi,k},t}} \mathbf{1}\{i^{\Delta^{\pi,k},t} \neq 0\} \right] \leq \|w\|_\infty b_T R_T.$$

By the two equalities above, the auxiliary fixed- Δ term has the same expectation as the revenue of π^Δ , while the strategy term has the same expectation as the revenue of the actual strategy π . Thus:

$$\mathbb{E}_{\mathbb{P}^{\pi^\Delta}} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta,t}} \mathbf{1}\{i^{\Delta,t} \neq 0\} \right] - \mathbb{E}_{\mathbb{P}^\pi} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta^{\pi,k},t}} \mathbf{1}\{i^{\Delta^{\pi,k},t} \neq 0\} \right] \leq \|w\|_\infty b_T R_T.$$

Adding the same quantity $J^*(F^{(\mathbb{N})}, T_0)$ from both sides of the previous inequality gives:

$$J^*(F^{(\mathbb{N})}, T_0) - \mathbb{E}_{\mathbb{P}^\pi} \left[\sum_{k=1}^{K_T} \sum_{t \in B^k} w_{i^{\Delta^{\pi,k},t}} \mathbf{1}\{i^{\Delta^{\pi,k},t} \neq 0\} \right] \leq \mathcal{R}^\Delta(F^{(\mathbb{N})}, T) + \|w\|_\infty b_T R_T.$$

The strategy runs $\mathcal{A}$ on the remaining periods $\{T_0 + 1, \dots, T\}$, if such periods exist. Since each one-period regret is at most $\|w\|_\infty$, and since $T - T_0 < b_T$, the remaining periods contribute at most $\|w\|_\infty b_T$ regret. By using a lemma similar to Lemma 13 and the definition of $J^\pi(F^{(\mathbb{N})}, T)$, we obtain:

$$J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) \leq \mathcal{R}^\Delta(F^{(\mathbb{N})}, T) + \|w\|_\infty b_T (3\sqrt{K_T A_T \log A_T} + 3 \log A_T + 1),$$

which completes in turn the proof of the lemma. $\blacksquare$

E.C.3.1 Proof of Theorem 5

Proof. Let $\pi \equiv \pi(\mathcal{A}, T)$ be the learning to restart strategy from Algorithm 4. Fix an arbitrary path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$. Let $T_0 := K_T b_T$ and $A_T := |\mathcal{D}_T|$. Since $b_T = \lceil T^{2/3} \rceil$ and $K_T = \lfloor T/b_T \rfloor$, one has $T - T_0 < b_T$. Then, by applying Lemma 8, we obtain:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\leq \mathcal{R}^\Delta(F^{(\mathbb{N})}, T) + 3\|w\|_\infty b_T \sqrt{K_T A_T \log A_T} \\ &\quad + 3\|w\|_\infty b_T \log A_T + \|w\|_\infty b_T. \end{aligned} \quad (\text{E.C.3.4})$$

It remains to bound $\mathcal{R}^\Delta(F^{(\mathbb{N})}, T)$. Fix $\Delta \in \mathcal{D}_T$. Running $\pi(\mathcal{A}, b_T, \Delta)$ repeatedly over the horizon $[T_0]$ is equivalent to partitioning $\{1, \dots, T_0\}$ into intervals $I_1, \dots, I_m$ such that a new copy of $\mathcal{A}$ is started at the beginning of each interval I_j , and $|I_j| \leq \Delta$ for all $j \in [m]$. Moreover, $m \leq K_T(b_T/\Delta + 1) \leq T/\Delta + T/b_T \leq 2T/\Delta$, where the last inequality uses $\Delta \leq b_T$.

For a fixed interval I_j , Corollary 5 gives the following inequality:

$$\sum_{t \in I_j} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I_j} r(S^{\mathcal{A},t}, F^t) \right] \leq \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + 2c_\zeta \|w\|_\infty |I_j|^2 \sqrt{\mathcal{M}_{I_j}(F^{(\mathbb{N})})}.$$

Summing over $j \in [m]$ gives the following inequality:

$$\mathcal{R}^\Delta(F^{(\mathbb{N})}, T) \leq \sum_{j=1}^m \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + 2c_\zeta \|w\|_\infty \sum_{j=1}^m |I_j|^2 \sqrt{\mathcal{M}_{I_j}(F^{(\mathbb{N})})}. \quad (\text{E.C.3.5})$$

Since $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), n)$ is non-decreasing in n , and since $|I_j| \leq \Delta$, we obtain:

$$\sum_{j=1}^m \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) \leq \frac{2T}{\Delta} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta). \quad (\text{E.C.3.6})$$

Because $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$, one has $\mathcal{M}(F^{(\mathbb{N})}, T) \leq M_T$. Therefore, one has:

$$\sum_{j=1}^m |I_j|^2 \sqrt{\mathcal{M}_{I_j}(F^{(\mathbb{N})})} \leq \Delta^2 \sum_{j=1}^m \sqrt{\mathcal{M}_{I_j}(F^{(\mathbb{N})})} \leq \Delta^2 \sqrt{\frac{2T}{\Delta}} M_T \leq \Delta^{3/2} \sqrt{2TM_T}.$$

Combining (E.C.3.5), and (E.C.3.6), we obtain, for every $\Delta \in \mathcal{D}_T$,

$$\begin{aligned} \mathcal{R}^\Delta(F^{(\mathbb{N})}, T) &\leq \frac{2T}{\Delta} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + 2\sqrt{2}c_\zeta \|w\|_\infty \Delta^{3/2} \sqrt{TM_T} \\ &\leq \tilde{c}_1((T/\Delta) \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \Delta^{3/2} \sqrt{TM_T}), \end{aligned} \quad (\text{E.C.3.7})$$

where $\tilde{c}_1 \equiv \tilde{c}_1(c_\zeta, \|w\|_\infty) = \max\{2, 2\sqrt{2}c_\zeta \|w\|_\infty\} > 0$.

Combining (E.C.3.4) and (E.C.3.7), and then minimizing over $\Delta \in \mathcal{D}_T$, gives:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\leq \min \{ \tilde{c}_1((T/\Delta) \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \Delta^{3/2} \sqrt{TM_T}) : \Delta \in \mathcal{D}_T \} \\ &\quad + 3\|w\|_\infty b_T \sqrt{K_T A_T \log A_T} + 3\|w\|_\infty b_T \log A_T + \|w\|_\infty b_T. \end{aligned} \quad (\text{E.C.3.8})$$

Finally, since $b_T = \lceil T^{2/3} \rceil$, one has $b_T \leq 2T^{2/3}$ and $K_T \leq T/b_T$. Consequently,

$$b_T \sqrt{K_T A_T \log A_T} \leq \sqrt{T b_T A_T \log A_T} \leq \sqrt{2} T^{5/6} \sqrt{A_T \log A_T},$$

and $b_T \log A_T \leq 2T^{2/3} \log A_T$ as $b_T \leq 2T^{2/3}$ and $A_T \leq \frac{5}{3} \log(T)$. Thus, (E.C.3.8) becomes:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\leq \min \{ \tilde{c}_1((T/\Delta) \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \Delta^{3/2} \sqrt{TM_T}) : \Delta \in \mathcal{D}_T \} \\ &\quad + \|w\|_\infty T^{5/6} (3\sqrt{2} \sqrt{A_T \log A_T} + 6 \log A_T + 2). \end{aligned}$$

Since $A_T \leq \frac{5}{3} \log(T) \leq 2 \log T$ for $T \geq 4$, and since $\log A_T \leq A_T \leq 2 \log T$, one has $\sqrt{A_T \log A_T} \leq 2 \log T$ and $\log A_T \leq 2 \log T$. Moreover, $2 \leq \log T$ for $T \geq 8$. Hence, for $T \geq 8$,

$$3\sqrt{2} \sqrt{A_T \log A_T} + 6 \log A_T + 2 \leq (6\sqrt{2} + 13) \log T \leq 25 \log T.$$

Thus, the inequality on the regret becomes:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\leq c_1 \left(\min \{ (T/\Delta) \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \Delta^{3/2} \sqrt{TM_T} : \Delta \in \mathcal{D}_T \} + T^{5/6} \log T \right), \end{aligned}$$

where $c_1 = \max\{\tilde{c}_1, 25\|w\|_\infty\}$. Since the preceding bound holds for every $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$, taking the supremum over $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$ proves the theorem. $\blacksquare$

E.C.3.2 Proof of Corollary 3

Proof. By Theorem 5 and the assumption on the regret bound for static preferences, one has:

$$\mathcal{R}^{\pi(\mathcal{A}, T)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq c_1 \left(\min \{ C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} + \Delta^{3/2} \sqrt{TM_T} : \Delta \in \mathcal{D}_T \} + T^{5/6} \log T \right).$$

It remains to bound the minimum term. First, suppose that the magnitude satisfies $M_T \geq C_{\mathcal{A}}^2/T^{5/3}$.

In that case, define $\Delta_0 = C_{\mathcal{A}}^{1/2} T^{1/4} M_T^{-1/4}$. The condition above gives $\Delta_0 \leq T^{2/3} \leq b_T$. If $\Delta_0 < 1$, then we set $\Delta = 1$; the claimed bound follows from the trivial bound $\mathcal{R}^\pi(\mathcal{F}_G, T) \leq \|w\|_\infty T$. Thus,

assume that $\Delta_0 \geq 1$. Let $\hat{\Delta} = 2^{\lfloor \log_2(\Delta_0) \rfloor}$. Then, $\hat{\Delta} \in \mathcal{D}_T$ and $\Delta_0/2 \leq \hat{\Delta} \leq \Delta_0$. Therefore, substituting $\hat{\Delta}$ in the expression from the minimum term gives the following two inequalities:
 $C_{\mathcal{A}} \frac{T}{\sqrt{\hat{\Delta}}} \leq \sqrt{2} C_{\mathcal{A}} \frac{T}{\sqrt{\Delta_0}} = \sqrt{2} C_{\mathcal{A}}^{3/4} T^{7/8} M_T^{1/8}$, and $\hat{\Delta}^{3/2} \sqrt{T M_T} \leq \Delta_0^{3/2} \sqrt{T M_T} = C_{\mathcal{A}}^{3/4} T^{7/8} M_T^{1/8}$.
Hence, the following inequality is satisfied:

$$\min \left\{ C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} + \Delta^{3/2} \sqrt{T M_T} : \Delta \in \mathcal{D}_T \right\} \leq (1 + \sqrt{2}) C_{\mathcal{A}}^{3/4} T^{7/8} M_T^{1/8}.$$

Absorbing the numerical constant into c_2 proves the first part of the corollary.

Now suppose that $M_T < C_{\mathcal{A}}^2/T^{5/3}$. Let $\hat{\Delta} = 2^{\lfloor \log_2(T^{2/3}) \rfloor}$. Since $T^{2/3} \leq b_T$, one has $\hat{\Delta} \in \mathcal{D}_T$. Moreover, $T^{2/3}/2 \leq \hat{\Delta} \leq T^{2/3}$. For this value, the following two inequalities are satisfied:

$$C_{\mathcal{A}} \frac{T}{\sqrt{\hat{\Delta}}} \leq \sqrt{2} C_{\mathcal{A}} T^{2/3}, \quad \text{and} \quad \hat{\Delta}^{3/2} \sqrt{T M_T} \leq T \sqrt{T M_T}.$$

The condition $M_T < C_{\mathcal{A}}^2/T^{5/3}$ implies that $\sqrt{T M_T} \leq C_{\mathcal{A}} T^{-1/3}$. Therefore, $T \sqrt{T M_T} \leq C_{\mathcal{A}} T^{2/3}$. Consequently, we obtain the following inequality:

$$\min \left\{ C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} + \Delta^{3/2} \sqrt{T M_T} : \Delta \in \mathcal{D}_T \right\} \leq (1 + \sqrt{2}) C_{\mathcal{A}} T^{2/3}.$$

Absorbing the numerical constant into c_2 proves the second inequality and concludes the proof. $\blacksquare$

E.C.3.3 Proof of Theorem 6

Proof. Fix $T \geq 2$, $M_T > 0$, $\gamma_T > 0$, and an arbitrary path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$. Let $T_0 := K_T b_T$ and $A_T := |\mathcal{D}_T|$. Since $b_T = \lceil T^{2/3} \rceil$ and $K_T = \lfloor T/b_T \rfloor$, one has $T - T_0 < b_T$. Also, define $\bar{c}_{\mathcal{A}} \equiv c_{\mathcal{A}} + 2$. The constant $c_{\mathcal{A}}$ is the one appearing in the near-stationary robustness condition of Definition 1. The additional 2 is the constant in Lemma 12 (see below).

Then, by applying Lemma 8, we obtain for all $\Delta \in \mathcal{D}_T$:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) &\leq \mathcal{R}^\Delta(F^{(\mathbb{N})}, T) + 3\|w\|_\infty b_T \sqrt{K_T A_T \log A_T} \\ &\quad + 3\|w\|_\infty b_T \log A_T + \|w\|_\infty b_T. \end{aligned} \tag{E.C.3.9}$$

It remains to bound $\mathcal{R}^\Delta(F^{(\mathbb{N})}, T)$. Fix $\Delta \in \mathcal{D}_T$. Running $\pi(\mathcal{A}, b_T, \Delta)$ over and over up to T_0 is equivalent to partitioning $\{1, \dots, T_0\}$ into intervals $I_1, \dots, I_m$ such that a new copy of $\mathcal{A}$ is started at the beginning of each I_j with $|I_j| \leq \Delta$ for all $j \in [m]$. Moreover, $m \leq K_T(b_T/\Delta + 1) \leq T/\Delta + T/b_T \leq 2T/\Delta$, where the last inequality uses $\Delta \leq b_T$.

We decompose the regret on the j -th interval I_j into the regret of the fixed assortment that maximizes cumulative expected revenue over I_j relative to the clairvoyant retailer and the regret of the strategy relative to that assortment. Thus,

$$\begin{aligned} \sum_{t \in I_j} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I_j} r(S^{\mathcal{A}, t}, F^t) \right] &\leq \sum_{t \in I_j} r(S^*(F^t), F^t) - \max \left\{ \sum_{t \in I_j} r(S, F^t) : S \in \mathcal{S} \right\} \\ &\quad + \max \left\{ \sum_{t \in I_j} r(S, F^t) : S \in \mathcal{S} \right\} - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I_j} r(S^{\mathcal{A}, t}, F^t) \right]. \end{aligned}$$

By Lemma 12, the first term is bounded by $2|I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})})$. By the near-stationary robustness condition in Definition 1, the second term is bounded by $c_{\mathcal{A}} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + c_{\mathcal{A}} |I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})})$.

Therefore, the following inequality is satisfied:

$$\sum_{t \in I_j} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I_j} r(S^{\mathcal{A},t}, F^t) \right] \leq \bar{c}_{\mathcal{A}} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + \bar{c}_{\mathcal{A}} |I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})}).$$

Summing over $j \in [m]$ gives the following inequality:

$$\mathcal{R}^{\Delta}(F^{(\mathbb{N})}, T) \leq \bar{c}_{\mathcal{A}} \sum_{j=1}^m \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) + \bar{c}_{\mathcal{A}} \sum_{j=1}^m |I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})}). \quad (\text{E.C.3.10})$$

Since $\mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), n)$ is non-decreasing in n , and since $|I_j| \leq \Delta$, we obtain:

$$\sum_{j=1}^m \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I_j|) \leq \frac{2T}{\Delta} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta). \quad (\text{E.C.3.11})$$

We now bound the movement term. For every interval $I = \{a, \dots, b\}$, define $m_t \equiv \max\{\mathcal{K}^t(S) : S \in \mathcal{S}\}$ for $t \in \{a+1, \dots, b\}$. For every $S \in \mathcal{S}$ and every $t \geq 2$, Pinsker's inequality (Tsybakov 2003) gives the following inequality:

$$|r(S, F^t) - r(S, F^{t-1})| \leq \|w\|_{\infty} \sum_{i \in S \cup \{0\}} |p_i(S, F^t) - p_i(S, F^{t-1})| \leq \sqrt{2} \|w\|_{\infty} \sqrt{\mathcal{K}^t(S)}.$$

Hence, $\max\{|r(S, F^t) - r(S, F^{t-1})| : S \in \mathcal{S}\} \leq \sqrt{2} \|w\|_{\infty} \sqrt{m_t}$. Since the intervals $I_1, \dots, I_m$ are disjoint, and since each $\mathcal{V}_{I_j}(F^{(\mathbb{N})})$ only sums one-period changes inside I_j , we obtain:

$$\sum_{j=1}^m \mathcal{V}_{I_j}(F^{(\mathbb{N})}) \leq \sqrt{2} \|w\|_{\infty} \sum_{t=2}^{T_0} \sqrt{m_t} \stackrel{(a)}{\leq} \sqrt{2} \|w\|_{\infty} (T \sum_{t=2}^T m_t)^{1/2} = \sqrt{2} \|w\|_{\infty} \sqrt{T \mathcal{M}(F^{(\mathbb{N})}, T)},$$

where (a) follows from the Cauchy-Schwarz inequality. Because $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$, one has $\mathcal{M}(F^{(\mathbb{N})}, T) \leq M_T$. Therefore, the following inequality is satisfied:

$$\sum_{j=1}^m |I_j| \mathcal{V}_{I_j}(F^{(\mathbb{N})}) \leq \Delta \sum_{j=1}^m \mathcal{V}_{I_j}(F^{(\mathbb{N})}) \leq \sqrt{2} \|w\|_{\infty} \Delta \sqrt{T M_T}. \quad (\text{E.C.3.12})$$

Combining (E.C.3.10), (E.C.3.11), and (E.C.3.12), we obtain, for every $\Delta \in \mathcal{D}_T$,

$$\mathcal{R}^{\Delta}(F^{(\mathbb{N})}, T) \leq \frac{2\bar{c}_{\mathcal{A}} T}{\Delta} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \bar{c}_{\mathcal{A}} \sqrt{2} \|w\|_{\infty} \Delta \sqrt{T M_T}. \quad (\text{E.C.3.13})$$

Combining (E.C.3.9) and (E.C.3.13), and then minimizing over $\Delta \in \mathcal{D}_T$, gives:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^{\pi}(F^{(\mathbb{N})}, T) &\leq \min \left\{ \frac{2\bar{c}_{\mathcal{A}} T}{\Delta} \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \bar{c}_{\mathcal{A}} \sqrt{2} \|w\|_{\infty} \Delta \sqrt{T M_T} : \Delta \in \mathcal{D}_T \right\} \\ &\quad + 3 \|w\|_{\infty} b_T \sqrt{K_T A_T \log A_T} + 3 \|w\|_{\infty} b_T \log A_T + \|w\|_{\infty} b_T. \end{aligned} \quad (\text{E.C.3.14})$$

Finally, since $b_T = \lceil T^{2/3} \rceil$, one has $b_T \leq 2T^{2/3}$ and $K_T \leq T/b_T$. Consequently,

$$b_T \sqrt{K_T A_T \log A_T} \leq \sqrt{T b_T A_T \log A_T} \leq \sqrt{2} T^{5/6} \sqrt{A_T \log A_T},$$

and $b_T \log A_T \leq 2T^{2/3} \log A_T$ together with $b_T \leq 2T^{2/3}$ and $A_T \leq \frac{5}{3} \log(T)$. Therefore, (E.C.3.14) can be simplified as follows:

$$\begin{aligned} J^*(F^{(\mathbb{N})}, T) - J^{\pi}(F^{(\mathbb{N})}, T) &\leq \tilde{c}_1 \min \left\{ (T/\Delta) \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \Delta \sqrt{T M_T} : \Delta \in \mathcal{D}_T \right\} \\ &\quad + \|w\|_{\infty} T^{5/6} \left(3\sqrt{2} \sqrt{A_T \log A_T} + 6 \log A_T + 2 \right), \end{aligned}$$

where $\tilde{c}_1 \equiv \tilde{c}_1(c_{\mathcal{A}}, \|w\|_{\infty}) \equiv \max\{2, \sqrt{2}\|w\|_{\infty}\}(2 + c_{\mathcal{A}})$.

Since $A_T \leq \frac{5}{3} \log(T) \leq 2 \log T$ for $T \geq 4$, and since $\log A_T \leq A_T \leq 2 \log T$, one has

$\sqrt{A_T \log A_T} \leq 2 \log T$ and $\log A_T \leq 2 \log T$. Moreover, $2 \leq \log T$ for $T \geq 8$. Hence, for $T \geq 8$,

$$3\sqrt{2}\sqrt{A_T \log A_T} + 6 \log A_T + 2 \leq (6\sqrt{2} + 13) \log T \leq 25 \log T.$$

Thus, the inequality on the regret becomes:

$$J^*(F^{(\mathbb{N})}, T) - J^\pi(F^{(\mathbb{N})}, T) \leq c_1 \left(\min \{ (T/\Delta) \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), \Delta) + \Delta \sqrt{TM_T} : \Delta \in \mathcal{D}_T \} + T^{5/6} \log T \right),$$

where $c_1 = \max\{\tilde{c}_1, 25\|w\|_\infty\}$. Since the preceding bound holds for every $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$, taking the supremum over $F^{(\mathbb{N})} \in \mathcal{F}_G(\gamma_T, M_T)$ proves the theorem. $\blacksquare$

E.C.3.4 Proof of Corollary 4

Proof. By Theorem 6 and the assumption on the regret bound for static preferences, one has:

$$\mathcal{R}^{\pi(\mathcal{A}, T)}(\mathcal{F}_G(\gamma_T, M_T), T) \leq c_1 \left(\min \{ C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} + \Delta \sqrt{TM_T} : \Delta \in \mathcal{D}_T \} + T^{5/6} \log T \right).$$

It remains to bound the minimum term. First, suppose that the magnitude satisfies $M_T \geq C_{\mathcal{A}}^2/T$.

In that case, we define $\Delta_0 = C_{\mathcal{A}}^{2/3} T^{1/3} M_T^{-1/3}$. The condition above gives $\Delta_0 \leq T^{2/3} \leq b_T$. If $\Delta_0 < 1$, then we set $\Delta = 1$; the claimed bound follows from the trivial bound $\mathcal{R}^\pi(\mathcal{F}_G, T) \leq \|w\|_\infty T$.

Thus, assume that $\Delta_0 \geq 1$. Let $\hat{\Delta} = 2^{\lfloor \log_2(\Delta_0) \rfloor}$. Then, $\hat{\Delta} \in \mathcal{D}_T$ and $\Delta_0/2 \leq \hat{\Delta} \leq \Delta_0$. Therefore, substituting $\hat{\Delta}$ in the terms inside the minimum gives the following two inequalities:

$$C_{\mathcal{A}} \frac{T}{\sqrt{\hat{\Delta}}} \leq \sqrt{2} C_{\mathcal{A}} \frac{T}{\sqrt{\Delta_0}} = \sqrt{2} C_{\mathcal{A}}^{2/3} T^{5/6} M_T^{1/6}, \quad \text{and} \quad \hat{\Delta} \sqrt{TM_T} \leq \Delta_0 \sqrt{TM_T} = C_{\mathcal{A}}^{2/3} T^{5/6} M_T^{1/6}.$$

Hence, the following inequality is satisfied:

$$\min \{ C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} + \Delta \sqrt{TM_T} : \Delta \in \mathcal{D}_T \} \leq (1 + \sqrt{2}) C_{\mathcal{A}}^{2/3} T^{5/6} M_T^{1/6}.$$

Absorbing the numerical constant into c_2 proves the first part of the corollary.

Now, suppose that $M_T < C_{\mathcal{A}}^2/T$. Let $\hat{\Delta} = 2^{\lfloor \log_2(T^{2/3}) \rfloor}$. Since $T^{2/3} \leq b_T$, one has $\hat{\Delta} \in \mathcal{D}_T$. Moreover, $T^{2/3}/2 \leq \hat{\Delta} \leq T^{2/3}$. For this value, the following two inequalities are satisfied:

$$C_{\mathcal{A}} \frac{T}{\sqrt{\hat{\Delta}}} \leq \sqrt{2} C_{\mathcal{A}} T^{2/3}, \quad \text{and} \quad \hat{\Delta} \sqrt{TM_T} \leq T^{2/3} \sqrt{TM_T}.$$

The condition $M_T < C_{\mathcal{A}}^2/T$ implies that $\sqrt{TM_T} \leq C_{\mathcal{A}}$. Therefore, $T^{2/3} \sqrt{TM_T} \leq C_{\mathcal{A}} T^{2/3}$. Consequently, we obtain the following inequality:

$$\min \{ C_{\mathcal{A}} \frac{T}{\sqrt{\Delta}} + \Delta \sqrt{TM_T} : \Delta \in \mathcal{D}_T \} \leq (1 + \sqrt{2}) C_{\mathcal{A}} T^{2/3}.$$

Absorbing the numerical constant into c_2 proves the second inequality and concludes the proof. $\blacksquare$

E.C.4 Preliminaries

We introduce notation that we use throughout the proofs. Let F and G be probability distributions on a common discrete measurable space $(\Omega, \mathcal{B})$. We define their KL divergence as $\mathcal{K}(F, G) := \sum_{\omega \in \Omega} F(\omega) \log(F(\omega)/G(\omega))$ with the conventions $0 \log(0/q) = 0$ for $q > 0$, and $\mathcal{K}(F, G) = +\infty$ if $F(\omega) > 0$ and $G(\omega) = 0$ for some $\omega \in \Omega$ (Cover and Thomas 2005). For an event $E \in \mathcal{B}$, we use $\mathcal{K}(F, G; E)$ for the KL divergence between $F(\cdot | E)$ and $G(\cdot | E)$, whenever the conditional distributions are well defined. We also use the total variation distance $\|F - G\|_{\text{TV}} := \sup\{|F(A) - G(A)| : A \in \mathcal{B}\}$, which, on a discrete space, satisfies $\|F - G\|_{\text{TV}} = \frac{1}{2} \sum_{\omega \in \Omega} |F(\omega) - G(\omega)|$; see Tsybakov (2003).

For $x \in \mathbb{R}^d$ and $p \in [1, \infty)$, we write $\|x\|_p := \left(\sum_{i=1}^d |x_i|^p\right)^{1/p}$ and $\|x\|_\infty := \max\{|x_i| : i \in [d]\}$. Also, $\mathbf{1}\{\cdot\}$ denotes the indicator function. Some statements are understood to hold almost surely under the relevant probability measure, and we omit the formal reference when the measure is clear from the context. Throughout the proofs, the *environment* refers to the adversary that selects the preferences path $F^{(\mathbb{N})}$, as introduced in Section 3 when defining adversarial regret.

The rest of this section collects preliminary results used in the theoretical analysis throughout the electronic companion. Table E.C.4.1 summarizes the lemmas and corollary proved below.

Lemma	Corollary
9, 10, 11, 12, 13, 14, 15, 16, 17	5

Table E.C.4.1: List of the results in Electronic Companion E.C.4

We next introduce notation used throughout. We distinguish paths of preferences with bounded magnitude, as in $\mathcal{F}_G$, from paths with a single abrupt change, as in $\mathcal{F}_A$. Throughout, references to $\mathcal{F}_S$ and $\mathcal{F}_A$ mean that $\mathcal{F}_S \equiv \mathcal{F}_S(\gamma_T)$ and $\mathcal{F}_A \equiv \mathcal{F}_A(\gamma_T)$ for some $\gamma_T > 0$; see Section 5.

General preferences paths. Recall from Section 3 that $\mathcal{F}$ denotes the set of all customers' preferences paths $F^{(\mathbb{N})}$ satisfying $p_i(S, F^t) \in [\varsigma, 1 - \varsigma]$ for all $i \in S \cup \{0\}$, $S \in \mathcal{S}$, $t \in \mathbb{N}$, and some fixed $\varsigma \in (0, 1)$. This assumption is a non-degeneracy condition. It is not restrictive and mainly rules out trivial consumers' purchasing behavior. For an interval $I = \{a, \dots, b\}$, with $a, b \in \mathbb{N}_{\geq 1}$, and a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}$, we define the magnitude of preferences over that interval by:

$$\mathcal{M}_I \equiv \mathcal{M}_I(F^{(\mathbb{N})}) := \sum_{t=a+1}^b \max\{\mathcal{K}^t(S) : S \in \mathcal{S}\},$$

where $\mathcal{K}^t(S)$ is defined in Section 4 for all $t \geq 2$ and any feasible assortment $S \in \mathcal{S}$.

Define $m_u := \max\{\mathcal{K}^u(S) : S \in \mathcal{S}\}$ for $u \in \{a+1, \dots, b\}$. For $t \in I$, we define $D_t := \sum_{u=a+1}^t \sqrt{m_u}$, with $D_a := 0$ by default. Similarly, for an interval $I = \{a, \dots, b\}$ and a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}$, define the revenue variation (between consecutive customers) over I by:

$$\mathcal{V}_I(F^{(\mathbb{N})}) := \sum_{t=a+1}^b \max\{|r(S, F^t) - r(S, F^{t-1})| : S \in \mathcal{S}\}.$$

Given an admissible strategy $\pi \in \mathcal{P}$ and a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}$, we denote

by $\mathbb{P}^\pi$ the probability distribution induced over histories by running π under $F^{(\mathbb{N})}$. We denote expectation with respect to $\mathbb{P}^\pi$ by $\mathbb{E}_{\mathbb{P}^\pi}[\cdot]$. In particular, when the policy is $\mathcal{A}$, we write $\mathbb{P}^\mathcal{A}$ and $\mathbb{E}_{\mathbb{P}^\mathcal{A}}[\cdot]$.

Abruptly changing preferences. We may sometimes consider paths of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_A$ in which a single abrupt shift occurs, as defined in Section 5. We refer to the preferences before the change as F^1 and to those after the change as F^τ . Unless otherwise specified, we assume that both F^1 and F^τ are chosen such that, for some $\tau \in \mathbb{N}$, the path $F^{(\mathbb{N})} \equiv (F^t : t \in \mathbb{N})$ satisfies $F^t \equiv F^1$ for all $t < \tau$, and $F^t \equiv F^\tau$ for all $t \geq \tau$, with the additional condition that $F^{(\mathbb{N})} \in \mathcal{F}_A$ (or a specified subset thereof). The variable $\tau \in \{2, \dots, T+1\}$ denotes the time at which the change occurs, where $\tau = T+1$ indicates that no change takes place by convention. Finally, we denote by $\mathbb{P}_\tau^\pi$ the distribution over the entire sales history $\mathcal{H}^{T+1} \equiv ((S^{\pi,t}, i^t) : t \in [T])$ induced by some admissible strategy $\pi \in \mathcal{P}$ when the change occurs at time τ , evaluated over the finite horizon T .

To formally introduce our analysis, we begin by defining the random variable $\mathcal{J}^\pi(F^{(\mathbb{N})}, T)$, which represents the expected revenue of a given strategy π over a selling horizon of T periods. Specifically:

$$\mathcal{J}^\pi(F^{(\mathbb{N})}, T) := \sum_{t=1}^{\tau-1} \sum_{i \in \psi_t(\mathcal{H}^t)} w_i p_i(\psi_t(\mathcal{H}^t), F^1) + \sum_{t=\tau}^T \sum_{i \in \psi_t(\mathcal{H}^t)} w_i p_i(\psi_t(\mathcal{H}^t), F^\tau).$$

Throughout the proofs, we use $\pi := (\psi_t(\cdot) : 1 \leq t \leq T)$ and $\pi := (S^{\pi,t} : 1 \leq t \leq T)$ interchangeably. We use the first notation when emphasizing the dependence of the strategy on the sales history, and the second when focusing on the assortments selected by the strategy.

E.C.4.1 Proof of Lemma 9

We start by deriving a complementary inequality to Pinsker's inequality (Tsybakov 2003) for categorical distributions that satisfy a non-degeneracy condition. Formally,

Lemma 9. *Let p and q be two categorical distributions defined on the same finite support. Suppose that there exists $\varsigma \in (0, 1)$ such that $q_i \geq \varsigma$ for every coordinate i in that support. Then:*

$$\mathcal{K}(p, q) \leq \varsigma^{-1} \left(\sum_i |p_i - q_i| \right)^2.$$

Proof. By the definition of the KL divergence, one has $\mathcal{K}(p, q) = \sum_i p_i \log(p_i/q_i)$. Using $\log(x) \leq x - 1$ for all $x > 0$, we obtain the following inequality:

$$\begin{aligned} \mathcal{K}(p, q) &\leq \sum_i p_i \left(\frac{p_i}{q_i} - 1 \right) \\ &= \sum_i \frac{p_i(p_i - q_i)}{q_i} = \sum_i \frac{(q_i + p_i - q_i)(p_i - q_i)}{q_i} = \sum_i (p_i - q_i) + \sum_i \frac{(p_i - q_i)^2}{q_i}. \end{aligned}$$

Since p and q are probability distributions on the same support, we have $\sum_i (p_i - q_i) = 0$. Therefore, we obtain $\mathcal{K}(p, q) \leq \sum_i (p_i - q_i)^2 / q_i$. Then, by using $q_i \geq \varsigma$, we obtain the following inequalities:

$$\mathcal{K}(p, q) \leq \varsigma^{-1} \sum_i (p_i - q_i)^2 \leq \varsigma^{-1} \left(\sum_i |p_i - q_i| \right)^2,$$

which completes the proof. ■

E.C.4.2 Proof of Lemma 10

We first state a lemma used to construct statistical tests for detecting changes in purchasing distributions. The result follows from the method of types for finite sets and its proof closely follows Chapter 12 of Cover and Thomas (2005). Let $\mathcal{X}$ be a finite set with $|\mathcal{X}| = d$. Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be two independent samples drawn from the same categorical distribution $p \in \Delta(\mathcal{X}) \equiv \{\tilde{p} \in [0, 1]^{\mathcal{X}} : \sum_{i \in \mathcal{X}} \tilde{p}_i = 1\}$. We define the empirical distributions of the two samples by:

$$\hat{p}_i^X = \frac{1}{m} \sum_{u=1}^m \mathbf{1}\{X_u = i\}, \quad \text{and} \quad \hat{p}_i^Y = \frac{1}{n} \sum_{u=1}^n \mathbf{1}\{Y_u = i\},$$

for $i \in \mathcal{X}$ and the pooled empirical distribution:

$$\hat{p}_i^0 = \frac{m\hat{p}_i^X + n\hat{p}_i^Y}{m+n},$$

for $i \in \mathcal{X}$. Define $G = m\mathcal{K}(\hat{p}^X, \hat{p}^0) + n\mathcal{K}(\hat{p}^Y, \hat{p}^0)$. Formally, one has the following.

Lemma 10. *For every $x > 0$, the following inequality is satisfied:*

$$\mathbb{P}[G > x] \leq (m+1)^d (n+1)^d \exp(-x) \leq (m+n)^{2d} \exp(-x).$$

In particular, if $d \leq K+1$, then the following inequality is satisfied:

$$\mathbb{P}[G > 8(K+1) \log(m+n) + 8 \log(T+1)] \leq (m+n)^{-2(K+1)} (T+1)^{-4}.$$

Proof. For every integer $r \geq 1$, let $\mathcal{P}_r(\mathcal{X})$ denote the set of empirical distributions on $\mathcal{X}$ that can be generated by a sample of size r . Fix $\mu^X \in \mathcal{P}_m(\mathcal{X})$ and $\mu^Y \in \mathcal{P}_n(\mathcal{X})$ and define the pooled empirical distribution associated with μ^X and μ^Y by $\mu^0 = (m\mu^X + n\mu^Y)/(m+n)$.

Next, we derive a type bound. This approach is standard and presented for completeness; see, e.g., Theorem 12.1.4 in Cover and Thomas (2005). For every $i \in \mathcal{X}$, we define $m_i = m\mu_i^X$. Since $\mu^X \in \mathcal{P}_m(\mathcal{X})$, each m_i is a non-negative integer and $\sum_{i \in \mathcal{X}} m_i = m$. Next, we define $\mathcal{X}_+ = \{i \in \mathcal{X} : \mu_i^X > 0\}$. Then, $m_i = 0$ for every $i \notin \mathcal{X}_+$, and $\sum_{i \in \mathcal{X}_+} \mu_i^X = 1$. Consequently, one has:

$$1 = \left(\sum_{i \in \mathcal{X}_+} \mu_i^X \right)^m = \sum_{\substack{r_i \geq 0, i \in \mathcal{X}_+ \\ \sum_{i \in \mathcal{X}_+} r_i = m}} \frac{m!}{\prod_{i \in \mathcal{X}_+} r_i!} \prod_{i \in \mathcal{X}_+} (\mu_i^X)^{r_i}.$$

All terms in the preceding sum are non-negative. Therefore, the term corresponding to $r_i = m_i$ for every $i \in \mathcal{X}_+$ is at most one, namely:

$$\frac{m!}{\prod_{i \in \mathcal{X}_+} m_i!} \prod_{i \in \mathcal{X}_+} (\mu_i^X)^{m_i} \leq 1.$$

Since $m_i = 0$ for every $i \notin \mathcal{X}_+$, this inequality is equivalent to the following inequality:

$$\frac{m!}{\prod_{i \in \mathcal{X}} m_i!} \leq \prod_{i \in \mathcal{X}_+} (\mu_i^X)^{-m_i}.$$

Using $m_i = m\mu_i^X$, we obtain the following inequality:

$$\frac{m!}{\prod_{i \in \mathcal{X}} m_i!} \leq \exp \left(- \sum_{i \in \mathcal{X}_+} m_i \log(\mu_i^X) \right) = \exp \left(- m \sum_{i \in \mathcal{X}_+} \mu_i^X \log(\mu_i^X) \right).$$

We now use this bound to control the probability of a type. The event $\{\hat{p}^X = \mu^X\}$ is the event that the sample $X_1, \dots, X_m$ contains exactly m_i observations equal to i , for every $i \in \mathcal{X}$. Hence, we have that $\mathbb{P}[\hat{p}^X = \mu^X] = \frac{m!}{\prod_{i \in \mathcal{X}} m_i!} \prod_{i \in \mathcal{X}} p_i^{m_i}$. If $p_i = 0$ for some i with $\mu_i^X > 0$, then the left-hand side is zero and the desired bound is immediate. Otherwise,

$$\begin{aligned} \mathbb{P}[\hat{p}^X = \mu^X] &\leq \exp\left(-m \sum_{i \in \mathcal{X}_+} \mu_i^X \log(\mu_i^X)\right) \prod_{i \in \mathcal{X}} p_i^{m \mu_i^X} \\ &= \exp\left(-m \sum_{i \in \mathcal{X}} \mu_i^X \log(\mu_i^X) + m \sum_{i \in \mathcal{X}} \mu_i^X \log(p_i)\right) \\ &= \exp\left(-m \sum_{i \in \mathcal{X}} \mu_i^X \log \frac{\mu_i^X}{p_i}\right) = \exp\left(-m \mathcal{K}(\mu^X, p)\right). \end{aligned}$$

The same argument, with n and μ^Y in place of m and μ^X , gives $\mathbb{P}[\hat{p}^Y = \mu^Y] \leq \exp(-n \mathcal{K}(\mu^Y, p))$. Importantly, we use the usual conventions $0 \log(0/p_i) = 0$ and $\mu_i \log(\mu_i/0) = +\infty$ when $\mu_i > 0$. It remains only to consider pairs (μ^X, μ^Y) whose joint type probability is positive. For such pairs, both μ^X and μ^Y are supported on the support of p , and all terms in the following decomposition are finite.

Since the two samples are independent, it follows that the following inequality is satisfied:

$$\mathbb{P}[\hat{p}^X = \mu^X, \hat{p}^Y = \mu^Y] \leq \exp\left(-m \mathcal{K}(\mu^X, p) - n \mathcal{K}(\mu^Y, p)\right).$$

We now decompose the exponent. Since $m \mu_i^X + n \mu_i^Y = (m+n) \mu_i^0$ for every $i \in \mathcal{X}$, one has:

$$\begin{aligned} m \mathcal{K}(\mu^X, p) + n \mathcal{K}(\mu^Y, p) &= m \sum_{i \in \mathcal{X}} \mu_i^X \log \frac{\mu_i^X}{p_i} + n \sum_{i \in \mathcal{X}} \mu_i^Y \log \frac{\mu_i^Y}{p_i} \\ &= m \sum_{i \in \mathcal{X}} \mu_i^X \log \frac{\mu_i^X}{\mu_i^0} + n \sum_{i \in \mathcal{X}} \mu_i^Y \log \frac{\mu_i^Y}{\mu_i^0} + \sum_{i \in \mathcal{X}} (m \mu_i^X + n \mu_i^Y) \log \frac{\mu_i^0}{p_i} \\ &= m \mathcal{K}(\mu^X, \mu^0) + n \mathcal{K}(\mu^Y, \mu^0) + (m+n) \mathcal{K}(\mu^0, p). \end{aligned}$$

By non-negativity of the KL divergence,

$$m \mathcal{K}(\mu^X, p) + n \mathcal{K}(\mu^Y, p) \geq m \mathcal{K}(\mu^X, \mu^0) + n \mathcal{K}(\mu^Y, \mu^0).$$

Therefore, whenever $m \mathcal{K}(\mu^X, \mu^0) + n \mathcal{K}(\mu^Y, \mu^0) > x$, one has $\mathbb{P}[\hat{p}^X = \mu^X, \hat{p}^Y = \mu^Y] \leq \exp(-x)$.

It remains to count empirical distributions. A sample of size m on a set of size d has at most $(m+1)^d$ empirical types, while a sample of size n has at most $(n+1)^d$ empirical types. Hence, the following inequality is satisfied:

$$\begin{aligned} \mathbb{P}[G > x] &\leq \sum_{\mu^X \in \mathcal{P}_m(\mathcal{X})} \sum_{\mu^Y \in \mathcal{P}_n(\mathcal{X})} \mathbf{1}\{m \mathcal{K}(\mu^X, \mu^0) + n \mathcal{K}(\mu^Y, \mu^0) > x\} \mathbb{P}[\hat{p}^X = \mu^X, \hat{p}^Y = \mu^Y] \\ &\leq (m+1)^d (n+1)^d \exp(-x). \end{aligned}$$

Since $m+1 \leq (m+n)$ and $n+1 \leq (m+n)$, we also have $(m+1)^d (n+1)^d \leq (m+n)^{2d}$. This inequality proves the first claim of the lemma.

For the second claim, take $x = 8(K+1) \log(m+n) + 8 \log(T+1)$. Using $d \leq K+1$, the first part gives the following inequality:

$$\mathbb{P}[G > x] \leq (m+n)^{2(K+1)} \exp(-x) = (m+n)^{-6(K+1)} (T+1)^{-8} \leq (m+n)^{-2(K+1)} (T+1)^{-4}.$$

The inequality completes in turn the proof of the lemma. $\blacksquare$

E.C.4.3 Proof of Lemma 11

We aim to evaluate the dynamic regret of a policy designed for static preferences when preferences evolve over time. We express such regret in terms of the regret incurred against static preferences.

Lemma 11. *Let $I = \{a, \dots, b\} \subseteq [T]$ be an interval, and let $F^{(\mathbb{N})} \in \mathcal{F}$ be some path of customers' preferences with $(F^t : s \in \mathbb{N}) \in \mathcal{F}_S(\gamma_T)$ for every $t \in I$. If the policy $\mathcal{A} \in \mathcal{P}$ is run on I , then:*

$$\sum_{t \in I} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I} r(S^{\mathcal{A},t}, F^t) \right] \leq \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I|) + c \|w\|_{\infty} \left(\sum_{t \in I} D_t + \sum_{t \in I} \left(\sum_{s=a}^{t-1} D_s^2 \right)^{1/2} \right),$$

where c depends only on the non-degeneracy parameter ς .

Proof. Let $n = |I|$ be the length of I . We compare the dynamic regret achieved by $\mathcal{A}$ on I (when started at period a), where customers' preferences evolve according to $(F^t : t \in I)$, with its regret under the static preferences F^a . Without loss of generality, we include the internal randomization of $\mathcal{A}$ in $\mathcal{H}^t$, the history generated by customers $\{a, \dots, t-1\}$, so that $S^{\mathcal{A},t}$ is $\mathcal{H}^t$ -measurable; in other words, $S^{\mathcal{A},t} \equiv \psi_t(\mathcal{H}^t)$ for a deterministic mapping $\psi_t(\cdot)$. We omit the dependence of $\mathcal{H}^t$ on a to simplify notation. Any randomized policy admits such a representation after augmenting the history with an auxiliary seed sampled at initialization. Throughout the proof, $\mathbb{P}_{[a:t]}^{\mathcal{A}}$ and $\mathbb{Q}_{[a:t]}^{\mathcal{A}}$ denote the distributions of these histories under the evolving preferences $(F^s : s \in I)$ and the static preferences F^a , respectively.

Step 1 (Revenue sensitivity to change in preferences). We first record a simple consequence of the definition of D_t . Fix a feasible assortment $S \in \mathcal{S}$ and some time period $t \in I$. By the triangle inequality and Pinsker's inequality (Tsybakov 2003), we obtain:

$$\begin{aligned} \sum_{i \in S \cup \{0\}} \left| p_i(S, F^t) - p_i(S, F^a) \right| &\leq \sum_{u=a+1}^t \sum_{i \in S \cup \{0\}} \left| p_i(S, F^u) - p_i(S, F^{u-1}) \right| \\ &\leq \sqrt{2} \sum_{u=a+1}^t \sqrt{\mathcal{K}^u(S)} \leq \sqrt{2} \sum_{u=a+1}^t \sqrt{m_u} = \sqrt{2} D_t. \end{aligned}$$

Consequently, for every $S \in \mathcal{S}$ and $t \in I$, one has $|r(S, F^t) - r(S, F^a)| \leq \sqrt{2} \|w\|_{\infty} D_t$.

Step 2 (Controlling the change in the history distribution). We next control how the distribution of the observed history changes when preferences evolve. Fix $t \in I$. Under $\mathbb{P}_{[a:t]}^{\mathcal{A}}$, conditional on $\mathcal{H}^s$, the purchase outcome in period s is generated by the assortment $S^{\mathcal{A},s} = \psi_s(\mathcal{H}^s)$ and the preferences F^s in period s . Under $\mathbb{Q}_{[a:t]}^{\mathcal{A}}$, the policy $\mathcal{A}$ selects assortments according to the same decision rule. Thus, after any given history, both processes select the same assortment, but the purchase outcome is generated under the static preferences F^a rather than the current preferences F^s . Hence, conditional on $\mathcal{H}^s$, the only difference between the two history distributions is the conditional purchasing distribution induced by the selected assortment. We now formalize this argument.

Let i^s denote the purchasing decision in period s . A realization of the history from period a to period $t - 1$ is written as $h_{[a:t]} = ((S^a, i^a), \dots, (S^{t-1}, i^{t-1}))$. The assortments appearing in this history must be consistent with the policy. That is, for each $s = a, \dots, t - 1$, $S^s = \psi_s(h_{[a:s]})$, with the convention that $h_{[a:a]}$ denotes the history available at the beginning of period a . If this consistency condition fails for at least one s , then both $\mathbb{P}_{[a:t]}^{\mathcal{A}}[h_{[a:t]}]$ and $\mathbb{Q}_{[a:t]}^{\mathcal{A}}[h_{[a:t]}]$ are equal to zero. We may therefore restrict attention to histories consistent with $\mathcal{A}$.

For such a history, the probability of $h_{[a:t]}$ under customers' evolving preferences is:

$$\mathbb{P}_{[a:t]}^{\mathcal{A}}[h_{[a:t]}] = \prod_{s=a}^{t-1} \pi^{\mathcal{A},s}(S^s \mid h_{[a:s]}) p_{i^s}(S^s, F^s),$$

where $\pi^{\mathcal{A},s}(S^s \mid h_{[a:s]})$ is the probability that policy $\mathcal{A}$ selects assortment S^s in period s , conditional on the history $h_{[a:s]}$; by the assumption made at the start of the proof, it is a Dirac mass. Similarly, the probability of the same history under the static preferences anchored at F^a is given by:

$$\mathbb{Q}_{[a:t]}^{\mathcal{A}}[h_{[a:t]}] = \prod_{s=a}^{t-1} \pi^{\mathcal{A},s}(S^s \mid h_{[a:s]}) p_{i^s}(S^s, F^a).$$

The factors $\pi^{\mathcal{A},s}(S^s \mid h_{[a:s]})$ are the same in both static and evolving preferences because the policy $\mathcal{A}$ only depends on the past history.

Hence, whenever the history has positive probability under $\mathbb{P}_{[a:t]}^{\mathcal{A}}$, these assortment-selection terms cancel in the likelihood ratio, and we obtain:

$$\log \left(\frac{\mathbb{P}_{[a:t]}^{\mathcal{A}}[h_{[a:t]}]}{\mathbb{Q}_{[a:t]}^{\mathcal{A}}[h_{[a:t]}]} \right) = \sum_{s=a}^{t-1} \log \left(\frac{p_{i^s}(S^s, F^s)}{p_{i^s}(S^s, F^a)} \right).$$

Taking expectations with respect to $\mathbb{P}_{[a:t]}^{\mathcal{A}}$, we obtain:

$$\mathcal{K}(\mathbb{P}_{[a:t]}^{\mathcal{A}}, \mathbb{Q}_{[a:t]}^{\mathcal{A}}) = \mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} \left[\sum_{s=a}^{t-1} \log \left(\frac{p_{i^s}(S^{\mathcal{A},s}, F^s)}{p_{i^s}(S^{\mathcal{A},s}, F^a)} \right) \right] = \sum_{s=a}^{t-1} \mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} \left[\log \left(\frac{p_{i^s}(S^{\mathcal{A},s}, F^s)}{p_{i^s}(S^{\mathcal{A},s}, F^a)} \right) \right].$$

Using the law of total expectations (Jacod and Protter 2004), and using that $S^{\mathcal{A},s}$ is $\mathcal{H}^s$ -measurable, we get the following sequence of equalities:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} \left[\log \left(\frac{p_{i^s}(S^{\mathcal{A},s}, F^s)}{p_{i^s}(S^{\mathcal{A},s}, F^a)} \right) \right] &= \mathbb{E}_{\mathbb{P}_{[a:s]}^{\mathcal{A}}} \left[\mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} \left[\log \left(\frac{p_{i^s}(S^{\mathcal{A},s}, F^s)}{p_{i^s}(S^{\mathcal{A},s}, F^a)} \right) \mid \mathcal{H}^s \right] \right] \\ &= \mathbb{E}_{\mathbb{P}_{[a:s]}^{\mathcal{A}}} \left[\sum_{i \in S^{\mathcal{A},s} \cup \{0\}} p_i(S^{\mathcal{A},s}, F^s) \log \left(\frac{p_i(S^{\mathcal{A},s}, F^s)}{p_i(S^{\mathcal{A},s}, F^a)} \right) \right] \\ &= \mathbb{E}_{\mathbb{P}_{[a:s]}^{\mathcal{A}}} \left[\mathcal{K} \left((p_i(S^{\mathcal{A},s}, F^s) : i \in S^{\mathcal{A},s} \cup \{0\}), (p_i(S^{\mathcal{A},s}, F^a) : i \in S^{\mathcal{A},s} \cup \{0\}) \right) \right]. \end{aligned}$$

The equality follows because $S^{\mathcal{A},s}$ is measurable with respect to $\mathcal{H}^s$.

Once the past history is fixed, the policy does not contribute to the conditional KL divergence. The conditional KL divergence associated with the realized assortment $S^{\mathcal{A},s}$ can be upper bounded by the largest such divergence over all feasible assortments in $\mathcal{S}$. Hence, one has:

$$\mathcal{K}(\mathbb{P}_{[a:t]}^{\mathcal{A}}, \mathbb{Q}_{[a:t]}^{\mathcal{A}}) \leq \sum_{s=a}^{t-1} \max \{ \mathcal{K}((p_i(S, F^s) : i \in S \cup \{0\}), (p_i(S, F^a) : i \in S \cup \{0\})) : S \in \mathcal{S} \}.$$

Using the inequality in Lemma 9 together with the definition of D_s and Step 1, we obtain, for

every $s \in I$ and $S \in \mathcal{S}$, the following inequality:

$$\mathcal{K}((p_i(S, F^s) : i \in S \cup \{0\}), (p_i(S, F^a) : i \in S \cup \{0\})) \leq 2\varsigma^{-1} D_s^2.$$

Hence, we obtain the following inequality:

$$\mathcal{K}(\mathbb{P}_{[a:t]}^{\mathcal{A}}, \mathbb{Q}_{[a:t]}^{\mathcal{A}}) \leq 2\varsigma^{-1} \sum_{s=a}^{t-1} D_s^2.$$

By applying Pinsker's inequality (Tsybakov 2003), we obtain:

$$\|\mathbb{P}_{[a:t]}^{\mathcal{A}} - \mathbb{Q}_{[a:t]}^{\mathcal{A}}\|_{\text{TV}} \leq \varsigma^{-1/2} \left(\sum_{s=a}^{t-1} D_s^2 \right)^{1/2}.$$

Step 3 (Sensitivity of revenue to the distribution of histories). We now compare the expected rewards generated by $\mathcal{A}$ under the two history distributions. For each period t , the random assortment $S^{\mathcal{A},t}$ is determined by the history before period t . Moreover, the mapping $S \mapsto r(S, F^a)$ is uniformly bounded by $\|w\|_{\infty}$. Therefore, by using Step 2, we obtain the following inequality:

$$|\mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] - \mathbb{E}_{\mathbb{Q}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)]| \leq \varsigma^{-1/2} \|w\|_{\infty} \left(\sum_{s=a}^{t-1} D_s^2 \right)^{1/2}.$$

Summing over $t \in I$ and using the triangle inequality, we obtain the following inequality:

$$\left| \sum_{t \in I} \mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] - \sum_{t \in I} \mathbb{E}_{\mathbb{Q}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] \right| \leq \varsigma^{-1/2} \|w\|_{\infty} \sum_{t \in I} \left(\sum_{s=a}^{t-1} D_s^2 \right)^{1/2}.$$

Step 4 (Decomposing dynamic regret). We next decompose the dynamic regret. Let $S^*(F^a)$ be an optimal assortment under the static preferences F^a . By Step 1, for every $t \in I$,

$$r(S^*(F^t), F^t) \leq r(S^*(F^t), F^a) + \sqrt{2} \|w\|_{\infty} D_t \leq r(S^*(F^a), F^a) + \sqrt{2} \|w\|_{\infty} D_t.$$

Hence, summing over $t \in I$, we obtain the following inequality:

$$\sum_{t \in I} r(S^*(F^t), F^t) \leq |I| r(S^*(F^a), F^a) + \sqrt{2} \|w\|_{\infty} \sum_{t \in I} D_t.$$

Similarly, applying the same argument to the realized assortment $S^{\mathcal{A},t}$, we have, pathwise:

$$\sum_{t \in I} r(S^{\mathcal{A},t}, F^t) \geq \sum_{t \in I} r(S^{\mathcal{A},t}, F^a) - \sqrt{2} \|w\|_{\infty} \sum_{t \in I} D_t.$$

Combining these two inequalities and taking the expectation with respect to $\mathbb{P}^{\mathcal{A}}$, we obtain:

$$\sum_{t \in I} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I} r(S^{\mathcal{A},t}, F^t) \right] \leq |I| r(S^*(F^a), F^a) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I} r(S^{\mathcal{A},t}, F^a) \right] + 2\sqrt{2} \|w\|_{\infty} \sum_{t \in I} D_t.$$

Step 5 (Regret bound). We now combine the preceding steps to prove the lemma. Since $\mathbb{P}_{[a:t]}^{\mathcal{A}}$ is the marginal law of the history up to period $t-1$ under the full law $\mathbb{P}^{\mathcal{A}}$, every $\mathcal{H}^t$ -measurable random variable Z_t satisfies $\mathbb{E}_{\mathbb{P}^{\mathcal{A}}} [Z_t] = \mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} [Z_t]$. The same identity holds under the static law $\mathbb{Q}^{\mathcal{A}}$. In particular, since $S^{\mathcal{A},t}$ is $\mathcal{H}^t$ -measurable, the random variable $r(S^{\mathcal{A},t}, F^a)$ may be evaluated under the corresponding marginal law $\mathbb{P}_{[a:t]}^{\mathcal{A}}$ or $\mathbb{Q}_{[a:t]}^{\mathcal{A}}$. Using Step 4, and then adding and subtracting the expected revenue under the static preferences, we obtain the following identity:

$$\begin{aligned} |I| r(S^*(F^a), F^a) - \sum_{t \in I} \mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] &= |I| r(S^*(F^a), F^a) - \sum_{t \in I} \mathbb{E}_{\mathbb{Q}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] \\ &\quad + \sum_{t \in I} \mathbb{E}_{\mathbb{Q}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] - \sum_{t \in I} \mathbb{E}_{\mathbb{P}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)]. \end{aligned}$$

Under $\mathbb{Q}_{[a:t]}^{\mathcal{A}}$, the policy $\mathcal{A}$ is run in an environment with static preferences F^a . Therefore, by the definition of the regret bound under static preferences, one has:

$$|I|r(S^*(F^a), F^a) - \sum_{t \in I} \mathbb{E}_{\mathbb{Q}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] \leq \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I|).$$

Moreover, by Step 3, the following inequality holds:

$$\left| \sum_{t \in I} \mathbb{E}_{\mathbb{Q}_{[a:t]}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] - \sum_{t \in I} \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} [r(S^{\mathcal{A},t}, F^a)] \right| \leq \varsigma^{-1/2} \|w\|_{\infty} \sum_{t \in I} \left(\sum_{s=a}^{t-1} D_s^2 \right)^{1/2}.$$

Therefore, combining the preceding bounds with Step 4, we obtain the following inequality:

$$\sum_{t \in I} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I} r(S^{\mathcal{A},t}, F^t) \right] \leq \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I|) + c \|w\|_{\infty} \left[\sum_{t \in I} D_t + \sum_{t \in I} \left(\sum_{s=a}^{t-1} D_s^2 \right)^{1/2} \right],$$

where we define $c := \max\{\varsigma^{-1/2}, 2\sqrt{2}\}$. This inequality proves the lemma. $\blacksquare$

E.C.4.4 Proof of Corollary 5

We now specialize Lemma 11 by expressing its bound in terms of the magnitude of customers' preferences over the segment of interest. Formally:

Corollary 5. *Let $I = \{a, \dots, b\} \subseteq [T]$ be an interval, and let $F^{(\mathbb{N})} \in \mathcal{F}$ be some path of customers' preferences with $(F^t : s \in \mathbb{N}) \in \mathcal{F}_S(\gamma_T)$ for every $t \in I$. If the policy $\mathcal{A} \in \mathcal{P}$ is run on I , then:*

$$\sum_{t \in I} r(S^*(F^t), F^t) - \mathbb{E}_{\mathbb{P}^{\mathcal{A}}} \left[\sum_{t \in I} r(S^{\mathcal{A},t}, F^t) \right] \leq \mathcal{R}^{\mathcal{A}}(\mathcal{F}_S(\gamma_T), |I|) + 2c \|w\|_{\infty} |I|^2 \sqrt{\mathcal{M}_I},$$

where c depends only on the non-degeneracy parameter ς .

Proof. Let $n = |I|$. The result follows from Lemma 11 by upper bounding the two terms involving D_t in that lemma. First, by the definition of D_t and the Cauchy–Schwarz inequality, for every $t \in I$:

$$D_t = \sum_{u=a+1}^t \sqrt{m_u} \leq (t-a)^{1/2} \left(\sum_{u=a+1}^t m_u \right)^{1/2} \leq n^{1/2} \sqrt{\mathcal{M}_I}.$$

Therefore, one has $\sum_{t \in I} D_t \leq n^{3/2} \sqrt{\mathcal{M}_I} \leq n^2 \sqrt{\mathcal{M}_I}$.

Second, using again the preceding bound on D_t , we obtain $\sum_{s=a}^{t-1} D_s^2 \leq \sum_{s \in I} D_s^2 \leq n(n\mathcal{M}_I) = n^2 \mathcal{M}_I$. Consequently, for every $t \in I$, one has $(\sum_{s=a}^{t-1} D_s^2)^{1/2} \leq n \sqrt{\mathcal{M}_I}$. Summing over $t \in I$ gives:

$$\sum_{t \in I} \left(\sum_{s=a}^{t-1} D_s^2 \right)^{1/2} \leq n^2 \sqrt{\mathcal{M}_I}.$$

The desired bound then directly follows. $\blacksquare$

E.C.4.5 Proof of Lemma 12

Lemma 12. *For every interval $I = \{a, \dots, b\}$ and every path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}$,*

$$\sum_{t \in I} r(S^*(F^t), F^t) - \max \left\{ \sum_{t \in I} r(S, F^t) : S \in \mathcal{S} \right\} \leq 2|I| \mathcal{V}_I(F^{(\mathbb{N})}).$$

Proof. Fix an arbitrary reference period $\tau_I \in I$. By the definition of $\mathcal{V}_I(F^{(\mathbb{N})})$ and the triangle inequality, for every $S \in \mathcal{S}$ and every $t \in I$, one has $|r(S, F^t) - r(S, F^{\tau_I})| \leq \mathcal{V}_I(F^{(\mathbb{N})})$. Indeed,

the left-hand side is bounded by the sum of one-step changes between t and τ_I , and this sum is bounded by $\mathcal{V}_I(F^{(\mathbb{N})})$. Since $S^*(F^{\tau_I})$ is the assortment that maximizes the expected revenue under preferences F^{τ_I} , we have, for every $t \in I$, $r(S^*(F^t), F^{\tau_I}) \leq r(S^*(F^{\tau_I}), F^{\tau_I})$.

Taken together, we obtain the following sequence of inequalities:

$$\begin{aligned} r(S^*(F^t), F^t) &\leq r(S^*(F^t), F^{\tau_I}) + \mathcal{V}_I(F^{(\mathbb{N})}) \leq r(S^*(F^{\tau_I}), F^{\tau_I}) + \mathcal{V}_I(F^{(\mathbb{N})}) \\ &\leq r(S^*(F^{\tau_I}), F^t) + 2\mathcal{V}_I(F^{(\mathbb{N})}). \end{aligned}$$

Therefore, for all $t \in I$, the following inequality is satisfied:

$$r(S^*(F^t), F^t) - r(S^*(F^{\tau_I}), F^t) \leq 2\mathcal{V}_I(F^{(\mathbb{N})}).$$

Summing over $t \in I$ gives the following inequality:

$$\sum_{t \in I} r(S^*(F^t), F^t) - \sum_{t \in I} r(S^*(F^{\tau_I}), F^t) \leq 2|I|\mathcal{V}_I(F^{(\mathbb{N})}).$$

Finally, since the best fixed assortment on I performs at least as well as $S^*(F^{\tau_I})$, we have:

$$\sum_{t \in I} r(S^*(F^{\tau_I}), F^t) \leq \max \left\{ \sum_{t \in I} r(S, F^t) : S \in \mathcal{S} \right\}.$$

Combining the last two inequalities proves the result. ■

E.C.4.6 Proof of Lemma 13

Lemma 13. $J^\pi(F^{(\mathbb{N})}, T) = \mathbb{E}_{\mathbb{P}_\pi}[\mathcal{J}^\pi(F^{(\mathbb{N})}, T)]$, for every $F^{(\mathbb{N})} \in \mathcal{F}_A$, every $\pi \in \mathcal{P}$, and every $T \geq 2$.

Proof. We use ψ_t instead of $\psi_t(\mathcal{H}^t)$ for $t \in [T]$ to simplify our notation. Without loss of generality, we assume that the internal randomization used by π to select $\psi_t(\mathcal{H}^t)$ is included in $\mathcal{H}^t$, so that $\psi_t(\mathcal{H}^t)$ is $\mathcal{H}^t$ -measurable. Otherwise, one may augment $\mathcal{H}^t$ with this randomization and follow the same proof. Fix $F^{(\mathbb{N})} \in \mathcal{F}_A$. Recall that, for each $t \in \mathbb{N}$, F^t denotes the distribution of customer t 's utilities over the products in $\mathcal{N}$ and the outside option, as defined in Section 3. Since preferences undergo at most one change at period $\tau \in \{2, \dots, T+1\}$, we denote by U^a a random utility vector distributed according to F^1 , and by U^b a random utility vector distributed according to F^τ . By convention, $\sum_{t=\tau}^T a_t = 0$ if $\tau = T+1$ for any sequence $(a_t : t \in \mathbb{N})$. Then:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_\pi}[\mathcal{J}^\pi(F^{(\mathbb{N})}, T)] &= \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=1}^T \sum_{i \in \psi_t} w_i \mathbf{1}\{i_t = i\} \right] = \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=1}^{\tau-1} \sum_{i \in \psi_t} w_i \mathbf{1}\{i_t = i\} \right] + \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=\tau}^T \sum_{i \in \psi_t} w_i \mathbf{1}\{i_t = i\} \right] \\ &= \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=1}^{\tau-1} \sum_{i \in \psi_t} w_i \mathbf{1}\{i_t = i \in \psi_t, U_{i_t}^a > U_j^a, \forall j \in \psi_t \cup \{0\} \setminus \{i_t\}\} \right] \\ &\quad + \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{t=\tau}^T \sum_{i \in \psi_t} w_i \mathbf{1}\{i_t = i \in \psi_t, U_{i_t}^b > U_j^b, \forall j \in \psi_t \cup \{0\} \setminus \{i_t\}\} \right] \\ &\stackrel{(a)}{=} \sum_{t=1}^{\tau-1} \mathbb{E}_{\mathbb{P}_\pi} \left[\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{i \in \psi_t} w_i \mathbf{1}\{i_t = i \in \psi_t, U_{i_t}^a > U_j^a, \forall j \in \psi_t \cup \{0\} \setminus \{i_t\}\} \mid \mathcal{H}^t, \psi_t \right] \right] \\ &\quad + \sum_{t=\tau}^T \mathbb{E}_{\mathbb{P}_\pi} \left[\mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{i \in \psi_t} w_i \mathbf{1}\{i_t = i \in \psi_t, U_{i_t}^b > U_j^b, \forall j \in \psi_t \cup \{0\} \setminus \{i_t\}\} \mid \mathcal{H}^t, \psi_t \right] \right] \end{aligned}$$

$$\begin{aligned}
&\stackrel{(b)}{=} \sum_{t=1}^{\tau-1} \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{i \in \psi_t} w_i \mathbb{E}_{\mathbb{P}_\pi} [\mathbf{1}\{i_t = i \in \psi_t, U_{i_t}^a > U_j^a, \forall j \in \psi_t \cup \{0\} \setminus \{i_t\}\} \mid \mathcal{H}^t, \psi_t] \right] \\
&\quad + \sum_{t=\tau}^T \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{i \in \psi_t} w_i \mathbb{E}_{\mathbb{P}_\pi} [\mathbf{1}\{i_t = i \in \psi_t, U_{i_t}^b > U_j^b, \forall j \in \psi_t \cup \{0\} \setminus \{i_t\}\} \mid \mathcal{H}^t, \psi_t] \right] \\
&\stackrel{(c)}{=} \sum_{t=1}^{\tau-1} \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{i \in \psi_t} w_i p_i(\psi_t, F^1) \right] + \sum_{t=\tau}^T \mathbb{E}_{\mathbb{P}_\pi} \left[\sum_{i \in \psi_t} w_i p_i(\psi_t, F^\tau) \right] = J^\pi(F^{(\mathbb{N})}, T),
\end{aligned}$$

where (a) follows from the law of total expectations (Jacod and Protter 2004), (b) uses finiteness and measurability of the assortment selected by the strategy π , and (c) is a consequence of the definition of the purchasing distribution $(p_i(S, F) : i \in \mathcal{N} \cup \{0\})$ and that the utility draw at period t is independent of the history $\mathcal{H}^t$ and the strategy's randomization (if any). $\blacksquare$

E.C.4.7 Proof of Lemma 14

We first derive a closed-form expression for the probability measure induced by a strategy over a finite selling horizon. Since policies may be randomized, we state the result for the full history. Specifically, we use $\mathcal{H}^{T+1} := ((S^t, i^t) : t \in [T])$, where $S^t \in \mathcal{S}$ is the assortment selected for customer t , and $i^t \in \mathcal{N}_0$ is the purchasing decision of customer t , to describe the history up to period T . For a complete realization $h^{T+1} := ((S^t, I^t) : t \in [T])$, we define its realization up to period $t-1$ by $h^t := ((S^s, I^s) : s \in [t-1])$, with h^1 denoting the empty history. Let $\pi \in \mathcal{P}$ be an admissible strategy with decision rules $\pi = (S^{\pi, t} : t \in [T])$. For a randomized strategy $\pi \in \mathcal{P}$, let $\pi^t(S \mid h^t)$ denote the probability that π selects assortment S in period t , conditional on the history h^t . If π is deterministic, then $\pi^t(S \mid h^t)$ is the indicator function for $S^{\pi, t} = S$.

Lemma 14. *Fix a path of customers' preferences $F^{(\mathbb{N})} \in \mathcal{F}_A$ with an abrupt change at period $\tau \leq T$. For any full history realization $h^{T+1} = ((S^t, I^t) : t \in [T])$, we have:*

$$\mathbb{P}_\pi^\pi[\mathcal{H}^{T+1} = h^{T+1}] = \prod_{t=1}^{\tau-1} \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^1) \prod_{t=\tau}^T \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^\tau),$$

Moreover, if $I^t \notin S^t \cup \{0\}$ for some $t \in [T]$, then $\mathbb{P}_\pi^\pi[\mathcal{H}^{T+1} = h^{T+1}] = 0$.

Proof. Fix a full-history realization $h^{T+1} = ((S^t, I^t) : t \in [T])$. By the chain rule, we have $\mathbb{P}_\pi^\pi[\mathcal{H}^{T+1} = h^{T+1}] = \prod_{t=1}^T \mathbb{P}_\pi^\pi[S^{\pi, t} = S^t, i^t = I^t \mid \mathcal{H}^t = h^t]$. For every $t \in [T]$, the conditional probability inside the product can be decomposed into the probability of selecting the assortment and the probability of observing the purchasing decision after the assortment has been selected. Thus,

$$\begin{aligned}
\mathbb{P}_\pi^\pi[S^{\pi, t} = S^t, i^t = I^t \mid \mathcal{H}^t = h^t] &= \mathbb{P}_\pi^\pi[S^{\pi, t} = S^t \mid \mathcal{H}^t = h^t] \mathbb{P}_\pi^\pi[i^t = I^t \mid \mathcal{H}^t = h^t, S^{\pi, t} = S^t] \\
&= \pi^t(S^t \mid h^t) \mathbb{P}_\pi^\pi[i^t = I^t \mid \mathcal{H}^t = h^t, S^{\pi, t} = S^t].
\end{aligned}$$

Conditional on the history h^t and the selected assortment S^t , the purchase outcome in period t is generated according to the customers' preferences prevailing in that period. Since the change

occurs at τ , customers' preferences are F^1 for $t < \tau$ and F^τ for $t \geq \tau$. Therefore, we have:

$$\mathbb{P}_\tau^\pi \left[i^t = I^t \mid \mathcal{H}^t = h^t, S^{\pi,t} = S^t \right] = p_{I^t}(S^t, F^1),$$

for $t < \tau$ and, for $t \geq \tau$, one has:

$$\mathbb{P}_\tau^\pi \left[i^t = I^t \mid \mathcal{H}^t = h^t, S^{\pi,t} = S^t \right] = p_{I^t}(S^t, F^\tau).$$

Substituting these two identities into the chain-rule decomposition gives:

$$\mathbb{P}_\tau^\pi[\mathcal{H}^{T+1} = h^{T+1}] = \prod_{t=1}^{\tau-1} \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^1) \prod_{t=\tau}^T \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^\tau).$$

This identity proves the product formula.

It remains to prove the zero-probability claim. Suppose that $I^t \notin S^t \cup \{0\}$ for some $t \in [T]$. By definition of the conditional purchasing probabilities, a customer cannot purchase a product that is not included in the selected assortment. For every preferences F , $p_{I^t}(S^t, F) = 0$ whenever $I^t \notin S^t \cup \{0\}$. If $t < \tau$, then the factor $p_{I^t}(S^t, F^1)$ in the product formula is equal to zero. If $t \geq \tau$, then the factor $p_{I^t}(S^t, F^\tau)$ in the product formula is equal to zero. In both cases, one factor in the product representation of $\mathbb{P}_\tau^\pi[\mathcal{H}^{T+1} = h^{T+1}]$ is zero. Thus, $\mathbb{P}_\tau^\pi[\mathcal{H}^{T+1} = h^{T+1}] = 0$, which concludes the proof. ■

E.C.4.8 Proof of Lemma 15

We now compare two probability measures induced by a strategy and the some pre- and post-change preferences, but with different candidate change times. The full-history formula in Lemma 14 is useful here because the assortment-selection probabilities are identical under the two measures. Hence, the terms involving $\pi^t(\cdot \mid h^t)$ cancel in the likelihood ratio. The remaining terms are the purchasing distributions, which differ only over the periods between the two candidate change times.

Lemma 15. *Let $\ell_j, \ell_{j+1} \in \{2, \dots, T\}$ with $\ell_j < \ell_{j+1}$. Then, one has:*

$$\mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) = \sum_{t=\ell_j}^{\ell_{j+1}-1} \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\log \frac{p_{i^t}(S^{\pi,t}, F^1)}{p_{i^t}(S^{\pi,t}, F^\tau)} \right].$$

Proof. By definition of the KL divergence, the following identity holds:

$$\mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) = \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\log(\mathbb{P}_{\ell_{j+1}}^\pi[\mathcal{H}^{T+1}] / \mathbb{P}_{\ell_j}^\pi[\mathcal{H}^{T+1}]) \right],$$

where $\mathcal{H}^{T+1} = ((S^{\pi,t}, i^t) : t \in [T])$ denotes the full (random) history. Fix a full-history realization $h^{T+1} = ((S^t, I^t) : t \in [T])$ with positive probability under $\mathbb{P}_{\ell_{j+1}}^\pi$. By Lemma 14, we have:

$$\mathbb{P}_{\ell_{j+1}}^\pi[\mathcal{H}^{T+1} = h^{T+1}] = \prod_{t=1}^{\ell_{j+1}-1} \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^1) \prod_{t=\ell_{j+1}}^T \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^\tau).$$

Similarly, by applying once again Lemma 14, we obtain:

$$\mathbb{P}_{\ell_j}^\pi[\mathcal{H}^{T+1} = h^{T+1}] = \prod_{t=1}^{\ell_j-1} \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^1) \prod_{t=\ell_j}^T \pi^t(S^t \mid h^t) p_{I^t}(S^t, F^\tau).$$

The factors $\pi^t(S^t \mid h^t)$ are the same in the two expressions, because the same strategy is used under the two candidate change times. Hence, these factors cancel in the likelihood ratio. The

purchasing probabilities also cancel for all periods $t < \ell_j$ and all periods $t \geq \ell_{j+1}$. Therefore,

$$\frac{\mathbb{P}_{\ell_{j+1}}^\pi[\mathcal{H}^{T+1} = h^{T+1}]}{\mathbb{P}_{\ell_j}^\pi[\mathcal{H}^{T+1} = h^{T+1}]} = \prod_{t=\ell_j}^{\ell_{j+1}-1} \frac{p_{I^t}(S^t, F^1)}{p_{I^t}(S^t, F^\tau)}.$$

This ratio is well-defined on the support of $\mathbb{P}_{\ell_{j+1}}^\pi$. Indeed, if $\mathbb{P}_{\ell_{j+1}}^\pi[\mathcal{H}^{T+1} = h^{T+1}] > 0$, then $I^t \in S^t \cup \{0\}$ for every $t \in [T]$. By the non-degeneracy condition from $\mathcal{F}$, the corresponding purchasing probabilities under F^1 and F^τ are strictly positive; recall Section 3.

Taking logarithms on both sides then gives the following identity:

$$\log \frac{\mathbb{P}_{\ell_{j+1}}^\pi[\mathcal{H}^{T+1} = h^{T+1}]}{\mathbb{P}_{\ell_j}^\pi[\mathcal{H}^{T+1} = h^{T+1}]} = \sum_{t=\ell_j}^{\ell_{j+1}-1} \log \frac{p_{I^t}(S^t, F^1)}{p_{I^t}(S^t, F^\tau)}.$$

Taking expectation with respect to $\mathbb{P}_{\ell_{j+1}}^\pi$, we obtain:

$$\mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) = \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\sum_{t=\ell_j}^{\ell_{j+1}-1} \log \frac{p_{I^t}(S^{\pi,t}, F^1)}{p_{I^t}(S^{\pi,t}, F^\tau)} \right] = \sum_{t=\ell_j}^{\ell_{j+1}-1} \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\log \frac{p_{I^t}(S^{\pi,t}, F^1)}{p_{I^t}(S^{\pi,t}, F^\tau)} \right],$$

which proves the result. $\blacksquare$

E.C.4.9 Proof of Lemma 16

We first define the *maximum* KL divergence between the pre-change and post-change preferences, conditional on a feasible assortment being offered. For any $S \in \mathcal{S}$, we introduce:

$$\mathcal{K}(F^1, F^\tau; S) := \sum_{i \in S \cup \{0\}} p_i(S, F^1) \log \left(\frac{p_i(S, F^1)}{p_i(S, F^\tau)} \right).$$

This is the KL divergence between the conditional purchasing distributions induced by F^1 and F^τ when assortment S is selected. By the non-degeneracy condition defining $\mathcal{F}$, all probabilities $p_i(S, F)$, with $i \in S \cup \{0\}$ and $F \in \{F^1, F^\tau\}$, are strictly positive and bounded away from zero. Hence, $\mathcal{K}(F^1, F^\tau; S)$ is finite for every $S \in \mathcal{S}$. Since $\mathcal{S}$ is finite, we may define with some abuse of notation:

$$\mathcal{K}(F^1, F^\tau) := \max \left\{ \mathcal{K}(F^1, F^\tau; S) : S \in \mathcal{S} \right\}.$$

Lemma 16. *If F^1 and F^τ are distinguishable on a feasible assortment, then $0 < \mathcal{K}(F^1, F^\tau) < \infty$.*

Proof. For every $S \in \mathcal{S}$, the non-negativity of KL divergence gives $\mathcal{K}(F^1, F^\tau; S) \geq 0$. Moreover, by the non-degeneracy condition, $p_i(S, F^\tau) \geq \varsigma$ for all $i \in S \cup \{0\}$. Therefore,

$$\mathcal{K}(F^1, F^\tau; S) = \sum_{i \in S \cup \{0\}} p_i(S, F^1) \log \left(\frac{p_i(S, F^1)}{p_i(S, F^\tau)} \right) < \infty.$$

Since $\mathcal{S}$ is finite, it follows that $0 \leq \mathcal{K}(F^1, F^\tau) < \infty$.

It remains to show strict positivity. Suppose, toward a contradiction, that $\mathcal{K}(F^1, F^\tau) = 0$. Then, $\mathcal{K}(F^1, F^\tau; S) = 0$ for every $S \in \mathcal{S}$. By the equality condition for KL divergence, we obtain $p_i(S, F^1) = p_i(S, F^\tau)$ for every $S \in \mathcal{S}$ and every $i \in S \cup \{0\}$. Hence, F^1 and F^τ induce the same purchasing distribution for every assortment. This conclusion contradicts the assumption that F^1 and F^τ are distinguishable on at least one assortment. Thus, $\mathcal{K}(F^1, F^\tau) > 0$, which completes the proof. $\blacksquare$

E.C.4.10 Proof of Lemma 17

Fix $T \geq 9$ and a subsegment length $\Delta \in [T-2]$. Let $\tilde{T} := \lfloor (T-2)/\Delta \rfloor + 1$ and $\ell_j := 2 + (j-1)\Delta$ for $j \in [\tilde{T}]$. Then, $\ell_{\tilde{T}} \leq T$, and each subsegment $\{\ell_j, \dots, \ell_{j+1}-1\}$ for $j \in [\tilde{T}-1]$, contains exactly Δ periods. Given two feasible assortments $S, \tilde{S} \in \mathcal{S}$, we denote by $\|S - \tilde{S}\|_1$ the Hamming distance between their binary encodings, namely the number of products on which they differ.

Lemma 17. *If F^1 and F^τ are equal conditional on $S^*(F^1)$, then, for $j \in [\tilde{T}-1]$,*

$$\mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\sum_{t=\ell_j}^{\ell_{j+1}-1} \mathbf{1} \left\{ \|S^{\pi,t} - S^*(F^1)\|_1 > 0 \right\} \right] \geq \frac{\mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi)}{\mathcal{K}(F^1, F^\tau)}.$$

Proof. By Lemma 15, we have:

$$\mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) = \sum_{t=\ell_j}^{\ell_{j+1}-1} \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\log \frac{p_{it}(S^{\pi,t}, F^1)}{p_{it}(S^{\pi,t}, F^\tau)} \right].$$

Fix $t \in \{\ell_j, \dots, \ell_{j+1}-1\}$. Under $\mathbb{P}_{\ell_{j+1}}^\pi$, the change occurs at period ℓ_{j+1} . Therefore, period t is still governed by the preferences F^1 before the change. Conditional on $\mathcal{H}^t$ and on the selected assortment $S^{\pi,t}$, the purchasing decision i^t is distributed according to the discrete probabilities $(p_i(S^{\pi,t}, F^1) : i \in S^{\pi,t} \cup \{0\})$. By the law of total expectations (Jacod and Protter 2004), one has:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\log \frac{p_{it}(S^{\pi,t}, F^1)}{p_{it}(S^{\pi,t}, F^\tau)} \right] &= \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\log \frac{p_{it}(S^{\pi,t}, F^1)}{p_{it}(S^{\pi,t}, F^\tau)} \mid \mathcal{H}^t, S^{\pi,t} \right] \right] \\ &= \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\sum_{i \in S^{\pi,t} \cup \{0\}} p_i(S^{\pi,t}, F^1) \log \left(\frac{p_i(S^{\pi,t}, F^1)}{p_i(S^{\pi,t}, F^\tau)} \right) \right] \\ &= \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\mathcal{K}(F^1, F^\tau; S^{\pi,t}) \right]. \end{aligned}$$

By assumption, F^1 and F^τ are equal conditional on $S^*(F^1)$. Therefore, we have $\mathcal{K}(F^1, F^\tau; S^*(F^1)) = 0$. Using this equality, we obtain:

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\mathcal{K}(F^1, F^\tau; S^{\pi,t}) \right] &= \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\mathcal{K}(F^1, F^\tau; S^{\pi,t}) \mathbf{1} \left\{ S^{\pi,t} \neq S^*(F^1) \right\} \right] \\ &\leq \mathcal{K}(F^1, F^\tau) \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\mathbf{1} \left\{ S^{\pi,t} \neq S^*(F^1) \right\} \right] \\ &= \mathcal{K}(F^1, F^\tau) \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\mathbf{1} \left\{ \|S^{\pi,t} - S^*(F^1)\|_1 > 0 \right\} \right]. \end{aligned}$$

The inequality follows from the definition of $\mathcal{K}(F^1, F^\tau)$.

Combining the preceding bounds and summing over $t \in \{\ell_j, \dots, \ell_{j+1}-1\}$, we obtain:

$$\begin{aligned} \mathcal{K}(\mathbb{P}_{\ell_{j+1}}^\pi, \mathbb{P}_{\ell_j}^\pi) &\leq \mathcal{K}(F^1, F^\tau) \sum_{t=\ell_j}^{\ell_{j+1}-1} \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\mathbf{1} \left\{ \|S^{\pi,t} - S^*(F^1)\|_1 > 0 \right\} \right] \\ &= \mathcal{K}(F^1, F^\tau) \mathbb{E}_{\mathbb{P}_{\ell_{j+1}}^\pi} \left[\sum_{t=\ell_j}^{\ell_{j+1}-1} \mathbf{1} \left\{ \|S^{\pi,t} - S^*(F^1)\|_1 > 0 \right\} \right]. \end{aligned}$$

Since $\mathcal{K}(F^1, F^\tau) > 0$ by Lemma 16, dividing both sides by $\mathcal{K}(F^1, F^\tau)$ gives the desired inequality. $\blacksquare$

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